Language as Visible Signal: Toward an Image-Native Language Model in Continuous State and Pixel Space

Imagized Language Model Project

August 2026

Abstract

Most language models make a finite symbolic encoding the primary interface to language. That interface is effective for modern encoded text but incomplete for historical, handwritten, regional, damaged, and unencoded writing, where visual form is part of the evidence. We study a stricter image-native contract: the deployed model receives writing pixels, maintains continuous visual states, predicts continuous visual futures, emits a raster answer, and rereads the visible pixels it generated. OCR and coded transcription remain optional post-hoc views rather than the model's language substrate.

We implement this contract through staged, falsifiable experiments. V34 is a codebook-free convolutional codec with 7.42 million parameters for 32×32 writing patches. It qualifies on held-font and held-family development and sealed audits, reaching 0.9978 rendered ink F1 and 0.9981 historical ink F1 on the sealed split with 2.38 GiB peak allocated memory. Reconstruction is only an interface result. V35 combines the frozen codec with a residual interface adapter, a 12-layer PIXAR-initialized causal transformer, deterministic anchor and conditional rectified-flow writers, a stop head, and a strict decode–threshold–re-encode feedback loop. PIXAR is an external causal foundation, not a project contribution; V35 is transfer-based rather than from-scratch pretraining.

The 129.09M-parameter V35 student completes 22,000 finite BF16 updates in 2.77 hours on one RTX 4090 with 2.899 GiB peak allocated memory. Its packaged runtime accepts only pixels and a visual-position mask and calls no tokenizer, Unicode or character IDs, OCR, lookup table, retrieval system, codebook, or teacher. Nevertheless, the fixed development audit rejects the mechanism. Autonomous copy accuracy is 0.3125%, retaining only 0.491% of its target-through-codec OCR ceiling; correct prompts improve over shuffled prompts by only 0.117 percentage point. Instruction accuracy is 0.1116% and is 0.445 point worse than the shuffled-prompt condition. The visual-causal and semantic-raster routes pass only 5/12 and 5/9 gates, respectively. Generated pixels are finite, nonblank, and prompt-responsive, but contain corrupted glyph-like marks rather than correctly bound answers. The decision is `not-qualified`, and the sealed split remains unopened. Thus V35 establishes a complete one-GPU raster-input/raster-output experimental runtime, not useful autonomous language, parity with text LLMs, or superior compute efficiency.

V36-P then isolates answer-level planning. Its 93.47M-parameter runtime maps only prompt pixels and a patch mask to five continuous plans and visual length; answer targets, candidates, text, OCR, and symbolic identifiers remain outside deployment. It completes 6,000 finite updates in 25.63 minutes with 1.541 GiB peak allocated memory. The fixed development gate nevertheless rejects it: EMA top-1/top-5 retrieval is 1.02/12.24%, raw top-1 is 2.04%, and only 13/23 conditions pass. Counterfactual assignment reaches 78.57%, but absolute retrieval, blank control, font and paraphrase transfer, and length fail. A post-result nonsealed audit finds that measuring occupancy after augmentation inflated mean train length to 37.31 patches versus

11.53 in development and left the train target space at effective rank 4.77/768. The sealed split stays closed and no renderer is opened.

V37 implements the resulting clean-mask, end-to-end visual semantic distillation test. Its 89.77M-parameter image-only student uses an exactly attributed Pixel-Linguist initialization and detached BGE-M3 targets prepared offline; neither text nor teacher is present at runtime. It completes 8,000 finite updates in 61.59 minutes with 2.748 GiB peak allocated memory. Prompt reading reaches 47.45/77.55% top-1/top-5, while candidate-independent answer plans reach 20.41/45.41% and 0.3377 MRR. Counterfactual assignment reaches 93.88% and paraphrase top-5 reaches 73.33%, but absolute answer retrieval, cosine margins, font and wording invariance, and length fail. The EMA route passes 20/33 frozen conditions and is `not-qualified`; sealed data and the raster renderer remain closed. V37 therefore establishes a practical one-GPU visual-semantic advance, not autonomous language or image generation.

V38 tests the diagnosed invariance and near-identity transition failures with five-font paired visual paths, 1,024 audited training paraphrases, semantically near negatives, and an unconstrained answer map. Its 90.75M-parameter image-only student completes 8,000 finite BF16 updates in 99.71 minutes with 2.969 GiB peak allocated memory on one RTX 4090 D. Prompt top-1/top-5 rises to 60.71/86.73%, held-font prompt consistency reaches 0.7925, and mean prompt-answer cosine falls from 0.9972 to 0.5769. Prompt-conditioned answer top-1/top-5 remains only 21.94/49.49%, however; answer cosine, held-font and paraphrase answer consistency, and length remain below gate. EMA passes 25/39 conditions and raw weights do not repair the result. V38 is `not-qualified`; zero sealed rows are rendered and no raster renderer is authorized.

V39.1 next corrects the bounded first-stop distribution, order negatives, length prior, loss balance, and short-run EMA in a matched 512-record trajectory pilot. Expected answer-span count becomes 6.303 against 6.313 target and count-mode accuracy reaches 15.22%, but answer MRR falls to 0.0741, segment MRR remains 0.0123, and transition-direction cosine is only 0.0065. We do not scale this unqualified semantic mechanism. V41 instead isolates the visible motor. The qualified V34 codec feeds exact, 32-pixel, clean-projected, and latent-perturbed glyph rasters into the pinned 22.76M-parameter MX-Font generator with four style-reference images. Clean projected ink F1 rises from 0.4318 to 0.5479; at latent noise $\sigma = 0.05$ it rises from 0.4317 to 0.5461 and retains 99.66% of clean projected-motor F1. All outputs are finite and nonblank. This qualifies an image-conditioned glyph motor, not autonomous language: the target glyph image is supplied to this audit and the model does not choose the next glyph.

V42 then removes that target supply and trains a 24.35M-parameter causal model directly on ordered $64 \times 1 \times 32 \times 32$ Chinese glyph rasters. On 2,048 development windows, full-history next-image top-1 is 19.97%, above image unigram (1.42%), symbolic bigram (12.26%), and shuffled earlier history (18.36%); ordered target log probability gains 0.174 nat over shuffled history. This is the first bounded positive natural-language result in the project, not a complete ILM: exact-suffix assignment is 53.03% and generated pixel F1 is 0.373. V43 adds same-suffix pair supervision and a 5.69M-parameter bank-free spatial rectified-flow writer. It preserves every language-control win and raises generated pixel F1 to 0.445, but counterfactual assignment is only 54.49%; the fixed 0.60 and 0.55 gates fail. A post-result diagnostic finds 99.22% seen-pair but only 58.30% unseen-train accuracy, while evaluator exact ink plans let the same writer reach 0.882 F1. V43 is partial: pair memorization and the autonomous visual plan, rather than writer capacity, are now the dominant bottlenecks. V44 freezes V42 and trains a 1.736M-parameter tangent residual in one pass over 24,000 unique raster pairs. Consumed and unseen-train accuracy become nearly equal (57.47% and 57.42%), removing the V43 memorization gap, but development binding reaches only 53.42% and matched natural top-1 falls from 19.43% to 17.43%. A post-result audit finds that the learned update is negatively aligned with target-image displacement and drifts toward the corpus-common image field. V44 passes 8/14 gates and is rejected-or-partial; the writer and frozen partition remain unopened. V45 then fits a fixed, zero-parameter, exactly invertible matrix-power field from 8,000 training-only rasters. It lowers weighted common resultant from 0.7190 to 0.0152, raises effective rank from 145.08 to 185.38, and improves all four displacement measures on the pinned 1,024-pair V44 holdout while preserving held-font and one-pixel-shift retrieval. V45 passes 13/13 preregistered representation gates in 66.51 seconds with 0.302 GiB peak allocated memory. A report-only V42 retrofit lowers natural top-1 from 19.43% to 15.04%, so V45 qualifies target geometry, not language; the next reader must train from scratch in these coordinates, and the writer and frozen partition remain unopened.

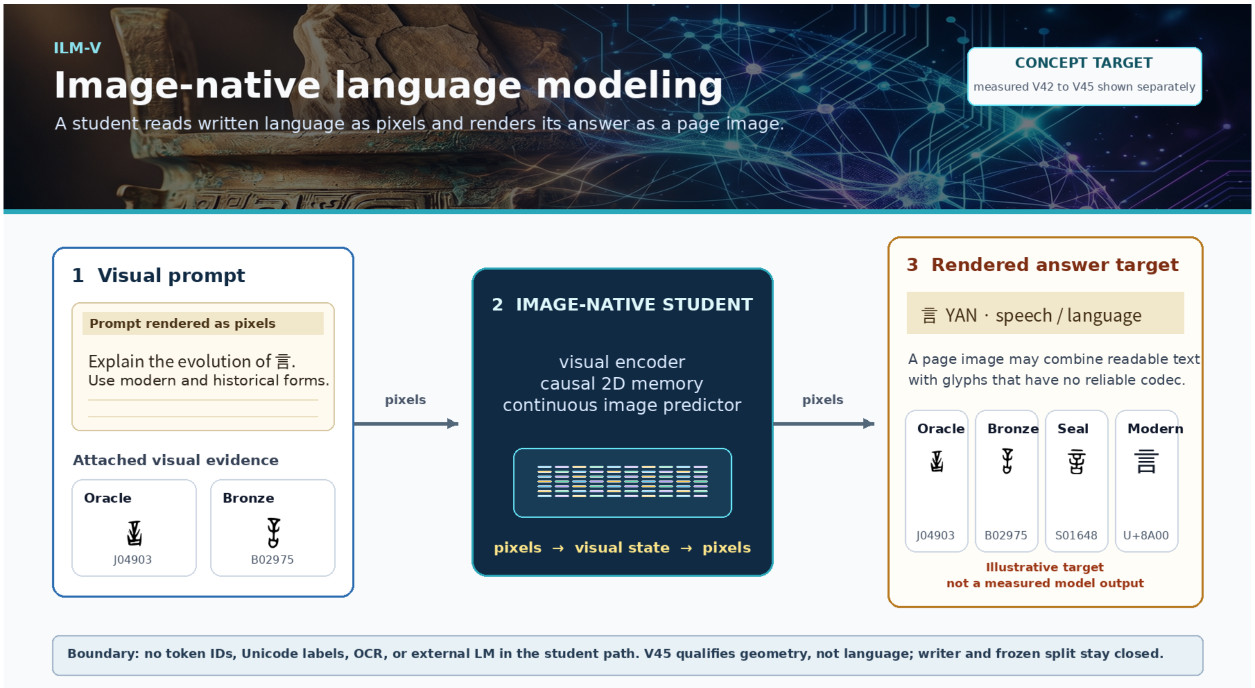

Figure 1: Image-native language modeling design target. A restrained AgInTi background is overlaid with deterministic typography and real local ziyuan glyph exemplars for `YAN` (U+8A00). The right panel is an illustrative answer-page target, not a measured model output. V42 supplies bounded positive ordered-raster language evidence; V43 and V44 remain partial or rejected binding results, while V45 qualifies only their replacement target geometry, as reported in Figures 29, 30, and 31.

1 Introduction: Writing as Language

1.1 Pain point

Human readers encounter language as marks on pages, screens, stone, bone, metal, and moving hands. Those marks vary with writer, era, medium, damage, typography, and reading direction. A finite token or Unicode interface is therefore not a generally accessible inventory of written language: it is an extremely useful codec for a documented subset. Historical and regional forms may have no code point, two visually different forms may share one code point, and an OCR transcript may erase the form whose evolution is the question.

Current language models are extremely capable, but their central language state is normally a sequence of symbolic tokens. Multimodal models can accept images, and modern APIs expose image generation [4, 5, 6]. These systems are important baselines, but attaching vision to a token-centered model does not make visible writing the predictive language substrate. An OCR system similarly solves a useful projection from image to available codes; it does not model all of the visual language that was projected away.

1.2 Goal and scope

The Imagized Language Model (ILM) instead treats written form as language:

$$\text{image input} \to \text{visual language state} \to \text{image output}.$$

The output should itself be a readable answer image. For a query such as "explain the evolution of character yan", the system should produce a page containing modern explanation and attested oracle, bronze, seal, and modern forms, including forms unavailable in computer fonts. Typed input is rasterized before the core. A photographed page can be transcribed, but the same model should also continue, translate, explain, compare, and generate writing. Its answer is an image first; coded transcription is an optional downstream view.

The canonical input/output can be written as a Visual Language Stream,

$$X \in [0,1]^{T\times D\times H\times W\times C}, \qquad p_\theta(X_{t+1:t+k} \mid X_{\le t}),$$

where T is reading sequence or visual time, D is optional geometric depth, H, W are spatial extent, and C is the sensor channel. A flat prompt has $T = 1, D = 1$; an ordered line or book has $T > 1, D = 1$; a 3D glyph string has $D > 1$; and a 3D character movie uses depth and time. Collapsed axes have length one, so these are shapes of one continuous field rather than separate vocabularies. The present proof fixes $D = 1$ and short T; depth and motion must earn separate causal and quality evidence after the 2D loop passes.

Prompt following partitions this stream into observed prompt frames and generated answer frames,

$$X_{\text{prompt}} \in [0,1]^{T_p\times D\times H\times W\times C} \longrightarrow Y_{\text{answer}} \in [0,1]^{T_a\times D\times H\times W\times C}.$$

A typed question is deterministically rendered into prompt frames; a scan or inscription enters natively. $T_a = 1$ is an answer page and $T_a > 1$ is an ordered text-image stream or movie. A language claim requires correct held-out changes in the generated answer when the prompt meaning changes. Input reconstruction, OCR accuracy, glyph classification, or attractive writing alone is insufficient.

The design is inspired by foveated visual reading, without claiming a biological simulation. Its broader hypothesis is sensory: the same perception–memory– prediction–motor–self-perception loop might later operate on speech waveforms or time-frequency fields. This paper tests visible writing only.

1.3 Contributions

The contribution is not the slogan *text as pixels*, nor a priority claim over earlier pixel language models. It is an implemented sequence of interfaces and controls that makes a stronger image-native language claim falsifiable:

1. A strict deployed-student boundary: after optional typed-prompt rasterization, model methods receive pixels and masks and return pixels. Token and Unicode IDs, strings, OCR, finite visual codebooks, glyph lookup, retrieval, and runtime teacher calls are excluded and audited.

2. A staged experimental record separating visual identity, continuous actuation, relational binding, generated-pixel rereading, natural continuation, and autonomous writing. Positive bounded mechanisms and negative natural-language results are retained rather than collapsed into a single capability claim.

3. V34, a qualified 7.42M-parameter continuous writing codec that preserves held-font modern Chinese and held-family historical forms without quantization. Its sealed reconstruction result establishes an interface, not language understanding.

4. V35, a 129.09M-parameter causal raster student combining that frozen codec with an explicitly credited PIXAR initialization, anchor and rectified-flow writers, and a visible decode–threshold–re-encode recurrent boundary. It trains and runs on one RTX 4090 without a symbolic deployed path.

5. V36-P, a 93.47M-parameter candidate-free visual semantic planner that maps prompt rasters to continuous answer plans. Its complete 6,000-update, one-4090 run passes counterfactual assignment but fails the conjunctive held-out semantic gate; the negative result exposes an occupancy-mask shift and low-rank target geometry before writer compute is spent.

6. V37, an 89.77M-parameter end-to-end visual semantic distillation student that fixes clean-mask construction, adapts the image reader, and learns a noncollapsed continuous answer-plan space from detached offline BGE-M3 targets. It raises prompt and answer-plan retrieval substantially on one RTX 4090, while its frozen font, wording, margin, and length failures keep the sealed split and renderer closed.

7. V38, a 90.75M-parameter paired-path visual reader and conditional answer-state model. Five distinct font paths, audited paraphrase views, nearest semantic negatives, and a full answer map improve prompt reading, held-font invariance, and transition geometry on one RTX 4090 D. The frozen 8,000-update result still fails absolute and transferred answer semantics; 25/39 conditions pass, so no sealed row or renderer is opened.

8. V39.1, a corrected ordered-answer pilot whose count process becomes calibrated while held-out answer and segment semantics remain weak. This negative matched result prevents an unjustified full-bank trajectory run.

9. V41, a hash-pinned bridge from the qualified V34 continuous codec to an exactly attributed MX-Font image-conditioned generator. It improves target-style ink F1 under clean and perturbed visual states on one RTX 4090 D, qualifying a robust output motor while making no language claim.

10. V42, a 24.35M-parameter causal Chinese raster model whose full ordered history beats image-unigram, symbolic-bigram, suffix, last-cell, and shuffled-history controls on a fixed development audit. This establishes a bounded image-native language signal while its counterfactual binding and direct raster-quality gates remain false.

11. V43, a 30.04M-parameter continuation with train-only exact-suffix pair supervision and a bank-free 5.69M-parameter spatial rectified-flow writer. It improves visible glyph coherence but fails its unchanged binding and pixel-F1 gates. Post-result controls separate pair-pool memorization from a high exact-plan writer ceiling, identifying the continuous reader plan as the next bottleneck.

12. V44, a frozen-V42, 1.736M-parameter long-history tangent residual trained once over 24,000 unique raster pairs. It removes V43's seen-pair memorization gap but fails held-out binding and matched natural-language retention. A scale-and-direction audit identifies drift toward a corpus-common image mode, motivating a representation test rather than a larger reader or writer.

13. V45, a zero-parameter, training-only noise-limited retinal field with an exact direction-plus-radius inverse. It passes 13/13 fixed reconstruction, common-mode, rank, held-pair displacement, font, translation, boundary, and resource gates. A failed report-only V42 retrofit prevents a calibration shortcut and authorizes only a new causal reader trained from initialization in V45 coordinates.

14. A preregistered development audit with correct, shuffled, blank, partial, paraphrased, and copy-counterfactual prompts; target-through-codec ceilings; writer selection; direct raster galleries; immutable hashes; and a sealed split that may open only after qualification. These controls reject V35's prompt-to-answer binding despite nonblank prompt-responsive pixels.

15. A rights-aware Visual Word-Origin Book target and Visual Language Stream interface for modern and historical writing. Historical answer generation, page-scale memory, depth, motion, speech, and broad multilingual capability remain prospective until they pass their own causal and quality controls.

2 Prior Art and Gap

Pixel language modeling is established prior art. PIXEL treats rendered text as patches for masked representation learning and cross-script transfer [10]. Rendering strategy is itself consequential: character-bigram rendering enabled a 22M PIXEL variant to match its 86M reference on the studied tasks while exposing patch-frequency anisotropy [11]. PIXEL-M4 extends encoder pretraining to English, Hindi, Ukrainian, and Simplified Chinese and reports multilingual semantic and syntactic transfer [12]. These are encoder-side results rather than autonomous raster answering.

PIXAR directly demonstrates decoder-only autoregressive language modeling from and to binary rendered patches [18]. MIXAR scales that family across eight languages and multiple scripts [19], while PixelGPT studies autoregressive pretraining across pixels and text [20]. SPIRAL studies cross-path inconsistency between text and rendered-image language paths and uses paired-path alignment without adding an inference-time teacher [21]. Accordingly, we make no priority claim for pixel language modeling or pixel generation. V35 deliberately reuses PIXAR's released causal weights so that one-GPU experiments test a new interface and binding hypothesis instead of attempting to reproduce industrial pretraining. The audited PIXAR source revision is MIT; the separately distributed weight archive contains no stated weight license, so the derived standalone artifact is local-research-only and is not redistribution-authorized.

OCR-free document systems such as Donut and Pix2Struct map document images to task outputs without a separate OCR engine [8, 9]. Token-free models such as CANINE, ByT5, MEGABYTE, BLT, and Scratchpad Patching remove subwords or allocate compute dynamically, but retain symbolic character or byte streams [22, 23, 24, 25, 26]. These are strong practical baselines. They differ from the stricter deployed ILM boundary in which visible writing is both the recurrent input and primary output.

Continuous image generation supplies a second foundation. MAR models each continuous image latent with a diffusion loss rather than vector quantization [34]; Continuous VAR derives

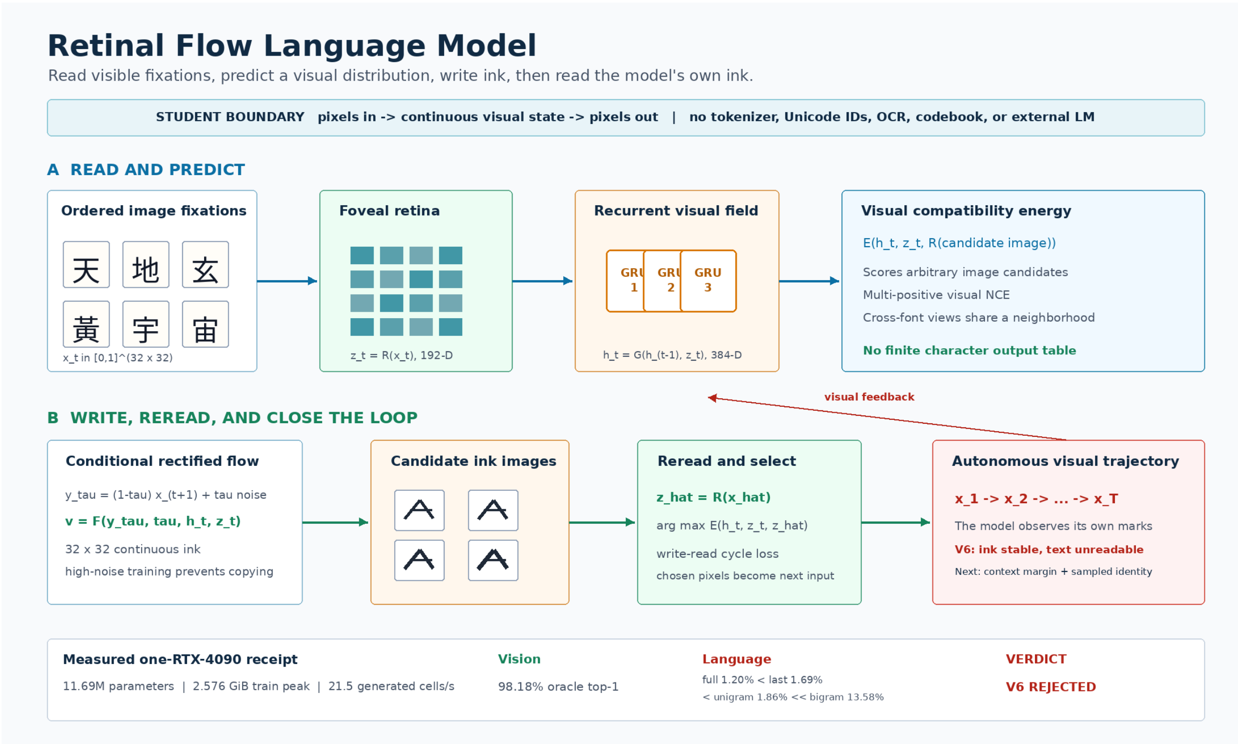

Figure 2: The implemented Retinal Flow Language Model. Training joins visual next-state energy, conditional rectified flow, a write–read cycle, V6 model-induced visual rollouts, and V7 calibrated context and sampled-image identity. Autonomous inference samples candidate ink images, rereads them, selects by visual energy, and feeds the selected pixels back.

non-quantized autoregression from strictly proper scoring rules [45]; and NextStep-1 pairs a large autoregressive backbone with a flow-matching image head [36]. Representation Autoencoders show that reconstructive fidelity alone does not guarantee a representation well suited to generation [37]. TextLDM reaches a parallel conclusion for continuous text latents: alignment to a pretrained language representation is critical, while reconstruction alone is insufficient [50]. V35's negative result is consistent with that warning: an almost lossless local glyph codec does not by itself supply an answer-level semantic plan.

The scoped gap tested here is therefore not *pixels instead of tokens*. It is the combination of a continuous non-quantized writing representation, a causal visual state, direct generated raster output, and a recurrent boundary in which the model must reread its own thresholded visible pixels. Correct, shuffled, blank, partial, paraphrased, and counterfactual prompts distinguish semantic binding from reconstruction, frequency, or merely changing pixels. Historical forms add a provenance constraint: generated marks must never be presented as attested source glyphs. External components are allowed when they work well, but their origin, license, training-time role, and presence or absence at runtime must remain explicit.

3 Architecture

Figure 2 gives the strict visual contract. For fixation images $x_t \in [0,1]^{H\times W}$, RFLM factors a visual sequence as

$$p_\Theta(x_{1:T}) = p(x_1)\prod_{t=1}^{T-1} p_\Theta(x_{t+1} \mid x_{\le t}).$$

The random variables are continuous image fields rather than entries in a linguistic or visual vocabulary.

Foveal retina. A convolutional reader maps each 32×32 image to a normalized 192-dimensional visual state:

$$z_t = R_\theta(x_t)/\|R_\theta(x_t)\|_2.$$

Independent font renderings are aligned with visual contrastive, invariance, and variance objectives. The target retina $R_{\bar\theta}$ is an exponential moving average of the online retina. Font coverage is checked from font cmap tables so missing-glyph boxes cannot become a spurious class.

Recurrent visual field and energy. A three-layer GRU integrates the fixation history,

$$h_t = G_\phi(h_{t-1}, z_t), \qquad h_t \in \mathbb{R}^{384}.$$

An energy function scores any candidate image c_j after rereading it:

$$s_{t,j} = E_\psi(h_t, z_t, R_{\bar\theta}(c_j)).$$

Multi-positive visual NCE admits multiple near-identical image views as valid targets. Unlike a softmax language head, this energy is not tied to a finite character inventory.

Conditional ink flow. For clean target x_{t+1}, noise ϵ, and $\tau \in [0,1]$, define

$$y_\tau = (1-\tau)x_{t+1} + \tau\epsilon, \qquad u_\tau = \epsilon - x_{t+1}.$$

The pixel writer predicts the path velocity

$$\hat u_\omega = F_\omega(y_\tau, \tau, h_t, z_t), \qquad \mathcal{L}_{\text{flow}} = \|\hat u_\omega - u_\tau\|_2^2.$$

Training emphasizes high-noise times, where target pixels cannot simply be copied, and weights sparse strokes and the predicted clean endpoint.

Write–read cycle. The one-step endpoint $\hat x_0 = y_\tau - \tau\hat u_\omega$ is reread through the frozen target retina. A visual NCE cycle objective aligns it to independent target renderings:

$$\mathcal{L}_{\text{cycle}} = \text{NCE}\Big(R_{\bar\theta}(\hat x_0), R_{\bar\theta}(x_{t+1}^{(1)}), R_{\bar\theta}(x_{t+1}^{(2)})\Big).$$

The complete loss combines flow, energy, cross-render invariance, retina contrast, endpoint, stroke, and cycle terms. At inference, the writer samples a candidate set $\mathcal{C}_t$ and selects

$$\hat x_{t+1} = \arg\max_{c\in\mathcal{C}_t} E_\psi(h_t, z_t, R_{\bar\theta}(c)).$$

The selected pixels become the next observation. This closes the visual loop and turns autonomous stability into a mandatory model property rather than a post-processing concern.

Model-induced visual rollout. V6 invokes the same sampler and energy reranker during training. From a clean prefix it generates two short-flow candidates, selects and detaches one bitmap, rereads it with the online retina, and advances the recurrent state. For two rollout steps it minimizes

$$\mathcal{L}_{\text{roll}} = 0.15\mathcal{L}_{\text{state}} + 0.35\mathcal{L}^{\text{roll}}_{\text{energy}} + 0.30\mathcal{L}_{\text{recovery}}.$$

The terms align clean and induced recurrent states, predict the real next image from the induced state, and recover clean next pixels after generated feedback. All targets remain continuous image fields. Candidate selection is stop-gradient, which bounds memory while exposing the trainable state and writer to deployed pixels.

Calibrated context and sampled identity. V7 corrects two training/evaluation mismatches. Raw target-energy differences can be increased by changing the scale or all competing scores, so V7 uses the normalized visual target log probability

$$\ell(y \mid h, \mathcal{C}) = s(y \mid h) - \log \sum_{c \in \mathcal{C}} \exp s(c \mid h)$$

and minimizes

$$\mathcal{L}_{\text{ctx}} = \max\{0, m - \ell(y \mid h_{\text{full}}, \mathcal{C}) + \ell(y \mid h_{\text{last}}, \mathcal{C})\}.$$

The last-only branch is detached. The training-only set $\mathcal{C}$ contains independent image renderings; positives are inferred by retinal similarity. Offline strings select the curriculum but anchor IDs and target indices do not enter the student, the pixels are disjoint from evaluation views, and the anchors are not deployed. V7 also differentiates through a two-step numerical pixel-flow endpoint,

$$\mathcal{L}_{\text{sample}} = \text{NCE}\left(R_{\bar{\theta}}(\hat{x}_0^{K=2}), \{R_{\bar{\theta}}(y^{(v)})\}_{v=1}^{V}\right).$$

This trains the sampled image path rather than only a one-step endpoint proxy.

4 Predictive Visual Field

Figure 3 separates linguistic uncertainty from stroke rendering. For target state $z_{t+1} = R(x_{t+1}) \in \mathbb{S}^{d-1}$, a deterministic proposal predicts

$$\mu_t = \frac{P_\psi([h_t, z_t])}{\|P_\psi([h_t, z_t])\|_2}.$$

The proposal is a continuous image-derived state, not a class index. It is trained with geodesic alignment, image contrast, and full-history advantage against a detached last-image branch.

In parallel, choose random unit source ϵ, let q_τ follow the shortest spherical geodesic from z_{t+1} to ϵ, and let u_τ be its tangent velocity. The conditional field learns

$$\mathcal{L}_{\text{state-flow}} = \mathbb{E}\, \|v_\eta(q_\tau, \tau, [h_t, z_t]) - u_\tau\|_2^2.$$

The loss is full tangent-vector energy, not coordinate-mean MSE. Sampling integrates backward with the spherical exponential map,

$$q_{k-1} = \text{Exp}_{q_k}\left(-\Delta\tau\, v_\eta(q_k, \tau_k, [h_t, z_t])\right),$$

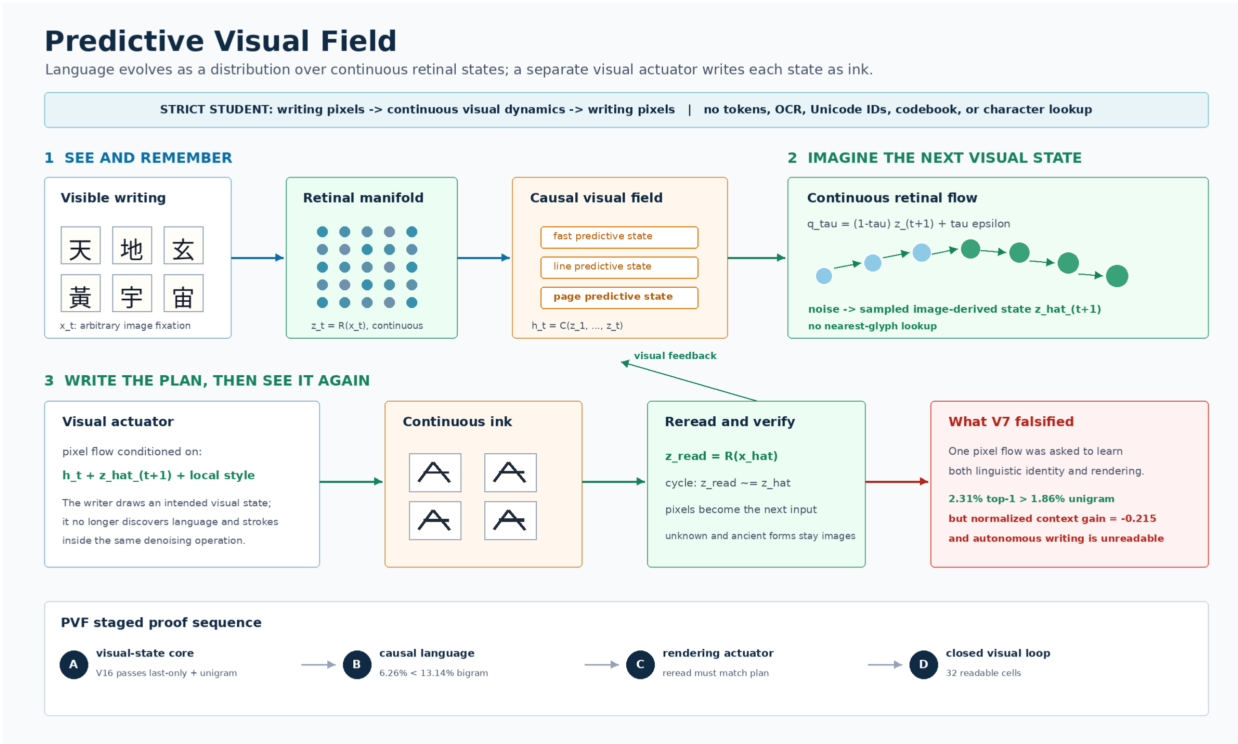

Figure 3: Predictive Visual Field factorization. A causal image field predicts a low-variance continuous visual proposal and a stochastic distribution over image-derived retinal states. A later separate actuator renders a selected state and the retina rereads its pixels.

yielding intended next visual states $\hat{z}_{t+1}$, not glyph IDs. Candidate images are used only by the evaluator: it rereads them with the frozen retina and computes kernel density against continuous samples. There is no learned 512-way output layer.

The training-only multiview anchor curriculum contains image tensors. Positives are inferred from retinal cosine similarity; labels and target indices do not enter the student, and the bank is not deployed. This visual contrast is necessary because geodesic endpoint alignment alone can reward a visual mean without separating language alternatives.

V16 augments the pretrained recurrent trajectory $h^R_{1:t}$ with three residual causal memory blocks. Each block combines a left-padded depthwise temporal field at dilation 1, 2, or 4, global causal attention, and a gated feed-forward residual. For memory M_ϕ, the corrected state is

$$h_t^{V16} = h_t^R + \sigma(g) W_o \, \mathrm{LN}\{M_\phi(h^R_{1:t}, z_{1:t})\}.$$

The per-channel gate is initialized at $g = -4$, preserving the trained V15 function at the start of the upgrade. The selected checkpoint has mean $\sigma(g) = 0.018023$; therefore the frozen result is a bounded memory pilot, not evidence that a large residual correction is necessary.

V17 implements a separate visual actuator that writes

$$\hat{x}_{t+1} = W_\omega(\epsilon_x, h_t, \hat{z}_{t+1}, R_\theta(x_t))$$

and is trained with pixel flow plus a reread constraint

$$\mathcal{L}_{\text{act}} = \mathcal{L}_{\text{pixel-flow}} + \lambda \left[1 - \cos\{R_{\bar{\theta}}(\hat{x}_{t+1}), z_{t+1}\}\right].$$

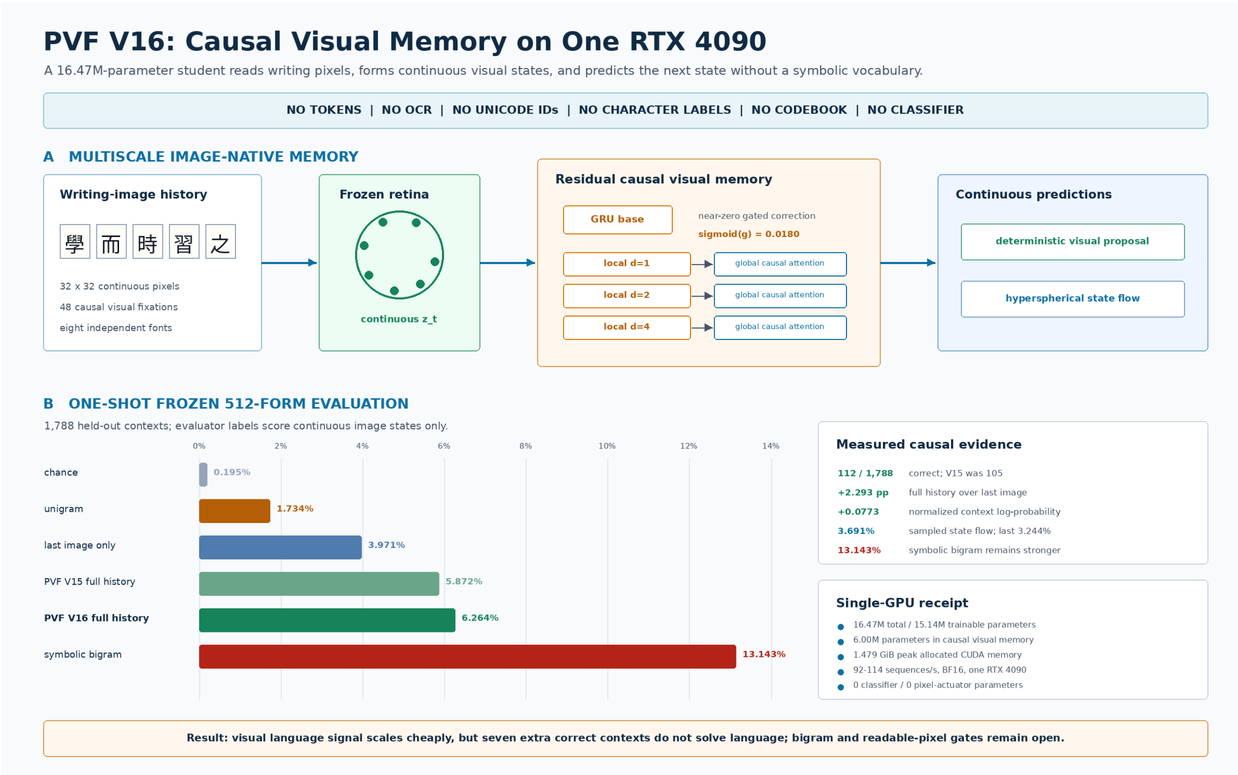

Figure 4: Measured PVF V16 memory result. Both deterministic proposal and stochastic hyperspherical field use full image history and beat random, last-only, and unigram controls. V16 improves proposal top-1 by seven frozen contexts over V15, but remains below the symbolic bigram and has no pixel actuator.

No nearest-glyph projection, character classifier, or token unembedding is permitted. In V17, $\hat{z}_{t+1}$ is supplied by a different-font image of the target form so actuation is isolated from language prediction. This differs from Embedded Language Flows, which demonstrates effective continuous flow dynamics but encodes and finally unembeds discrete tokens [44]. It combines the next-fixation representation objective of recurrent JEPA [42] with the prediction/generation separation explored by D-JEPA [43], and uses intrinsic flow matching on a Riemannian manifold [39, 40], while retaining an image-derived language state.

5 Isolated Visual Actuation and Motor Planning

The V17 actuator tests whether an image-derived continuous state can control new pixels without a glyph table. A frozen retina produces $z = R(x^{\text{semantic}}) \in \mathbb{S}^{191}$ from one font view, and a convolutional encoder produces style vector $s = E_\omega(x^{\text{style}})$ from a different character in the desired target render. A 5.73M-parameter conditioned U-Net integrates rectified flow from Gaussian noise to a 32×32 ink image. The target image enters losses only; the spatial condition to the writer is a constant blank plane.

Training combines stroke-weighted flow matching, an endpoint loss, retinal cosine, multi-positive retinal contrast, and gradients through a two-step deployed sampler. At evaluation, correct and shuffled branches receive identical style images and initial noise. Only the intended state is rolled

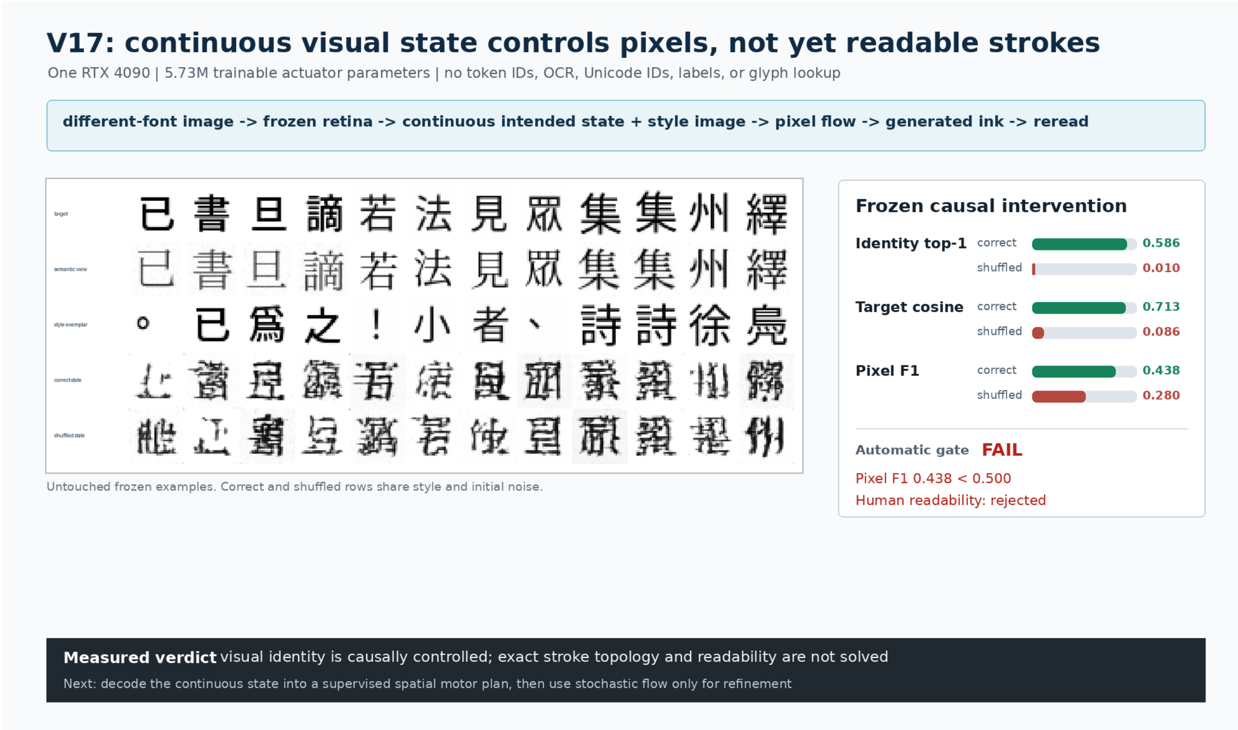

Figure 5: Measured V17 frozen result. Continuous intended state strongly controls recovered visual identity under a same-noise shuffled-state intervention, but pixel F1 and human readability fail. The correct-state row contains state-dependent pseudo-characters, not reliably readable target forms.

within the batch. The global evaluator retrieves among all 512 generated examples, with duplicate positives inferred from image-state similarity rather than labels.

V17 was selected at step 1,600 by a preregistered development rule and queried the frozen record split once. Correct-state top-1 is 58.594%, versus 0.977% shuffled; target cosine is 0.71301 versus 0.08612; and style-copy cosine is only 0.05858. These measurements reject style copying as the explanation for the causal signal. Pixel F1 is 0.43849 versus 0.28001 shuffled, failing the required 0.50, and human review rejects most outputs as unreadable. Training takes 339.67 seconds and peaks at 1.588 GiB allocated CUDA memory on one RTX 4090.

The result identifies a topology gap rather than an identity-control gap.

5.1 Topology-first visual motor plan V18

V18 decodes state and style into a directly supervised spatial visual motor plan,

$$p_\omega = D_\omega(z, E_s(x^{\text{style}})) \in [0,1]^{H\times W}.$$

A context MLP modulates a learned 4×4 seed, conditioned residual blocks upsample it to 32×32, and a sigmoid emits continuous ink. The loss joins stroke-weighted binary cross entropy, soft Dice, pixel and Sobel-edge distances, retinal identity and contrast, and a correct-versus-shuffled state margin. Target pixels supervise the output but never enter the condition. The plan remains continuous and image-derived; it is not copied target ink, an ID lookup, or a discrete visual codebook. Stochastic flow is reserved for later residual style and surface variation rather than stroke placement.

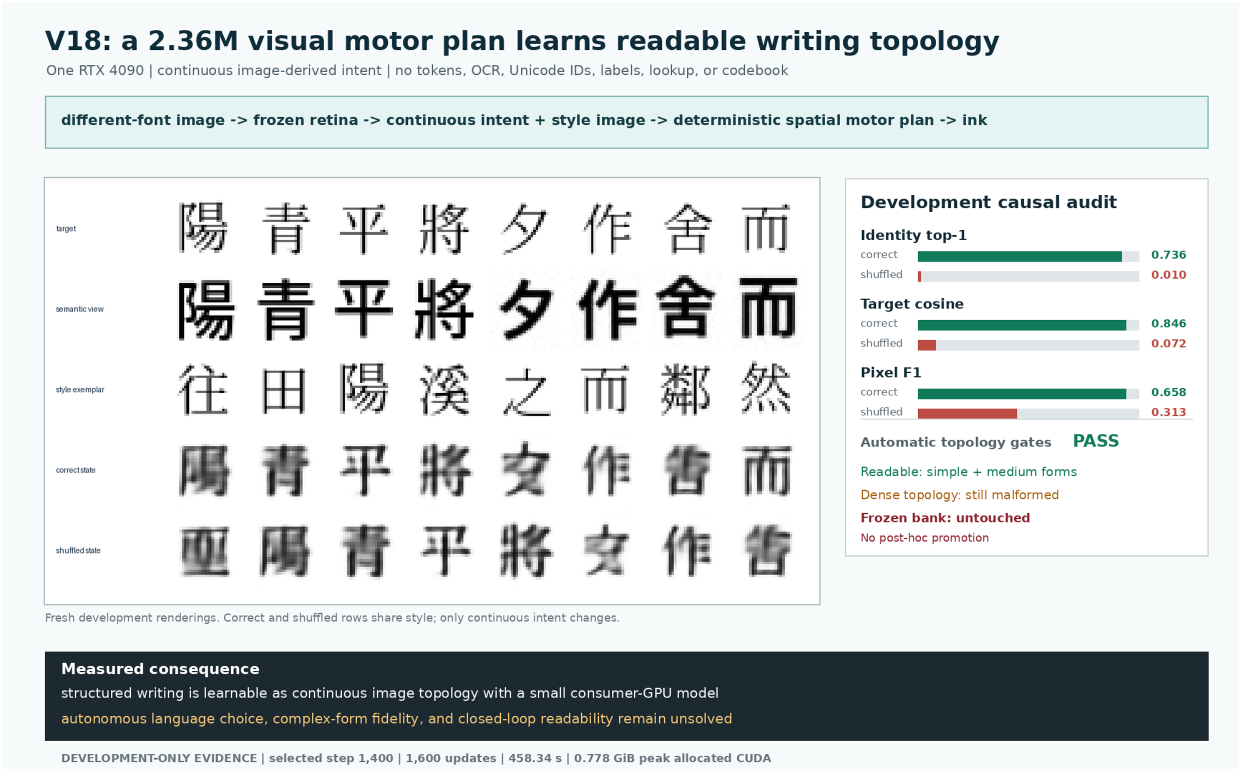

Figure 6: Measured V18 development result. A 2.36M-parameter deterministic motor planner turns continuous image-derived intent into recognizable writing. Shuffling only intended state destroys identity and topology while preserving mean ink occupancy. This is development-only evidence; the frozen bank remains untouched.

The salted partition contains 6,573 training, 223 development, and 221 frozen records. Selection maximized development correct-state pixel F1 subject to fixed topology, state-dependence, retinal-similarity, and non-copying gates. Step 1,400 was selected at pixel F1 0.6890 versus 0.3368 shuffled and global identity top-1 78.32% versus 1.17%. A fresh development seed yields 512 examples: identity top-1 is 73.63% versus 0.98%, target cosine is 0.8462 versus 0.0716, and pixel F1 is 0.6577 versus 0.3129. Mean ink fraction is 0.17505 in both branches and style-copy cosine is 0.0686.

Full-resolution review finds clear simple and medium forms but merged strokes in dense forms. Because the preregistration required human readability without defining a numeric blinded threshold, post-hoc inspection cannot authorize a frozen query. V18 therefore accepts readable topology emergence on development, rejects broad-complexity readability, and leaves the frozen identifier bank sealed. The next experiment must define complexity strata and a recognition rubric prospectively.

5.2 Spatial-retinal residual V19 and its rejection

V18 compresses the semantic image into one global vector before drawing. V19 tests the narrower hypothesis that this pooling discards component location in dense forms. The frozen retina now exposes both its existing global state and its pre-pooling spatial field,

$$(z, F) = R(x^{\text{semantic}}), \qquad z \in \mathbb{S}^{191}, \quad F \in \mathbb{R}^{192\times4\times4}.$$

V19 development gate	Measured	Required	Result
Overall pixel F1	0.6710	> 0.68	fail
Dense pixel F1	0.7278	> 0.58	pass
Dense gain over shuffled field	0.0088	> 0.12	fail
Dense gain over zero field	0.0054	> 0.03	fail
Identity top-1	72.66%	$> 75\%$	fail
Target cosine	0.8416	> 0.84	pass

Table 1: Fresh 512-candidate V19 development audit. Four fixed automatic gates fail, so the blinded human review and frozen evaluation are not authorized.

Let $c = [z, E_s(x^{\text{style}})]$ and let $q_0 = S(c)$ be the learned global 4×4 seed of a V18-architecture planner. A conditioned convolutional adapter supplies a spatial residual,

$$a = A_\varphi(F, c), \qquad q = q_0 + \sigma(g)a, \qquad p = D(q, c).$$

The adapter's final projection is initialized exactly to zero, so V19 initially equals the selected global planner for every input even though its scalar gate is trainable. All global-planner and retina parameters remain frozen. This makes a gain attributable to spatial information rather than unrestricted baseline fine-tuning.

A clean causal comparison requires a new global baseline trained from scratch on the V19 partition. Loading V18 would contaminate V19 because V18's training split can intersect the new holdout. The selected clean global checkpoint is therefore frozen before the spatial adapter is trained; no V17 or V18 learned weight enters the experiment.

Complexity is computed from target image geometry, never a character label. For thresholded ink b, define

$$C(b) = \bar{b} + \tfrac{1}{2}\left(\overline{|\Delta_x b|} + \overline{|\Delta_y b|}\right) + 0.1\,\overline{\mathbf{1}[\mathrm{AvgPool}_4(b) > 0.05]}.$$

The preregistered strata are simple $C < 0.24$, medium $0.24 \le C < 0.35$, and dense $C \ge 0.35$, weighted 1, 1.25, and 2 during spatial training. The objective is

$$\begin{aligned}\mathcal{L}_{19} = {} & \mathcal{L}_{\text{WBCE}} + \mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_1 + 0.25\mathcal{L}_{\text{edge}} \\ & + 0.05\mathcal{L}_{\text{id}} + 0.05\mathcal{L}_{\text{NCE}} + 0.1\mathcal{L}_{\text{spatial-margin}} + 0.1\mathcal{L}_{\text{zero-margin}}.\end{aligned}$$

Rowwise terms use the complexity weights. The margins require correct fields to outperform batch-shuffled fields by 0.03 and the zero-field branch by 0.01 in target pixel error.

Evaluation fixes target and style while comparing five branches: correct global state plus correct field; correct global plus shuffled field; shuffled global plus correct field; both shuffled; and correct global plus zero field. The prospective gate requires overall pixel F1 above 0.68, dense F1 above 0.58, dense spatial-shuffle and zero-field advantages, global identity above 0.75, and 38/48 blinded recognitions with stratum-specific minima. The implementation passes exact-equivalence, gradient-isolation, field-shape, intervention, complexity, and serialization tests.

A clean global planner was trained from scratch for 1,600 updates on the V19 partition, selected at step 1,600, and frozen. It reached development F1 0.6754, identity top-1 77.93%, and target cosine 0.8339. The V19 intervention then trained only 764,545 adapter parameters for 1,600 updates in 329.51 seconds, using 0.899 GiB peak allocated CUDA memory on one RTX 4090. A separate deterministic audit rendered 512 new development candidates. Table 1 records the fixed decision.

This result rejects the additive routing mechanism, not the possibility that the field contains useful information. The global planner can already draw the whole form, so ordinary reconstruc-

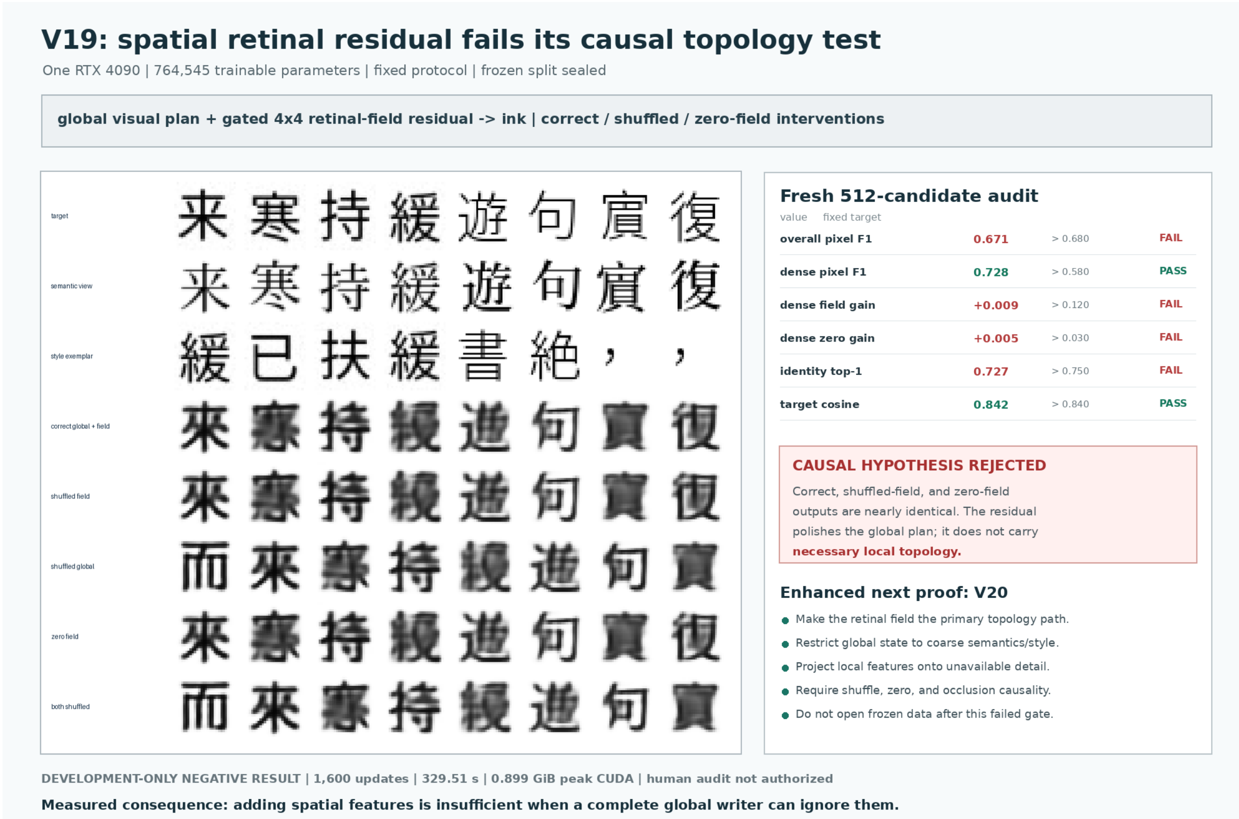

Figure 7: Measured V19 result. Correct, shuffled-field, and zero-field rows are nearly identical, while shuffling the global state changes identity. The spatial adapter learned a shared residual polish rather than necessary local topology.

tion gradients need not use the field. The next bounded candidate must make local visual state structurally necessary.

5.3 Retinal topology router V20: local causality without writer selection

V20 implements the preregistered coarse/detail decomposition

$$p = \sigma\left(U(G(z,s)) + (I - UD)H(F,s)\right),$$

where G emits only one logit per 8×8 output block and H maps each 4×4 retinal-field cell pointwise to its corresponding 8×8 detail patch. The fixed projection $I - UD$ removes each patch mean, so the global route cannot represent within-block detail. An exactly capacity-matched control replaces F with tile(z) while retaining the same local decoder, losses, and operations. Each arm has 506,448 trainable parameters.

The new salted split contains 6,611 training, 196 development, and 210 frozen records. Both arms train for 1,600 updates against the same frozen V16 retina. The fixed interventions hold target and style constant while shuffling the local field, zeroing it, shuffling global state, shuffling both, or occluding each field quadrant. Validation renders 512 development candidates, including 146 dense forms. Table 2 records the final endpoints; neither is a selected checkpoint.

The candidate's correct dense F1 exceeds shuffled-field by 0.12177 and zero-field by 0.36128; its quadrant locality is exactly 1.0. These measurements establish that the continuous field carries

V20 development gate	Candidate	Required	Result
Overall pixel F1	0.6361	> 0.66	fail
Dense pixel F1	0.7013	> 0.68	pass
Dense gain over shuffled field	0.1218	> 0.12	pass
Dense gain over zero field	0.3613	> 0.10	pass
Identity top-1	75.00%	$> 70\%$	pass
Target cosine	0.8152	> 0.82	fail
Occlusion locality	1.0000	> 0.40	pass
Detail block-mean magnitude	2.03×10^{-6}	$< 10^{-6}$	fail
Dense gain over matched control	0.0179	> 0.03	fail

Table 2: Final V20 development endpoints. The field candidate passes every local-causality gate but fails writer selection and the matched-control margin. Because neither arm selects, the formal paired audit, blinded review, and frozen evaluation remain forbidden.

necessary, topographically aligned detail. They do not establish a selected writer. Overall F1 and reread cosine remain below threshold. Both arms also exceed the strict 10^{-6} detail-mean tolerance after detail magnitude grows, indicating that floating-point recentering is a poor way to encode an algebraic invariant. Candidate dense F1 exceeds the control by only 0.01791 at the final endpoint, below the paired 0.03 margin. Candidate and control train in 325.90 and 327.11 seconds and peak at 0.323 and 0.397 GiB allocated CUDA memory, respectively.

5.4 Field-complete writer V21: complete local route, rejected rasterizer

V21 tests the field-complete correction rather than widening V20. Global state and image-derived style emit only spatially uniform modulation. Each local field cell independently emits one coarse coefficient and 63 detail coefficients,

$$(\gamma, \beta) = M(z, s), \qquad (c_{ij}, a_{ij}) = A_F(\gamma \odot F_{ij} + \beta), \qquad \ell_{ij} = c_{ij}\mathbf{1} + B_0 a_{ij},$$

where B_0 contains the 63 non-DC columns of a normalized Walsh–Hadamard basis. Its column sums and Gram error are exactly zero in the recorded FP32 computation. The 16 nonoverlapping 8×8 patches make the 32×32 logit field. There is no coordinate input, position parameter, cell mixing, global spatial projection, or learned glyph table. The matched control tiles z into all 16 source cells while retaining every parameter and operation. Each arm has 582,336 trainable parameters.

The preregistered salted split has 6,613 training, 195 development, and 209 frozen records. Both arms train for exactly 1,600 updates. No candidate checkpoint passes every selection gate. Table 3 therefore reports step 1,400 only as the best diagnostic snapshot under the fixed ranking, not as a selected model.

The tiled-global control selects its structural step 1,200 and reaches only 0.1473 overall F1, 0.3074 dense F1, 25.00% identity top-1, and 0.4332 target cosine. Candidate step 1,400 exceeds those values by 0.4564 overall F1 and 0.3979 dense F1. This comparison is descriptive rather than the formal paired audit: the snapshots use different development draws and steps, and the paired evaluator requires a selected candidate before constructing a shared audit. It must therefore refuse V21.

Candidate and control train in 327.04 and 333.41 seconds at 0.325 and 0.400 GiB peak allocated CUDA memory. Frozen-image count remains zero. V21 therefore establishes complete, necessary, topographic use of continuous local state, but rejects nonoverlapping pointwise patch emission as

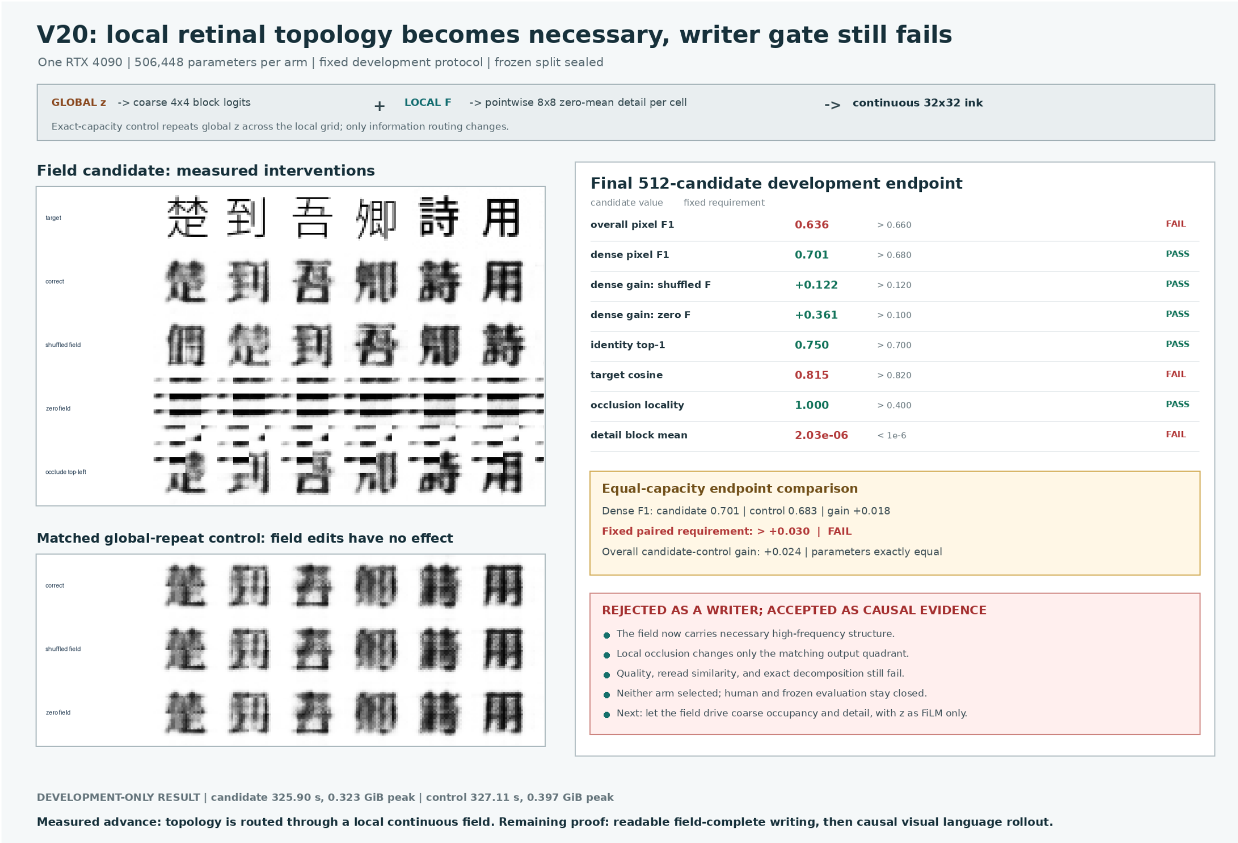

Figure 8: Measured V20 result. Shuffling or zeroing the candidate's local field destroys detail, and a quadrant occlusion changes only its corresponding output quadrant. The equal-parameter global-repeat control is invariant to field interventions. Recognizable endpoints do not pass the fixed writer and paired-control gates.

a broadly readable writer. The next writer test should add overlapping local support with a fixed partition-of-unity blend, or an equivalently bounded multiscale local operator, while retaining an exact-parameter tiled-global control and a measured influence radius.

5.5 Visual Binding Stream V22: image prompt, rejected relation

V22 removes supplied semantic intent and asks the smallest prompt-following question that also requires arbitrary visible form. Every episode is a six-frame image stream,

$$X = (l_1, g_1, l_2, g_2, o, q),$$

where two visible labels bind two distinct, previously unseen glyph images. The operation image is either the visible form for same or other, and the final query is one of the two label images. A paired counterfactual changes only q; the correct canonical answer image must switch. Source glyphs use independent noncanonical faces, so exact pixel copying cannot satisfy the target.

The frozen V16 retina maps each frame to global and local continuous states. A four-block visual Transformer forms a query from its final hidden state and a softmax over all six frames. Weighted global/local values feed an overlapping 12×12, stride-8 local writer with a fixed positive partition-of-unity blend. The query-blind control replaces only the final query retinal state with a learned

V21 candidate development gate	Measured	Required	Result
Overall pixel F1	0.6038	> 0.66	fail
Simple pixel F1	0.5648	> 0.58	fail
Medium pixel F1	0.5945	> 0.60	fail
Dense pixel F1	0.7053	> 0.70	pass
Dense gain over shuffled field	0.1739	> 0.15	pass
Dense gain over zero field	0.3465	> 0.20	pass
Identity top-1	79.10%	$> 74\%$	pass
Target cosine	0.8331	> 0.82	pass
Occlusion pixel change	0.0926	> 0.03	pass
Occlusion locality	1.0000	> 0.95	pass
Detail block-mean magnitude	3.87×10^{-7}	$< 5\times10^{-6}$	pass
Decomposition error	0.0	$< 10^{-6}$	pass

Table 3: Best V21 candidate diagnostic snapshot. It passes every causal, identity, reread, and algebraic gate but fails overall, simple, and medium pixel fidelity. It is not a selected checkpoint.

V22 candidate development gate	Measured	Required	Result
Binary choice	0.4824	> 0.85	fail
Counterfactual switch	0.0078	> 0.80	fail
Held-out-combination switch	0.0113	> 0.75	fail
Identity top-1	0.1592	> 0.45	fail
Identity gain over query shuffle	0.0225	> 0.20	fail
Target cosine	0.4601	> 0.78	fail
Pixel F1	0.5107	> 0.58	fail
Oracle-writer F1	0.6020	> 0.64	fail
Paired output L1	0.0089	> 0.08	fail
Frozen images instantiated	0	0	pass

Table 4: V22 query-aware endpoint. The anti-copy margins and structural receipts pass, but every prompt-dependence and output-quality gate fails. The query-blind control ends at 0 switch accuracy, 0.1533 identity top-1, 0.5125 pixel F1, and exactly zero paired-output difference.

continuous null state. Both arms contain exactly 3,410,128 trainable parameters with identical shapes. The student path contains no string, token or Unicode ID, OCR, character/operation label, answer index, glyph lookup, finite codebook, or external language model.

The fixed identity bank contains 815 training, 104 development, and 105 frozen Chinese forms under a salted split; two operation/label combinations are held out compositionally. Each arm trains for exactly 1,600 updates. Table 4 reports the final development endpoints. Candidate and control writer/identity trajectories nearly overlap, while prompt switching remains absent.

The mechanism audit makes the rejection more specific than failure to converge. Candidate mean attention is 1.0 on the operation frame, 2.49×10^{-12} and 5.53×10^{-12} on the two glyph frames, and 1.37×10^{-13} on the query. The entropy penalty rewards this confident shortcut. The task, however, is a relation among operation, query, labels, and glyphs; no single frame contains the answer. Candidate training takes 258.70 seconds at 0.315 GiB peak allocated CUDA memory, so model/GPU capacity is not the identified bottleneck.

No candidate checkpoint selects. The paired evaluator refuses before fresh data construction, blinded review is unauthorized, and no frozen image is instantiated. V22 establishes a real image-prompt to image-answer interface and weak unseen-form writing only; it does not establish prompt

Figure 9: Measured V21 result. The candidate uses local state for complete spatial structure: shuffling or zeroing the field corrupts the form and a local occlusion stays local. The equal-parameter tiled-global source repeats one patch. Disjoint candidate patches remain blurred and seamed, causing the preregistered broad-fidelity rejection.

understanding.

V23 removes unrestricted single-frame selection. A shared visual comparator must match query and label images, while an operation-image reader provides a continuous gate,

$$a_i = C_\theta(R(q), R(l_i)), \quad m_i = \mathrm{softmax}_{i\in\{1,2\}}(a_i), \quad s = \sigma(U_\theta(R(o))),$$

$$w_i = sm_i + (1-s)(1-m_i), \quad (\hat{z}, \hat{F}) = \sum_{i=1}^{2} w_i V(R(g_i)).$$

This relation circuit composes multiple visible roles before writing. Its operation semantics and equality function are learned from answer images; no symbolic target index enters the student. Exact query-blind, operation-blind, pair-swap, query-counterfactual, and operation-counterfactual interventions are required before any longer answer stream. A qualified writer should be frozen first so binder/writer co-adaptation cannot hide another average-glyph route.

5.6 Visual Relation Circuit V23: bounded prompt relation accepted

V23 fixes the V22 causal defect while preserving the same strict visual interface. Its bank contains 817 training, 109 development, and 98 frozen Chinese identities under the preregistered salt

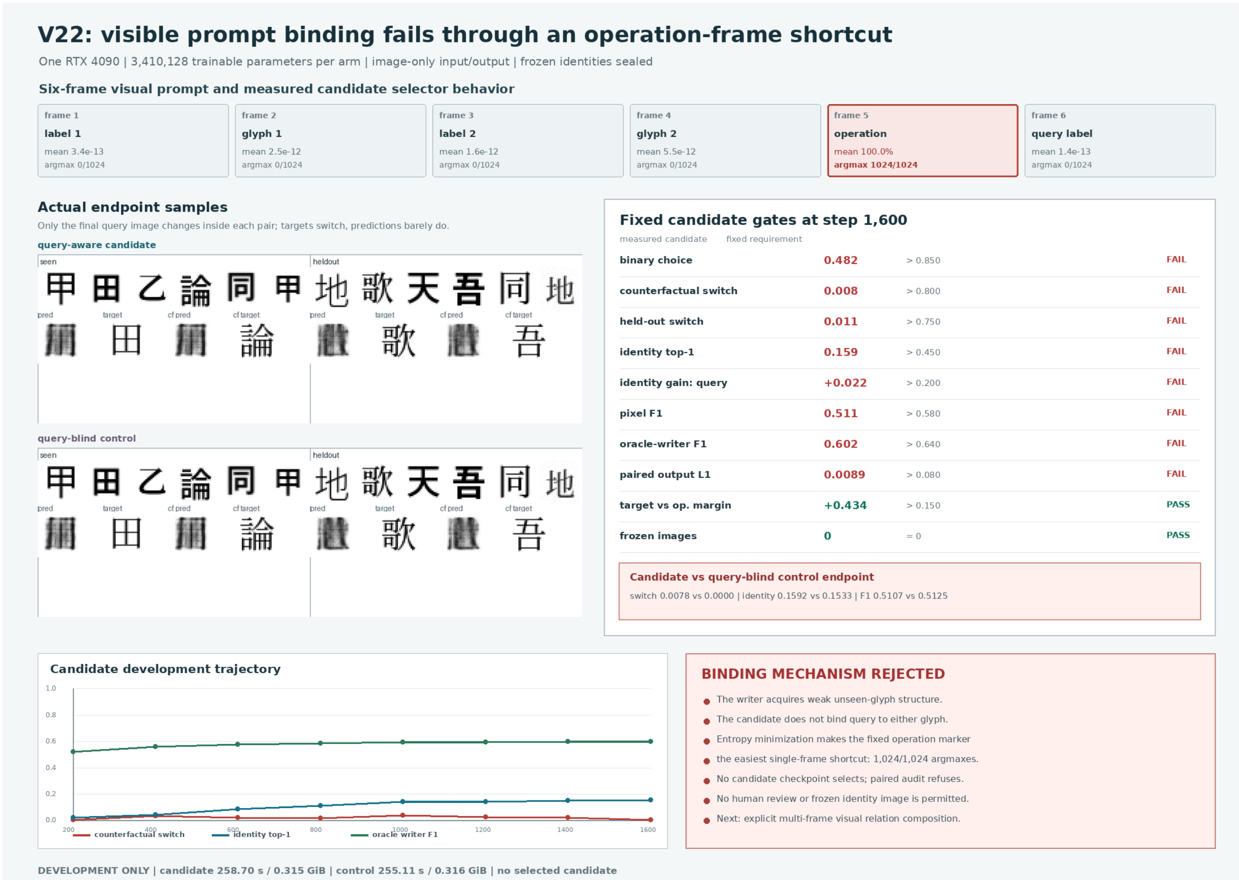

Figure 10: Measured V22 result. Actual endpoint samples show that targets switch when only the final visual query changes, while candidate predictions barely move and match the query-blind control. The candidate selector assigns every one of 1,024 development argmaxes to the operation frame. Writer metrics improve without relational prompt binding.

`visual-relation-circuit-v23`. The two held-out operation/label compositions never occur during training, although every individual operation and label pair does. Every model call receives only floating image tensors or continuous states derived from those tensors; renderer strings and episode metadata are deleted before `forward`.

Stage A independently trains a 1,122,081-parameter convolutional U-Net to map a noncanonical source-glyph image to a canonical target image. It has no identity embedding, target-font ID, coordinate channel, or character table. The selected step-1,000 checkpoint reaches 0.7583 pixel F1, 0.9951 identity top-1, and 0.9375 target cosine on all 109 unseen development identities. Its pixel-F1 gains over the raw source and a shuffled source are 0.2173 and 0.4237. The canonicalizer is then frozen before relation training.

Stage B uses normalized frozen-retina states. For each label image,

$$c_i = \langle R(q), R(l_i) \rangle, \qquad m_i = \mathrm{softmax}_{i \in \{1,2\}}(\tau c_i),$$

where the positive temperature τ is learned. A small operation-image reader produces

$$s = \sigma(U_\theta(R(o))),$$

Fresh paired metric	Relation-aware	Query-blind	Operation-blind
Binary choice	0.9990	0.5010	0.5002
Query switch	0.9980	0.0000	0.0615
Operation switch	0.9980	0.0156	0.0000
Identity top-1	0.9951	0.2947	0.2993
Pixel F1	0.7563	0.5407	0.5404
Query-output L1	0.1915	0.0000	0.0109
Operation-output L1	0.1912	0.0017	0.0000
Pair-swap output L1	0.0000	0.0000	0.0000

Table 5: V23 fresh paired development audit. A control has exactly zero response to its blinded factor by construction, while the relation-aware arm retains both dependencies with matched parameter shapes.

and composes match and operation before generation:

$$w_i = sm_i + (1-s)(1-m_i), \qquad x_r = w_1 g_1 + w_2 g_2, \qquad \hat{y} = C_\psi(x_r).$$

Thus same routes the visually matched source and other routes its complement. The operation semantics are learned from answer-image loss, not supplied as a class label. The routed variable is a continuous source-pixel field, and C_ψ is the frozen Stage A canonicalizer. The trainable relation circuit has 25,602 parameters: the operation reader, one temperature scalar, and two continuous null states.

Candidate, query-blind, and operation-blind arms have exactly identical parameter names and shapes. Query-blind replaces only $R(q)$ with its learned continuous null state; operation-blind replaces only $R(o)$. All arms process base, query-counterfactual, operation-counterfactual, and pair-swapped image prompts jointly for 600 fixed updates. The selected candidate reaches 0.9946 binary choice, 0.9902 query switch, 0.9902 operation switch, 0.9873 identity top-1, and 0.7548 pixel F1 on checkpoint-selection development data.

A fresh paired audit then constructs 1,024 new development episodes and 4,096 prompt variants. Table 5 reports the exact interventions. The candidate changes identity and pixels when either visible causal factor changes. Each blind control is exactly invariant to the factor it cannot see, and all arms remain exactly invariant to pair order.

After the paired gate passes, an opaque development review presents 48 image cards without transcriptions, targets, correctness labels, or frozen examples. The reviewing agent chooses which of the two visible source forms the generated answer depicts before the scorer opens the sealed key. It obtains 48/48 overall and 12/12 on held-out compositions, above the fixed 44/48 and 11/12 gates. This is an agent visual review, not a human-subject study.

Only then does the frozen evaluator instantiate the 98 held-out identities. It runs once on 1,024 episodes, 4,096 prompt variants, and four identity-bank views, with no model selection or threshold changes. Every gate in Table 6 passes. The run takes 4.74 seconds and peaks at 0.498 GiB allocated CUDA memory. Stage A training takes 214.10 seconds at 0.700 GiB; candidate relation training takes 117.18 seconds at 1.406 GiB, all on one RTX 4090.

V23 therefore accepts a bounded image-only relation: multiple visible roles are composed, unseen source forms are routed, and the answer is emitted as pixels. It does not establish arbitrary syntax or free-form generation. Frame roles and the two-pair same/other algebra remain architecturally fixed, and the answer is a canonicalized version of one visible source form. V24 implements the preregistered next test: remove absolute frame roles, consume a variable-length visual instruc-

V23 frozen gate	Measured	Required	Result
Binary choice	0.9983	> 0.95	pass
Query switch	0.9961	> 0.90	pass
Operation switch	0.9971	> 0.90	pass
Held-out minimum switch	0.9961	> 0.85	pass
Identity top-1	0.9946	> 0.75	pass
Pixel F1	0.7848	> 0.68	pass
Target cosine	0.9399	> 0.82	pass
Query-label visual match	0.9995	> 0.98	pass
Operation-gate accuracy	1.0000	> 0.98	pass
Pair-swap consistency	1.0000	> 0.99	pass

Table 6: Single authorized V23 frozen evaluation over previously unseen identities. Every threshold was fixed before implementation and remained unchanged.

tion, emit two answer-image frames, and make frame 2 causally depend on rereading the generated pixels of frame 1.

5.7 Visual Packet Reread Stream V24: causal two-frame stream accepted

V24 replaces V23's six absolute roles with a variable sequence of three-frame visual packets. Its prompt and autonomous answer have shapes

$$X \in [0,1]^{B\times T\times 1\times 32\times 32}, \quad T \in \{15, 18, 21, 24\}, \qquad \hat{Y} \in [0,1]^{B\times 2\times 1\times 32\times 32}.$$

Each packet is *header, content A, content B.* The visible header glyphs for pair (U+5951), operation (U+6CD5), query (U+95EE), distractor (U+65C1), and stop (U+6B62) denote their respective roles. Packets before the terminating packet are shuffled. Training uses 15, 18, and 21 active frames; 24 frames with three distractors are held out by length. Dynamic batching adds only all-zero packet images and supplies neither active lengths nor padding masks.

The identity bank contains 829 training, 88 development, and 107 frozen Chinese forms under the fixed salt `visual-packet-reread-stream-v24`. Two operation/label-pair compositions are held out while every constituent operation and label pair appears separately in training. As in V23, renderer strings and episode metadata are deleted before every model call.

For packet j, the frozen V16 retina yields normalized header and content states H_j, A_j, B_j. Three learned normalized prototypes locate roles from header pixels:

$$e_{jr} = \tau_r \langle H_j, p_r\rangle, \qquad r \in \{\text{pair}, \text{operation}, \text{query}\}.$$

Straight-through hard routing selects the top two pair packets and top one operation and query packet; its backward surrogate is a softmax. Scores inspect only header images, not packet contents. With located query q, operation o, pair labels A_j, and pair-glyph images B_j^{image}, frame 1 uses the frozen V23 comparison temperature, operation reader, and canonicalizer:

$$m_j = \text{softmax}_{j\in\text{pair}}\left(\tau_{23}\langle q, A_j\rangle\right), \qquad s = \sigma(U_{23}(o)),$$

$$w_j = sm_j + (1-s)(1-m_j), \qquad x_1 = \sum_{j\in\text{pair}} w_j B_j^{\text{image}}, \qquad \hat{y}_1 = C_{23}(x_1).$$

The second step rereads the actual generated pixels:

$$r_1 = R(\hat{y}_1), \qquad b_j = \text{softmax}_{j\in\text{pair}}\left(\tau_{23}\langle r_1, B_j\rangle\right),$$

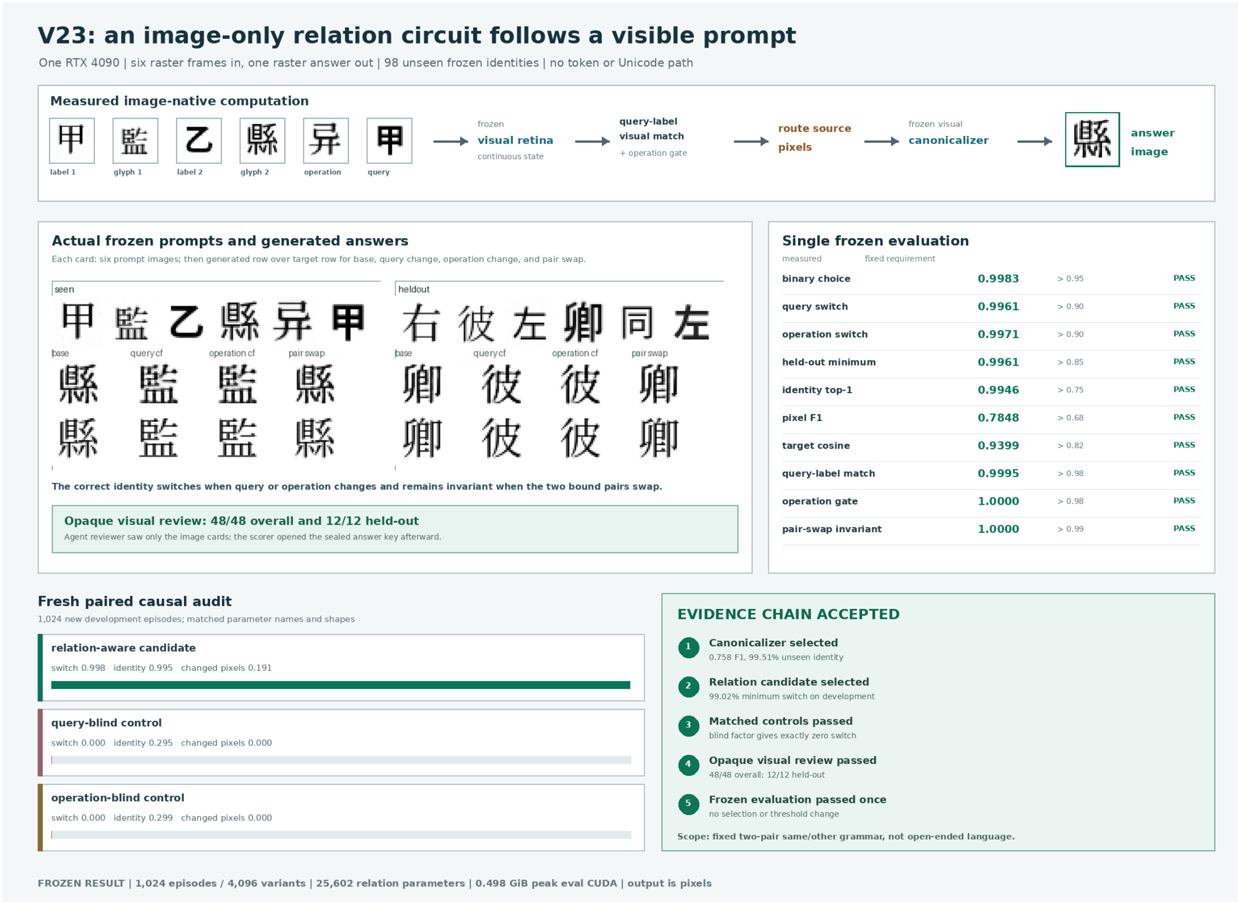

Figure 11: Measured V23 result. Actual frozen samples show generated answers above their targets for base, query-counterfactual, operation-counterfactual, and pair-swapped prompts. The matched blind controls localize both causal dependencies, and the single frozen run passes every fixed gate.

$$x_2 = \sum_{j \in \text{pair}} b_j A_j^{\text{image}}, \qquad \hat{y}_2 = C_{23}(x_2).$$

Thus no pre-render identity state crosses the frame-1/frame-2 boundary. All weighted reductions accumulate in FP64.

V16 retina parameters and every inherited V23 parameter are frozen. V24 trains only three 192-dimensional role prototypes, four 192-dimensional continuous null states, and three role temperatures: 1,347 parameters total. The packet-aware, header-blind, query-blind, operation-blind, and history-blind arms have identical parameter names, shapes, and counts. Each receives exactly 800 AdamW updates in BF16 with batch 64 and eight workers on one RTX 4090. The candidate selects step 100, then completes the fixed run in 394.18 seconds at 2.322 GiB peak allocated CUDA memory.

A fresh audit uses 1,024 newly rendered development episodes at seed 22260832. Table 7 shows the paired causes. A blind arm is exactly invariant to the intervention it cannot observe. Header blindness prevents reliable role localization, while history blindness leaves frame 1 intact but breaks frame 2.

After every paired gate passes, an opaque agent visual audit presents 48 development cards containing only the packet images, two visible source choices, and two generated frames. It exposes

Fresh paired metric	Packet-aware	Corresponding blind
Query switch	0.9922	0.0000
Query-output L1	0.1677	0.0000
Operation switch	0.9932	0.0000
Operation-output L1	0.1676	0.0000
Frame-2 history switch	0.9961	0.0000
History-output L1	0.1421	0.0000
Minimum role localization	1.0000	0.0820
Frame-1 identity top-1	0.9961	0.3410 / 0.3197 / 0.2740
Frame-2 label top-1	0.9969	0.4873 / 0.3717

Table 7: V24 fresh paired development audit. The last two control cells list query/operation/header-blind and history/header-blind results, respectively. All arms have exactly 1,347 trainable parameters.

V24 frozen gate	Measured	Required	Result
Frame-1 binary choice	0.9980	> 0.95	pass
Query switch	0.9766	> 0.90	pass
Operation switch	0.9727	> 0.90	pass
Generated-history switch	0.9961	> 0.90	pass
Held-out minimum switch	0.9635	> 0.85	pass
Frame-1 identity top-1	0.9867	> 0.75	pass
Frame-2 label top-1	0.9973	> 0.95	pass
Frame-1 pixel F1	0.8371	> 0.68	pass
Frame-2 pixel F1	0.7286	> 0.58	pass
Generated-history consistency	0.9863	> 0.92	pass
Held-out $T = 24$, frame 1	0.9847	> 0.90	pass
Held-out $T = 24$, frame 2	1.0000	> 0.90	pass

Table 8: Single authorized V24 frozen evaluation. The sealed report records no model selection, no threshold change, and no repeated frozen run.

no targets, transcriptions, correctness labels, or frozen forms. The committed judgments score 47/48 for frame 1 and 48/48 for frame 2, including 12/12 for each frame at held-out $T = 24$. This is an agent visual audit, not a human-subject study.

Only then does the evaluator instantiate frozen images. It runs exactly once on 1,024 episodes, 107 unseen identities, and four identity-bank views, without model selection, threshold changes, or repetition. Table 8 reports the primary gates. All remaining role, distractor, packet-permutation, teacher-forcing, and image-boundary gates also pass. The run takes 8.13 seconds and peaks at 0.542 GiB allocated CUDA memory.

V24 therefore accepts a variable-input, two-frame image stream under one fixed visual grammar. It establishes that frame 2 can depend causally on rereading generated frame-1 pixels, not merely on a hidden state. It does not establish arbitrary sentence parsing, factual retrieval, page continuation, historical form synthesis, free-form output length, or movie generation. V25 implements the next narrower question: remove packet algebra, train on ordinary rendered Chinese, and require the resulting visual history to beat causal and frequency controls before a writer is authorized.

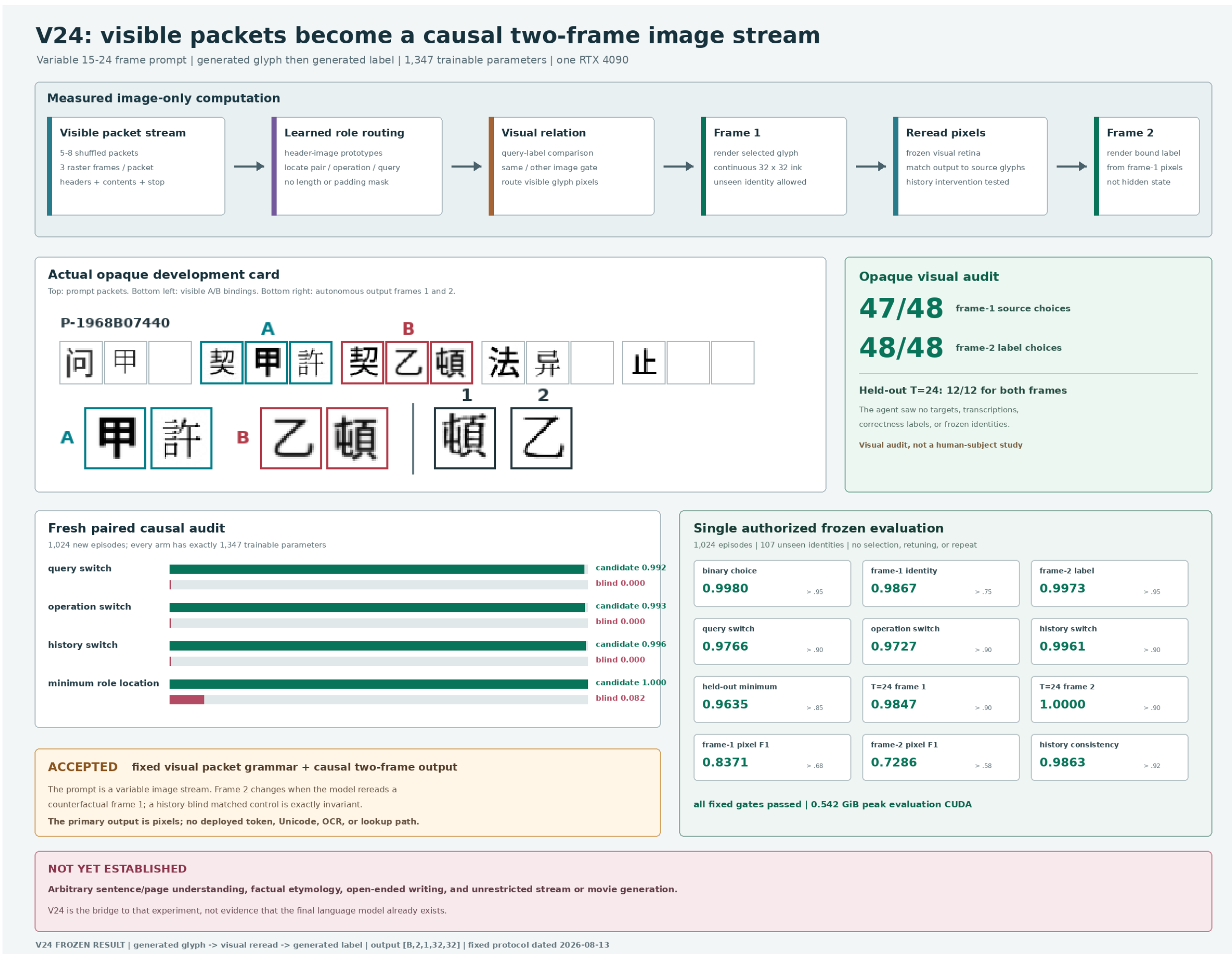

Figure 12: Measured V24 result. An actual opaque review card shows a variable-length image prompt and two autonomous answer frames. Matched controls localize header, query, operation, and generated-history causality; every gate in the single frozen evaluation passes.

5.8 Visual Cell Stream V25: natural Chinese prediction rejected

V25 treats each visible writing unit as an inspectable retinal observation and stacks observations along visual time:

$$X \in [0,1]^{B\times T\times 1\times 32\times 32}, \qquad T = 64.$$

The third nontrivial axis of one sample is reading order, not a character ID or geometric depth. The student receives only these grayscale images, their order, continuous corruption and flow times, random image noise, and its own generated pixels. Strings, bytes, token and Unicode IDs, OCR transcripts, character labels, candidate-bank rows, and external models are excluded from the learned path.

The fixed manifest contains 7,017 records from 16 public-domain Chinese works, with SHA-256 `76048753...f6866c03`. Hash partitioning by original record gives 6,608 training, 190 development, and 219 frozen groups. Training and development fonts are disjoint. Two independently rendered views of every 65-cell span reverse their context/target roles within a batch; their strings and renderer metadata are deleted before each model call. The frozen partition is counted but never rendered. An evaluator-only 1,024-form image bank and symbolic frequency tables are unavailable to the student and absent from deployment.

A frozen V16 retina maps each image to a continuous unit state $z_t \in \mathbb{S}^{191}$. Eight causal Transformer blocks of width 384 use rotary reading position and continuously corrupted prior images:

$$h_t = C_\theta(\tilde{z}_{\le t}), \qquad \hat{z}_{t+1} = \frac{P_\theta(h_t)}{\|P_\theta(h_t)\|_2}.$$

There is no character embedding or vocabulary projection. The fixed language objective combines independent-view cosine prediction with multi-positive in-batch visual contrast,

$$\mathcal{L}_{\text{language}} = 1 - \langle \hat{z}_{t+1}, z'_{t+1}\rangle + 0.5\mathcal{L}_{\text{visual contrast}}.$$

The complete model contains 25,549,714 parameters; 15,810,241 are trainable in the language stage. The run performs 2,400 BF16 AdamW updates with physical batch three, two view directions, and four-step accumulation, for 24 effective streams per update on one RTX 4090.

Development evaluation uses 2,048 fixed windows. The same model is queried with full history, only the last cell, shuffled prior history with the last cell fixed, and blank history. The evaluator separately computes image-unigram and symbolic-bigram controls from training frequencies. It also forms 512 pairs with the same last cell but different histories and different true next cells. Table 9 reports every selection-relevant measure.

Thus the causal field uses ordered history, but does not use it accurately enough to select the correct next form. Its top-1 changes on 77.93% of paired counterfactuals, yet changes to the correct target only 12.89% of the time. Context sensitivity alone is not understanding. The language stage is rejected, writer training is not authorized by the fixed path, and no frozen evaluation is opened.

For diagnosis only, a separate command explicitly marks itself exploratory and continues from the rejected language checkpoint. It freezes the causal field and trains the 7,073,617-parameter continuous flow writer for 1,200 updates. For clean target x, noise ϵ, and time τ it learns the velocity of $y_\tau = (1-\tau)x + \tau\epsilon$, conditions on h_t and $\hat{z}_{t+1}$, and rereads its sampled pixels. This run cannot change the evidence verdict. On 256 targets and 16 autonomous continuations, generated identity top-1 is zero, reread target cosine is 0.08015, and pixel F1 is 0.32210. Blank rate is zero and position-16 density is 0.97657 times target density. The writer therefore learns ink occupancy, not the intended continuation.

V25 development measure	Measured	Required	Result
Full-history top-1	0.01123	diagnostic	–
Last-only top-1	0.00146	diagnostic	–
Shuffled-history top-1	0.00342	diagnostic	–
Image-unigram top-1	0.01611	diagnostic	below
Symbolic-bigram top-1	0.12158	benchmark	below
Full minus last top-1	+0.00977	> 0.030	fail
Full minus unigram top-1	−0.00488	> 0.030	fail
Full minus last log probability	+0.04882	> 0.050	fail
Full minus shuffled top-1	+0.00781	> 0.015	fail
Counterfactual switch accuracy	0.12891	> 0.550	fail
Full-history target cosine	0.27513	> 0.550	fail
Student boundary / peak CUDA	clean / 0.598 GiB	clean / < 18	pass

Table 9: V25 fixed natural-Chinese development audit. Full history has a small advantage over last-only and shuffled history, but remains below the image unigram and symbolic bigram. Six semantic and causal gates fail.

V25 localizes the next problem to the predictive state rather than page geometry. Cross-font visual appearance, exact identity, and context-predictive meaning share one target state; cosine can improve toward a frequent visual centroid without selecting the next glyph. A subsequent preregistered model should separate an exact appearance field from a context-predictive residual and require both under matched state-shuffle interventions.

Long context has a separate exact representation. For a lattice width K, sequence index t maps serpentine-wise to

$$r = \lfloor t/K \rfloor, \qquad c = \begin{cases} t \bmod K, & r \text{ even}, \\ K - 1 - (t \bmod K), & r \text{ odd}. \end{cases}$$

This bijection preserves adjacency at row turns and maps clean $N \times 1 \times 32 \times 32$ cells to either an inspectable flat page or a learned $C \times R \times K$ retinal lattice with a validity mask. A 256×256 lattice holds 65,536 cells, but was not used in V25. Scaling this representation before repairing the 64-cell objective would increase context without establishing better language.

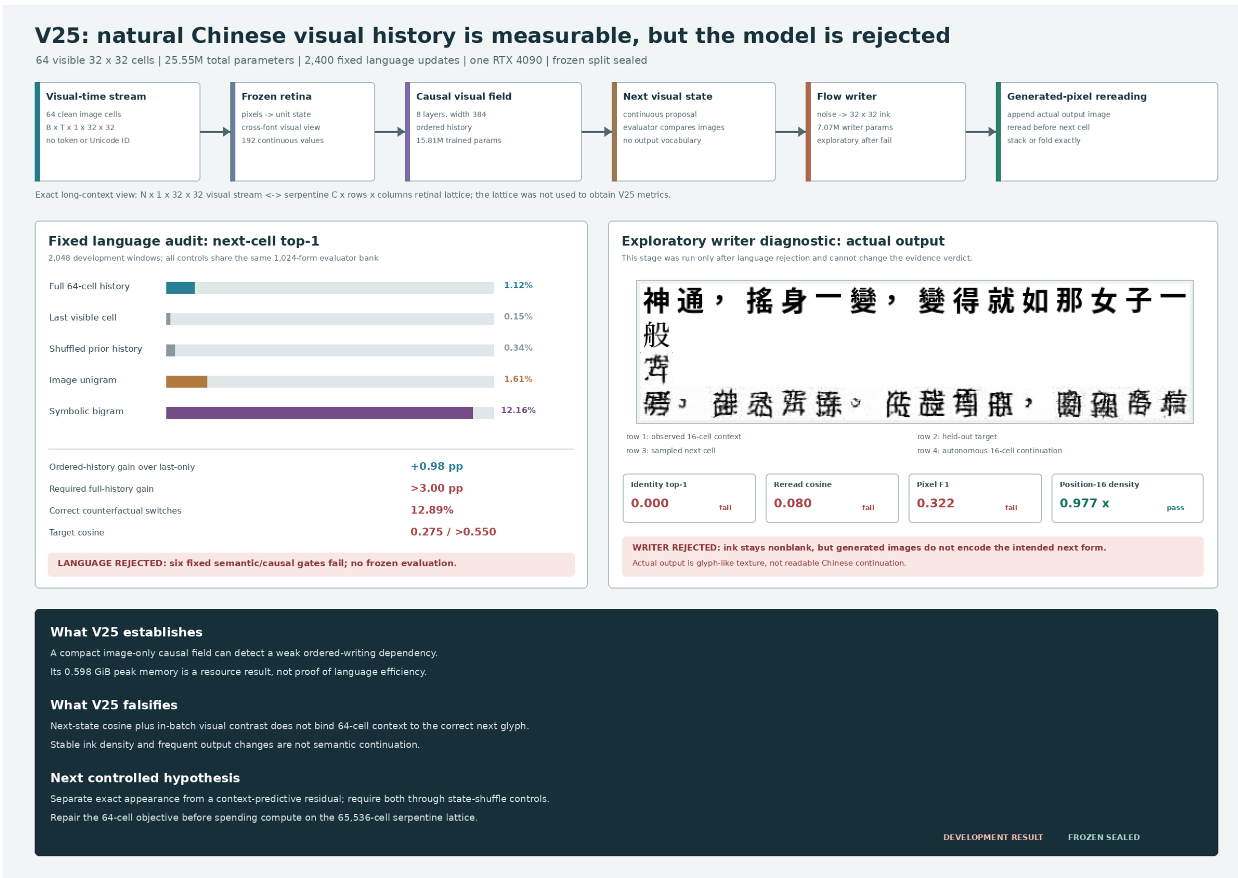

Figure 13: Measured V25 development result generated directly from the fixed language report, separately labeled exploratory writer report, and actual writer sample. Ordered visual history has a weak effect, but the unigram and bigram controls remain stronger and generated images are not the intended Chinese forms. The frozen partition stays sealed.

5.9 Factorized Visual Context V26: changed state, failed binding

V26 tests the repair proposed by V25 while preserving the same ordinary-Chinese corpus, record-group partitions, disjoint font partitions, 64-cell horizon, frozen V16 retina, and strict student boundary. It changes the supervised unit. Every natural training example predicts only its full-context endpoint and future offsets 2, 4, and 8; losses are no longer averaged over all short prefixes. A fixed set of 16,384 cross-record training pairs shares the same four-character suffix but has different earlier histories and different next glyphs. Pair suffixes are rendered with identical font and augmentation draws, so their final four image tensors are exactly equal.

For $X = (x_1, \ldots, x_{64})$, the appearance route receives only the last visible image and the history route receives only the preceding images:

$$a = A_\theta(R(x_{64})), \qquad r = C_\theta(R(x_{1:63}))\,, \qquad s = \mathrm{RMSNorm}(a + \sigma(G_\theta([a; r])) \odot r)\,.$$

Zeroing r is therefore a true last-only intervention. For two suffix-matched contexts, swapping r changes only earlier-history state while preserving the last-glyph appearance route.

For horizon $h \in \{1, 2, 4, 8\}$ and independent Gaussian noise ϵ_k, $k = 1, \ldots, 8$, a shared conditional head emits unit retinal particles

$$q_{h,k} = \mathrm{normalize}\, P_\theta(s + e_h + N_\theta(\epsilon_k)) \in \mathbb{S}^{191}.$$

These samples are continuous predictions, not vocabulary rows or persistent glyph prototypes. For target state y_h, the empirical energy score is

$$\mathcal{E}(Q_h, y_h) = \frac{1}{8}\sum_k \|q_{h,k} - y_h\|_2 - \frac{1}{128}\sum_{k,j}\|q_{h,k} - q_{h,j}\|_2.$$

Training combines next- and multi-horizon energy scores, continuous candidate ranking against an 8,192-entry FIFO of detached image states, and a softplus pair-ranking loss with fixed margin 0.10. The queue contains image observations but no IDs and is absent from deployed inference. This design follows the calibrated continuous-prediction motivation of proper energy scores [45]; it does not assume that particle diversity itself is semantic.

The complete model contains 19,142,721 parameters, of which 16,476,865 are trainable. Its fixed run performs 8,000 BF16 AdamW updates in 1,863.27 seconds on one RTX 4090, with 0.888403 GiB peak allocated CUDA memory. The corpus manifest SHA-256 is `76048753...f6866c03`, the preregistered protocol SHA-256 is `dcfa5b97...b899df1`, and the final checkpoint SHA-256 is `065f84e1...db6e1d`. The frozen partition is counted but never rendered.

The fixed natural audit uses 2,048 development windows and the same evaluator-only 1,024-form image bank for every branch. Table 10 shows the preregistered mechanism and language comparisons.

The retina-bank oracle is exactly 1.0, so the frozen visual sensor and evaluator can distinguish every audited target. The model nevertheless assigns almost no top-1 advantage to history. Its full-history target log probability is −6.932756, versus −6.952485 for suffix-4 and −6.936448 after shuffling cells 1–60 while preserving the suffix. Only the full-versus-last log-probability gate passes.

The primary causal audit uses 512 cross-record suffix-4 pairs. Its result is shown in Table 11.

The pair result localizes the failure. The history branch is not constant: pixel-identical suffix contexts produce substantially different residuals. Those differences do not bind to the correct next-glyph distribution. Because suffix-matched pair members have exactly the same appearance state, ordinary two-way pair ranking and the residual-swap target test are algebraically equivalent in this implementation; both are retained to preserve the preregistered intervention interface rather

V26 natural development measure	Measured	Required	Result
Full-history top-1	0.000488	diagnostic	–
Last-only / suffix-4 top-1	0.000488 / 0.000488	diagnostic	–
Shuffled-prefix top-1	0.000488	diagnostic	–
Image-unigram top-1	0.014160	control	below
Symbolic-bigram top-1	0.135254	benchmark	below
Full minus last top-1	0.000000	> 0.020	fail
Full minus unigram top-1	-0.013672	> 0.030	fail
Full minus last log probability	$+0.170451$	> 0.10	pass
Full minus suffix-4 log probability	$+0.019729$	> 0.03	fail
Full minus shuffled log probability	$+0.003692$	> 0.03	fail
Full target cosine / particle spread	0.160451 / 1.210599	diagnostic	–
Retina oracle / student boundary	1.0 / clean	≥ 0.99 / clean	pass
Peak allocated CUDA	0.888403 GiB	< 18 GiB	pass

Table 10: V26 fixed natural-Chinese development audit. Full history improves target log probability over last-only, but not over the same four-cell suffix or suffix-preserving history shuffle. It remains below image unigram and symbolic bigram controls.

V26 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Mean appearance-state difference	0.000000	exact	pass
Mean history-residual difference	4.628258	diagnostic	changed
Correct two-way pair ranking	0.500000	> 0.65	fail
Swapped-residual target accuracy	0.500000	> 0.65	fail
Mean correct-versus-other margin	0.0000677	diagnostic	near zero
Top-1 output switch rate	0.212891	diagnostic	–
Both pair targets top-1 correct	0.025391	diagnostic	–

Table 11: V26 pixel-identical suffix intervention. Earlier history materially changes the continuous residual, but neither ordinary nor swapped-residual scoring prefers the associated target above chance.

than counted as independent evidence. A suffix-8 diagnostic agrees, with 0.501953 ranking and swap accuracy.

V26 therefore rejects this factorization and objective, not image-native language modeling in general. A broad particle cloud and a changing hidden state are not evidence of useful conditional language. The next attributable experiment should remain at 64 cells and make matched target choice conditionally identifiable during optimization. It must beat suffix-only, history-shuffled, image-unigram, and symbolic-bigram controls before a pixel writer, serpentine page lattice, geometric depth, or motion is authorized.

We further localize this failure with a post-hoc diagnostic that cannot revise the V26 result or authorize frozen evaluation. Every V26 parameter is frozen. Three visual scorers, totaling 1.11M trainable parameters, receive appearance, history residual, or fused state and score two arbitrary candidate images using normalized projected compatibility. One pass over all 16,384 train suffix pairs optimizes symmetric 2×2 assignment. The fixed 512 development pairs yield 1,024 cross-font assignments and 2,048 scored arms. Appearance-only accuracy is exactly 0.500000, history accuracy is 0.506836, and fused-state accuracy is 0.503418. Their mean margins are 0.000108 and 0.000371. In contrast, direct frozen-retina cosine assigns the same cross-font target images at 0.999512 with

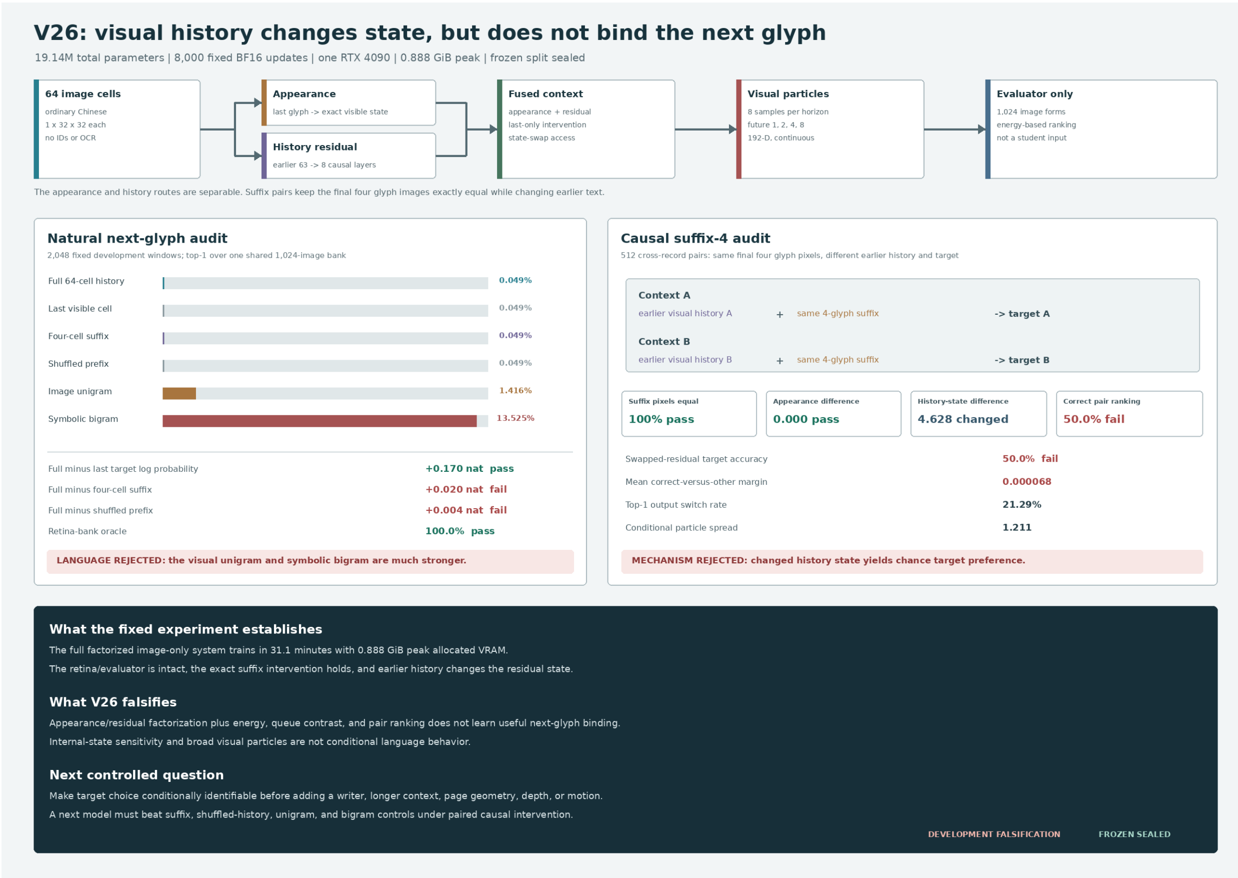

Figure 14: Measured V26 development result generated directly from the fixed audit. The visual retina and exact suffix intervention pass, and earlier history changes the residual state, but natural next-glyph accuracy remains below frequency controls and matched target preference is exactly chance. The frozen partition stays sealed and no writer is trained.

a mean margin of 0.747237. Thus candidate visibility is intact, but V26's history state does not expose a transferable next-glyph relation to this deterministic nonlinear decoder.

The resulting V27 direction is therefore stricter than replacing an output head. A trainable image retina and causal context field must be optimized jointly with deterministic candidate compatibility under matched alternatives. The primary score remains image-to-image and vocabulary-free; an appearance-only chance arm, raw-retina identity control, prefix shuffle, unigram, and symbolic bigram remain required controls. Stochastic form generation follows only after this relation passes.

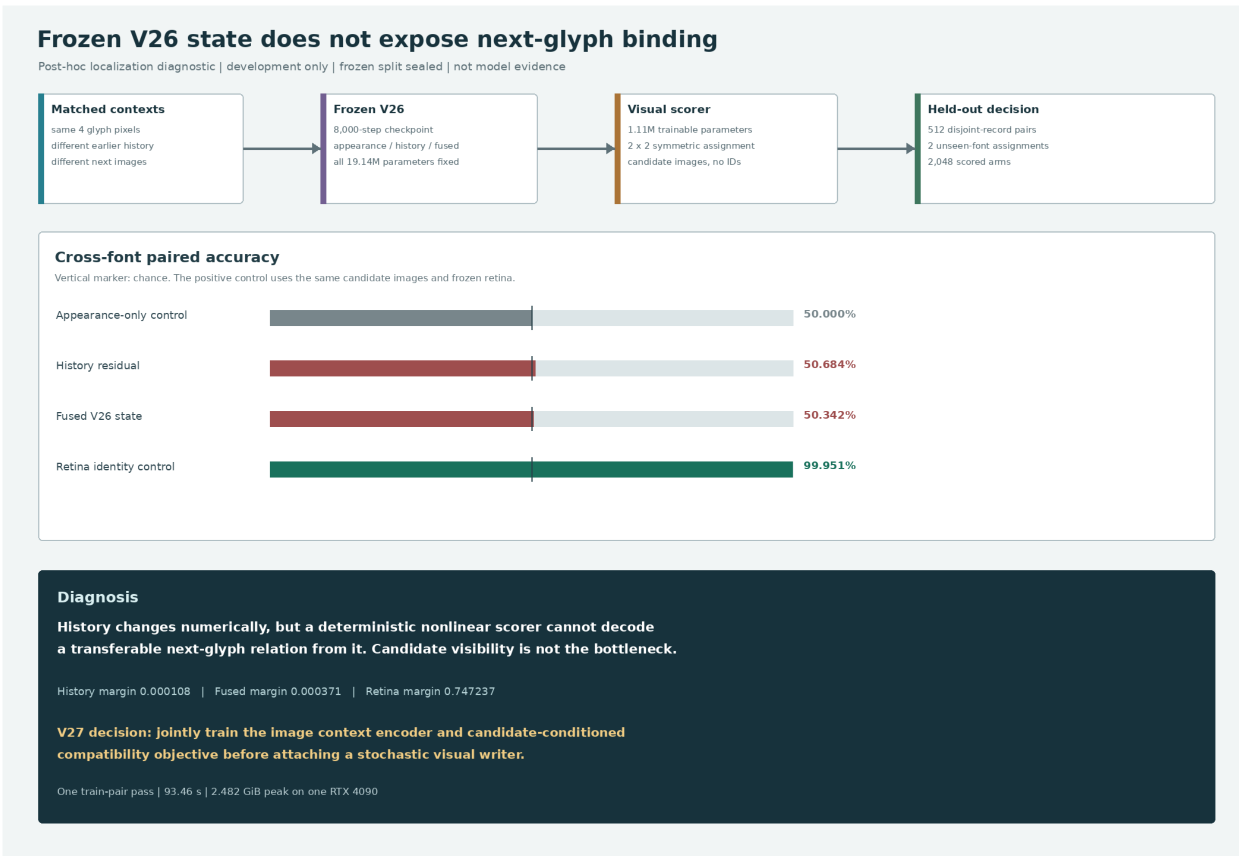

Figure 15: Post-hoc frozen-V26 localization. The raw retina nearly perfectly distinguishes paired target images across unseen fonts, while deterministic candidate scorers over frozen history and fused states remain at chance. This is a development diagnostic, not preregistered model evidence.

5.10 Joint Visual Compatibility V27: visible candidates, failed relation

V27 executes that joint test without changing the corpus, record partitions, font separation, 64-cell horizon, or student boundary. Its online retina initializes from V16 and is optimized at one tenth the main learning rate. An eight-layer width-384 causal rotary field maps the ordered context images to a normalized query. An exponential-moving-average retina and residual image projector map an arbitrary candidate image to a normalized key:

$$q(X) = \operatorname{normalize} Q(F(R_o(X))_{64}), \qquad k(Y) = \operatorname{normalize} G_t(R_t(Y)),$$

$$s(X,Y) = \exp(\alpha) q(X)^\top k(Y), \qquad \exp(\alpha) \le 100.$$

The candidate collection is not part of the model or checkpoint. At deployed inference the scorer accepts any floating candidate image. Every paired training view independently permutes its two candidate images, and the evaluator undoes only that visible-array permutation. This makes fixed diagonal position unusable as an identity channel.

The fixed objective combines multi-positive cross-font natural-image contrast, symmetric row-and-column assignment on suffix-matched pairs, cross-font candidate identity, and VICReg variance/covariance regularization:

$$\mathcal{L} = \mathcal{L}_{\text{natural}} + 2\mathcal{L}_{\text{pair}} + 0.5\mathcal{L}_{\text{identity}} + 0.05\mathcal{L}_{\text{VC}}.$$

No vocabulary classification, symbolic target, reconstruction decoder, stochastic particle head, or writer is trained. The complete model has 18,599,553 parameters, of which 17,118,401 are trainable. The single fixed run performs 8,000 BF16 AdamW updates. Training and the complete development audit take 2,347.47 seconds on one RTX 4090 with 2.267916 GiB peak allocated CUDA memory. The protocol SHA-256 is `0c386dd2...cb223310`; the final checkpoint SHA-256 is `49dc07f2...1e77e539`. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and the same evaluator-only 1,024-image bank for every branch. Table 12 reports the fixed language comparisons.

V27 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.016113 / 0.023926	diagnostic	–
Last-only / suffix-4 top-1	0.001953 / 0.008301	diagnostic	–
Shuffled-prefix top-1	0.013184	diagnostic	–
Image-unigram top-1	0.020508	control	below
Symbolic-bigram top-1	0.125977	benchmark	below
Full minus unigram top-1	−0.004395	> 0.03	fail
Full minus bigram top-1	−0.109863	> 0.01	fail
Full minus bigram log probability	−1.856770	> 0.05	fail
Full minus suffix-4 log probability	+0.059248	> 0.03	pass
Full minus shuffled log probability	+0.002726	> 0.03	fail
Learned 1,024-bank cross-font identity	0.948730	≥ 0.99	fail
Student boundary / peak CUDA	clean / 2.267916 GiB	clean / < 18	pass

Table 12: V27 fixed natural-Chinese development audit. Full context improves target log probability over the four-cell suffix, but the advantage almost vanishes when the same prefix images are shuffled. The model remains below image unigram and symbolic bigram controls.

Only one of five natural-language gates passes. Full target log probability is −6.876778, versus −6.936025 for suffix-4 and −6.879503 after shuffling cells 1–60. Thus the apparent context gain is not attributable to the order of the earlier writing. Full top-1 is also lower than both frequency controls.

The primary causal audit contains 512 cross-record suffix-4 pairs in two unseen development fonts, yielding 1,024 assignments and 2,048 scored arms. Table 13 reports the fixed mechanism controls.

V27 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Candidate permutation score error	0.000000	$< 10^{-5}$	pass
Candidate permutation agreement	1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	≥ 0.99	pass
Learned 1,024-bank identity	0.948730	≥ 0.99	fail
Full-context arm accuracy	0.507080	> 0.65	fail
Full both-correct rate	0.088867	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.505615	control	near chance
Full minus suffix-4 accuracy	+0.007080	> 0.15	fail
Full minus shuffled accuracy	+0.001465	> 0.05	fail
Full minus shuffled margin	+0.002265	> 0.02	fail

Table 13: V27 pixel-identical suffix intervention. Candidate visibility, permutation equivariance, suffix equality, and chance controls pass. The learned context/candidate relation remains at chance and does not materially exceed a suffix-preserving prefix shuffle.

The raw V16 retina reaches 0.999512 cross-font target identity with 0.736124 mean cosine margin. Candidate pixels and the evaluator are therefore adequate. This is a two-candidate pair control. The learned identity gate instead uses 1,024 candidates, reaches 0.948730, and therefore cannot be subtracted from the raw number as a before/after change. Full assignment remains only 0.1465 percentage point above the shuffled-prefix arm. Seven of thirteen mechanism gates pass, but all five context-binding gates and the learned full-bank identity gate fail.

V27 rejects this global query/key compatibility mechanism, not image-native language modeling in general. It also removes a tempting explanation for V26: the main failure is not that arbitrary candidate images are invisible to the retina. The next controlled experiment should preserve the raw retinal geometry exactly and supervise dense, explicitly ordered future visual fields rather than compressing all language evidence into one global dot product. The authoritative representation remains an ordered $N \times 1 \times 32 \times 32$ stream of full glyph images. A reversible 2D fold may expose parallel spatial computation, but must invert exactly and pass an order intervention. A writer, longer lattice, geometric depth, and motion remain unauthorized until ordered context beats suffix, shuffle, unigram, and bigram controls.

V28 executes this dense-future test below without changing the corpus, partitions, student boundary, or frozen-data rule.

5.11 Dense Visual Future Energy V28: order signal without binding

V28 consumes an authoritative ordered volume $X \in \mathbb{R}^{64\times1\times32\times32}$ and predicts future writing images at offsets $d \in \{1, 2, 4\}$. The fixed V16 retina is never updated. For image x, a residual semantic adapter, initialized as the identity, and its stop-gradient EMA copy produce

$$r(x) = \text{normalize}\, R_0(x), \qquad z(x) = \text{normalize}\, A_\theta(r(x)), \qquad \bar{z}(x) = \text{EMA}(z(x)).$$

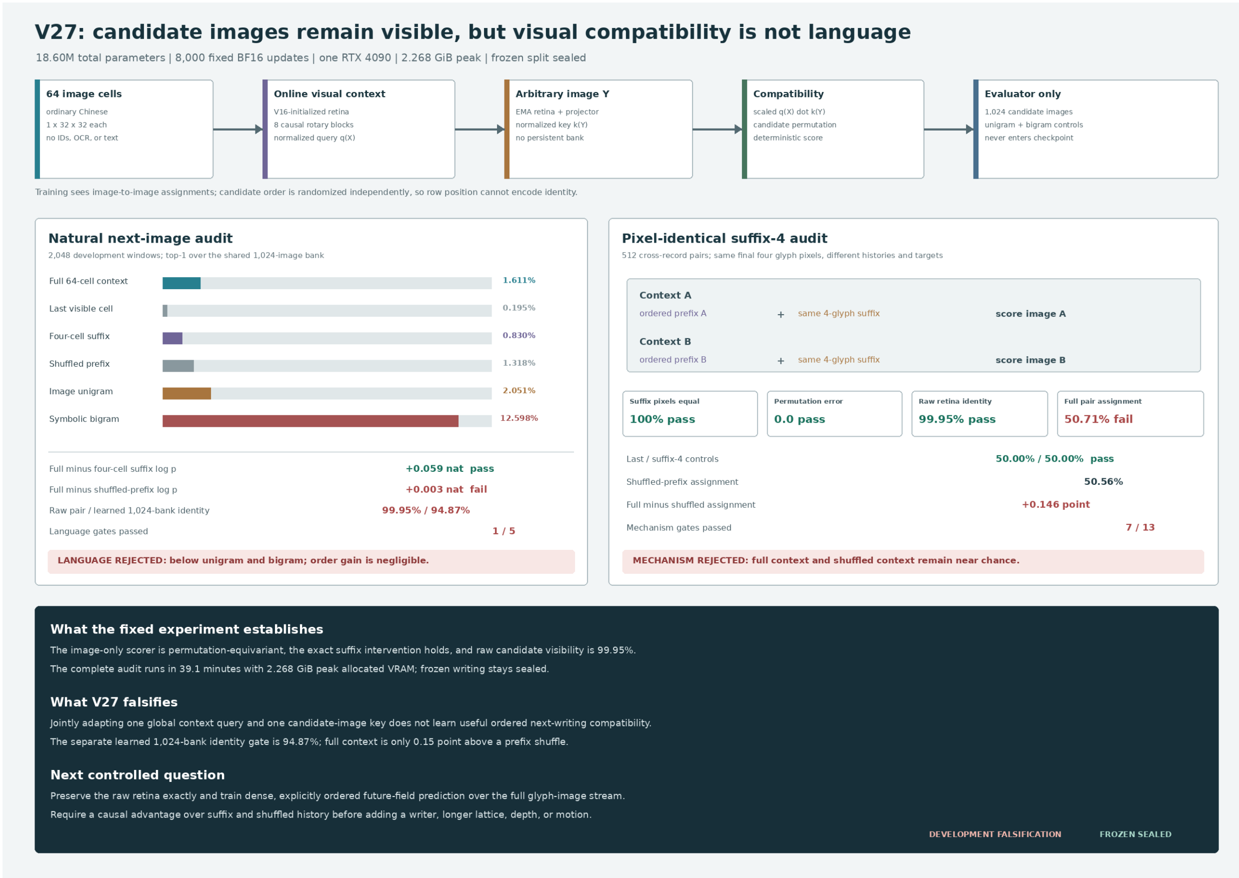

Figure 16: Measured V27 development result generated directly from the fixed audit. The image-only boundary, candidate permutation, exact suffix intervention, and raw-retina identity controls pass. Jointly learned visual compatibility remains near chance, misses its separate full-bank identity gate, and does not use prefix order. The frozen partition stays sealed and no writer is trained.

An eight-layer width-384 causal rotary field reads the concatenated raw and semantic states. At each selected position and horizon it emits four normalized raw hypotheses, four normalized semantic hypotheses, and mixture logits:

$$H_{1:t} = F([r(x_1); z(x_1)], \ldots, [r(x_t); z(x_t)]),$$
$$\{q^r_{d,k}, q^z_{d,k}, \pi_{d,k}\}_{k=1}^4 = P_d(H_t).$$

The score of an arbitrary candidate image Y is continuous and contains no vocabulary table:

$$s_d(X, Y) = \log \sum_{k=1}^{4} \pi_{d,k} \exp\Big(\tau_r q^{r\top}_{d,k} r(Y) + \tau_z q^{z\top}_{d,k} \bar{z}(Y)\Big).$$

Each natural microbatch samples 16 stratified causal positions, including the endpoint, and uses opposite-font targets at horizons 1, 2, and 4. The fixed objective is

$$\mathcal{L} = \mathcal{L}_{\text{dense}} + 0.25\mathcal{L}_{\text{ES}} + 0.5\mathcal{L}_{\text{id}} + 2\mathcal{L}_{\text{pair}} + \mathcal{L}_{\text{natural-order}} + \mathcal{L}_{\text{pair-order}}.$$

Here $\mathcal{L}_{\text{dense}}$ is weighted multi-positive cross-font contrast, $\mathcal{L}_{\text{ES}}$ is a strictly proper empirical energy score over the continuous raw hypotheses, and $\mathcal{L}_{\text{id}}$ trains the semantic route across fonts. The final three terms use identical candidate pixels under symmetric suffix-pair assignment and under explicit full, suffix-only, and suffix-preserving shuffled-prefix interventions. Temporary equality groups are derived from canonical pixels by the host loss; they are not model inputs or inference state.

The complete model has 17,859,142 parameters, of which 16,377,990 are trainable. Its single fixed run performs 10,000 BF16 AdamW updates and the full development audit in 7,134.82 seconds (118.91 minutes) on one RTX 4090, with 1.144400 GiB peak allocated CUDA memory. The protocol SHA-256 is `b8e515d2...a7780f7f`, the final checkpoint SHA-256 is `22503464...2c7f64f`, and the audit SHA-256 is `2cb73707...8f0fb3d`. The frozen partition is never rendered.

The natural audit uses 2,048 development windows and one fixed evaluator-only 1,024-image bank. Table 14 reports the result.

V28 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.014160 / 0.035156	diagnostic	–
Last-only / suffix-4 top-1	0.004395 / 0.015625	diagnostic	–
Shuffled-prefix top-1	0.002441	diagnostic	–
Image-unigram top-1	0.018555	control	below
Symbolic-bigram / trigram top-1	0.131348 / 0.193359	controls	below
Full minus unigram top-1	-0.004395	> 0.03	fail
Full minus bigram top-1	-0.117188	> 0.01	fail
Full minus bigram log probability	-1.801083	> 0.05	fail
Full minus suffix-4 log probability	$+0.030369$	> 0.03	pass
Full minus shuffled log probability	$+0.215112$	> 0.03	pass
Raw / EMA 1,024-way identity	0.920410 / 0.964355	EMA ≥ 0.95	pass
EMA minus raw identity	$+0.043945$	≥ 0.02	pass
Student boundary / peak CUDA	clean / 1.144400 GiB	clean / < 18	pass

Table 14: V28 fixed natural-Chinese development audit. Full context changes target probability relative to suffix and shuffled controls, and the semantic route improves same-scope identity. Correct future-image rank remains below the unigram, bigram, and trigram controls.

Full target log probability is -6.829707, versus -6.860076 for suffix-4 and -7.044819 after shuffling cells 1–60. This is an attributable order effect in probability, but suffix-4 top-1 is slightly higher than full-context top-1 and the symbolic bigram is more than nine times as accurate. Four of six language gates fail.

The primary causal audit uses 512 cross-record context pairs whose final four glyph images are bitwise equal but whose earlier histories and targets differ. Candidate order is independently permuted in two unseen development fonts. Table 15 reports the fixed mechanism controls.

V28 is a clean negative result. It establishes that the $N \times 1 \times 32 \times 32$ visual-time interface, frozen-retina semantic adaptation, dense multi-horizon prediction, and continuous candidate scoring are executable within 1.15 GiB peak allocated memory. It does not establish a useful language model or a capability-normalized efficiency advantage. The failure is now narrower: ordered prefix information changes scores, but the training objective does not distribute that evidence reliably enough to bind each context to its correct future image.

V29 executes that candidate-conditioned incremental-evidence test below without changing the corpus, partitions, student boundary, or frozen-data rule.

5.12 Conditional Visual Density Ratio V29: order without assignment

V29 retains the authoritative image stream $X \in [0,1]^{B \times N \times 1 \times 32 \times 32}$, where N is reading time. It loads the V28 context input, eight causal blocks, context normalization, frozen V16 retina, and both

V28 suffix-4 pair measure	Measured	Required	Result
Suffix pixel equality	1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw V16 two-candidate identity	0.999512	diagnostic	–
Full-context arm accuracy	0.495605	> 0.65	fail
Full both-correct rate	0.096680	> 0.40	fail
Last-only / suffix-4 accuracy	0.500000 / 0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.499512	control	chance
Full minus suffix-4 accuracy	−0.004395	> 0.15	fail
Full minus shuffled accuracy	−0.003906	> 0.05	fail
Full minus shuffled mean margin	+0.022070	> 0.02	pass

Table 15: V28 pixel-identical suffix intervention. The full score has a larger mean margin than the shuffled score on a subset of examples, but this movement does not yield reliable correct rank or context-to-target binding.

frozen semantic adapters. V28 future heads are discarded. The causal field returns one retained state per visible image:

$$H_T = C_\theta([r(x_1); z(x_1)], \ldots, [r(x_T); z(x_T)]) \in \mathbb{R}^{T\times 384}.$$

An arbitrary candidate image y forms a continuous query $q_0 = W_y[r(y); \bar{z}(y)]$. Two six-head cross-attention layers update this query from all context states. The scalar relation head receives the updated query, its displacement, and its elementwise interaction with the original query:

$$q_{l+1} = \mathrm{EvidenceLayer}_l(q_l, H_T), \qquad \rho_\theta(H_T, y) = M_\theta([q_2; q_2 - q_0; q_2 \odot q_0]).$$

There is no candidate-ID embedding or bank-sized output layer.

For prefix P, exact four-image suffix S, and candidate y, define

$$F(P, S, y) = \rho_\theta(H(P, S), y), \qquad B(S, y) = \rho_\theta(H(S), y),$$

$$G(P, S, y) = F(P, S, y) - B(S, y).$$

Rows of G are centered over candidates. Natural training contrasts all 1,024 host-side candidate images for full, suffix, and incremental scores and adds an ordered-versus-shuffled increment margin. Exact-suffix pairs add symmetric two-by-two full and incremental assignment, per-row positive margin, and per-row ordered-versus-shuffled margin. Candidate images are encoded once by the frozen image path, but their bank is neither serialized nor required at inference. The student receives no bank index, source string, token, Unicode ID, character ID, OCR result, symbolic probability, or external-model state.

The complete model has 20,080,961 parameters, of which 18,451,585 are trainable. Its single fixed run performs 8,000 BF16 AdamW updates and the full development audit in 4,956.14 seconds (82.60 minutes) on one RTX 4090, with 3.356465 GiB peak allocated CUDA memory. The protocol SHA-256 is `4cb17b79...9ffad0f`, the final checkpoint SHA-256 is `a8ec9919...d508b8a`, and the audit SHA-256 is `16645844...4a5108`. Frozen images are never instantiated.

The fixed natural audit uses 2,048 development windows and one evaluator-only 1,024-image bank. Table 16 reports the result.

The primary causal audit uses 512 cross-record pairs whose final four writing images are bitwise equal while earlier histories and next images differ. Candidate order is independently permuted in two unseen development fonts. Table 17 reports the fixed mechanism controls.

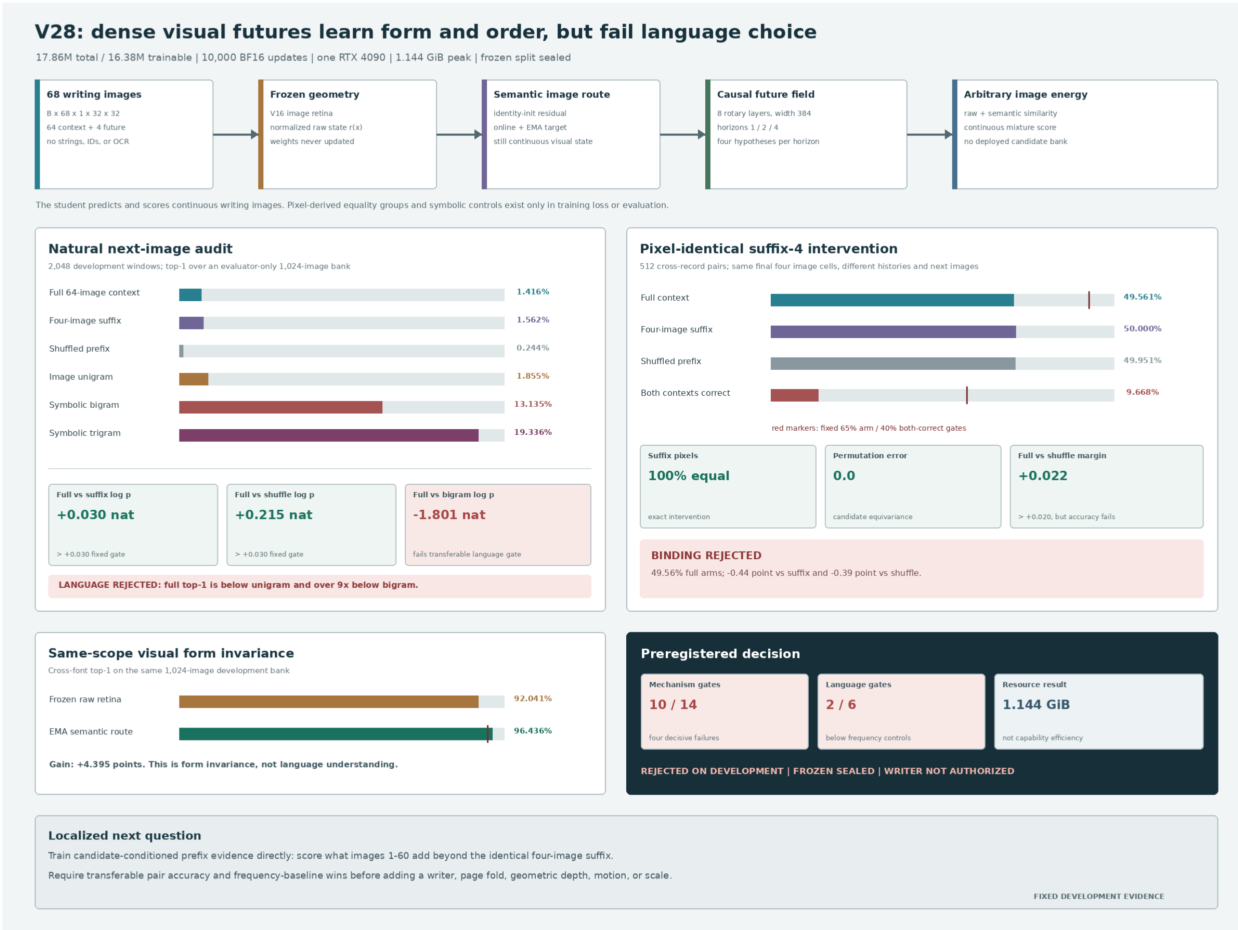

Figure 17: Measured V28 development result generated directly from the fixed audit. Frozen raw geometry and the learned semantic route preserve candidate visibility, and ordered history affects probability and mean margin. Natural selection remains below frequency controls and matched pair assignment stays at chance. Ten of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

The exact suffix exposes a structural limit of the subtraction. Both context rows share one baseline vector (b_1, b_2). For the diagonal assignment, the mean incremental margin is

$$
\begin{aligned}
&\tfrac{1}{2}[(G_{11} - G_{12}) + (G_{22} - G_{21})] \\
&= \tfrac{1}{2}[(F_{11} - b_1 - F_{12} + b_2) + (F_{22} - b_2 - F_{21} + b_1)] \\
&= \tfrac{1}{2}[(F_{11} - F_{12}) + (F_{22} - F_{21})].
\end{aligned}
$$

The suffix terms cancel exactly, and row centering leaves within-row margins unchanged. Subtraction can redistribute margin between rows, but cannot improve the aggregate assignment margin beyond the full critic. The fixed audit confirms that full and incremental mean margins are both 0.005103277.

V29 is a clean negative result. It rejects this candidate-query critic and density-ratio loss, not image-native language modeling or the ordered visual cell interface. The failure is now more specific: a global scalar critic can learn whether a history resembles natural order without learning which candidate image belongs to which history. A bounded successor should predict a continuous next-

V29 natural development measure	Measured	Required	Result
Full-context top-1 / top-5	0.023438 / 0.063965	top-1 ≥ 0.15	fail
Suffix-4 / shuffled top-1	0.013672 / 0.011719	diagnostic	–
Increment / shuffled increment top-1	0.023438 / 0.008789	diagnostic	–
Image-unigram top-1	0.013672	control	full above
Symbolic-bigram / trigram top-1	0.138672 / 0.202637	controls	full below
Full minus unigram top-1	+0.009766	> 0.03	fail
Full minus bigram top-1	−0.115234	> 0.01	fail
Full minus bigram log probability	−1.516622	> 0.05	fail
Full minus suffix-4 log probability	+0.179446	> 0.03	pass
Full minus shuffled log probability	+3.102479	> 0.03	pass
Raw / frozen-semantic identity	0.920410 / 0.964355	semantic ≥ 0.95	pass
Boundary / bank absence / peak CUDA	clean / clean / 3.356465 GiB	required	pass

Table 16: V29 fixed natural-Chinese development audit. Earlier ordered images strongly affect target probability, but full 1,024-way ranking remains far below a symbolic bigram. Order detection is not conditional visual choice.

V29 suffix-4 pair measure	Measured	Required	Result
Suffix pixels / score rows	1.000000 / 1.000000	exact	pass
Permutation error / agreement	0.000000 / 1.000000	exact	pass
Raw two-candidate identity	0.999512	≥ 0.99	pass
Full arm / both-correct	0.497314 / 0.038086	> 0.65 / > 0.40	fail
Suffix-only arm accuracy	0.500000	exact chance	pass
Shuffled-prefix arm accuracy	0.501221	control	chance
Increment arm / both-correct	0.507080 / 0.089844	> 0.65 / > 0.40	fail
Shuffled-increment arm accuracy	0.495605	control	chance
Increment minus shuffled accuracy	+0.011475	> 0.10	fail
Increment minus shuffled margin	+0.003167	> 0.05	fail

Table 17: V29 pixel-identical suffix intervention. Perception, exactness, and candidate equivariance pass. Full and incremental candidate assignment remain at chance and fail every learned binding threshold.

image patch field from context, encode an arbitrary candidate into the same field, and compare corresponding patches before scalar aggregation. A parameter- and compute-matched bilinear visual scorer should control whether spatial interaction, rather than capacity, provides any gain. The existing suffix, shuffle, frequency, permutation, identity, and boundary controls must pass before a writer, page fold, geometric depth, motion, or larger model is authorized.

5.13 Spatial Visual Next-Field V30: local sensitivity without language

V30 changes the predicted random variable rather than the scalar-score baseline. Both arms retain the V29 causal context field and frozen visual perception, discard the candidate-query critic, and emit a candidate-independent continuous next-image field:

$$P_\theta(X) \in \mathbb{R}^{16\times 192}.$$

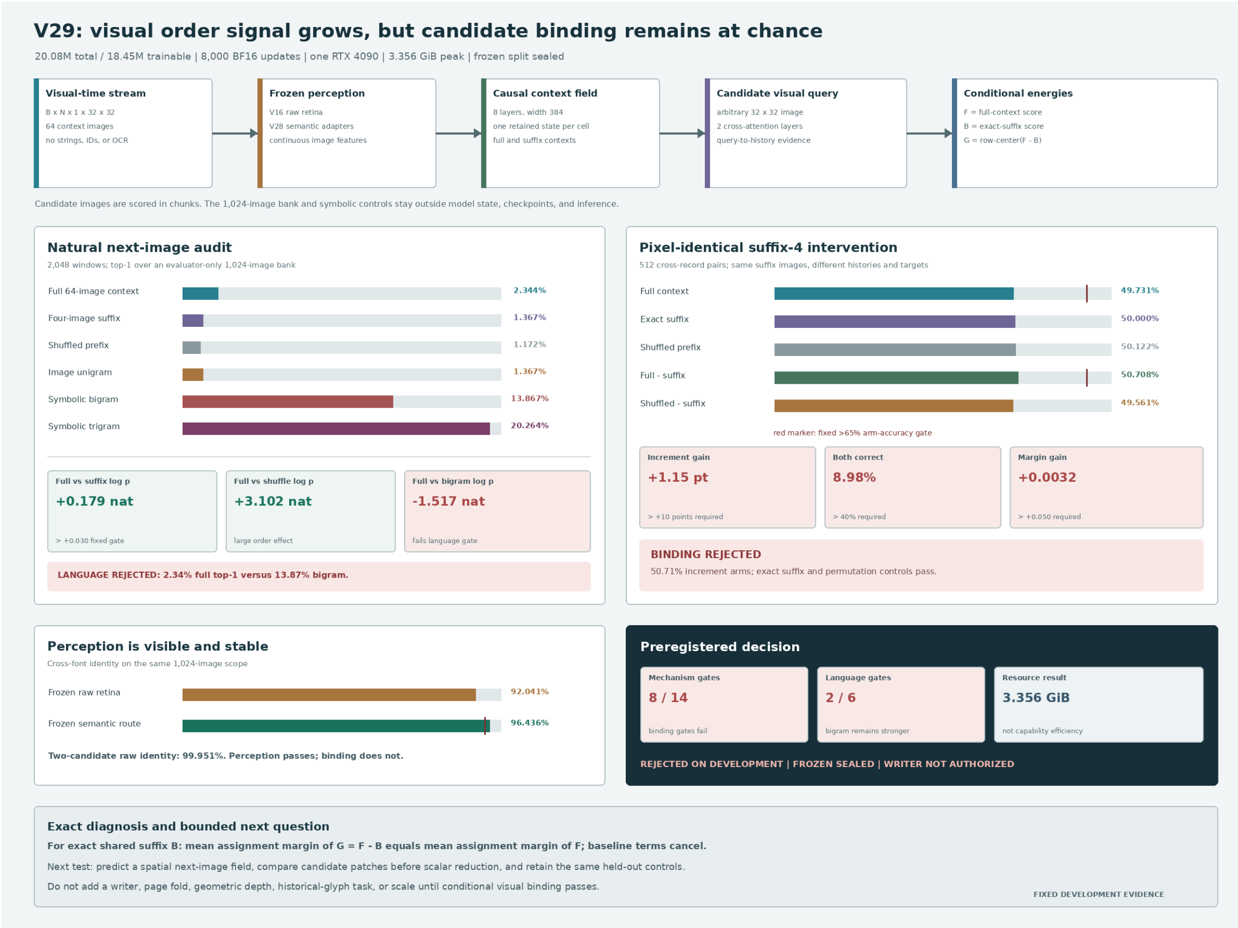

Figure 18: Measured V29 development result generated directly from the fixed audit. The critic detects a large natural order effect while exact-suffix candidate binding stays at chance. All perception, equality, permutation, leakage, and resource controls pass. Eight of fourteen mechanism gates and two of six language gates pass, so frozen evaluation and writer training remain unauthorized.

The 16 rows preserve the frozen retina's 4×4 topology. For an arbitrary candidate image y, the spatial arm compares corresponding normalized retinal cells before reduction,

$$s_{sp}(X, y) = \tau \frac{1}{16} \sum_{p=1}^{16} P_{\theta,p}(X)^\top u_p(y).$$

The matched control repeats the candidate's frozen global-semantic vector over the same rows,

$$s_{gl}(X, y) = \tau \frac{1}{16} \sum_{p=1}^{16} P_{\theta,p}(X)^\top z(y).$$

It preserves candidate visibility, trainable capacity, score reduction, and compute topology while removing local correspondence. A 180-degree reversal of the candidate cells intervenes on the spatial route and leaves the global route exactly invariant.

The arms have 18,641,153 total and 17,011,777 trainable parameters. They start from tensors with the same keys, shapes, dtypes, values, and SHA-256 (`507d875d...d9547ca8e`). Each independently completes 8,000 finite BF16 AdamW updates on one RTX 4090. The spatial and global runs take 2,693.27 and 3,789.99 seconds and peak at 1.595290 and 1.597071 GiB allocated memory. The joint audit takes 39.33 seconds. These are resource measurements for a rejected mechanism, not capability-normalized efficiency claims. The protocol, spatial checkpoint, global checkpoint, and audit SHA-256 values begin `81d2b2af`, `11a3a7e9`, `66378d4b`, and `2d0a3a08`, respectively.

Table 18 reports the fixed 2,048-window natural audit. Both routes detect ordered context, but the global route is stronger and both remain far below the symbolic bigram.

V30 natural development measure	Spatial	Global	Spatial requirement
Full-context top-1	0.012695	0.024902	$\geq$ 0.15; fail
Suffix-4 / shuffled top-1	0.006348 / 0.000488	0.015625 / 0.001465	diagnostic
Patch-reversed top-1	0.000488	0.024902	diagnostic
Image-unigram top-1	0.013184	0.013184	spatial below
Symbolic-bigram top-1	0.117188	0.117188	spatial below
Symbolic-trigram top-1	0.200195	0.200195	diagnostic
Full minus shuffled log probability	+0.315340	+0.650060	> 0.03; pass
Full minus patch-reversed log probability	+0.015412	0.000000	> 0.05; fail
Spatial minus global top-1	−0.012207		> 0.01; fail
Spatial minus global log probability	−0.190402		> 0.05; fail
Cross-font candidate identity	0.963379	0.964355	$\geq$ 0.95; pass

Table 18: V30 fixed natural-Chinese development audit. Patch alignment affects the spatial rank, but its target-probability effect misses the causal gate and the parameter-identical global control is stronger.

The 512-pair intervention keeps the final four context images bitwise equal while changing earlier history and the next image. Table 19 shows that neither route binds the correct candidate to the earlier context.

V30 suffix-4 pair measure	Spatial	Global	Spatial requirement
Suffix pixels / score-row error	1.000000 / 0	1.000000 / 0	exact; pass
Full arm accuracy	0.500488	0.505859	> 0.65; fail
Full both-correct rate	0.050781	0.081055	> 0.40; fail
Suffix-only arm accuracy	0.500000	0.500000	exact chance
Shuffled-prefix arm accuracy	0.492920	0.491943	control
Patch-reversed arm accuracy	0.496582	0.505859	control
Full minus shuffled accuracy	+0.007568	+0.013916	> 0.10; fail
Full minus shuffled mean margin	+0.010235	+0.034522	> 0.05; fail
Full minus patch-reversed accuracy	+0.003906	0.000000	> 0.05; fail
Candidate-column error / agreement	0 / 1.000000	0 / 1.000000	exact; pass
Spatial minus global full accuracy	−0.005371		> 0.05; fail

Table 19: V30 pixel-identical suffix intervention. Candidate pixels, exact suffix equality, and column equivariance are controlled. Local patch alignment does not produce reliable context-to-candidate assignment.

The field's sensitivity is real but insufficient. Reversal changes spatial scores by as much as 3.297, yet it improves target log probability by only 0.0154 nat and pair correctness by 0.391 percentage point. A single deterministic field with bilinear candidate energy is encouraged toward an average next retinal state. Natural next writing is multimodal; the fixed contrastive and pair losses do not make this one field represent the required conditional alternatives. This interpretation does not reject all spatial continuous distributions. It rejects this field, decoder, and objective on the fixed benchmark.

V30 passes 12/18 spatial common gates, 12/12 global integrity gates, 5/9 matched-arm gates, and 0/8 spatial language gates. Frozen images remain uninstantiated. This failure motivates the separately preregistered V31 multimodal continuous-field test below; it does not authorize a writer, page fold, depth axis, or historical-glyph task.

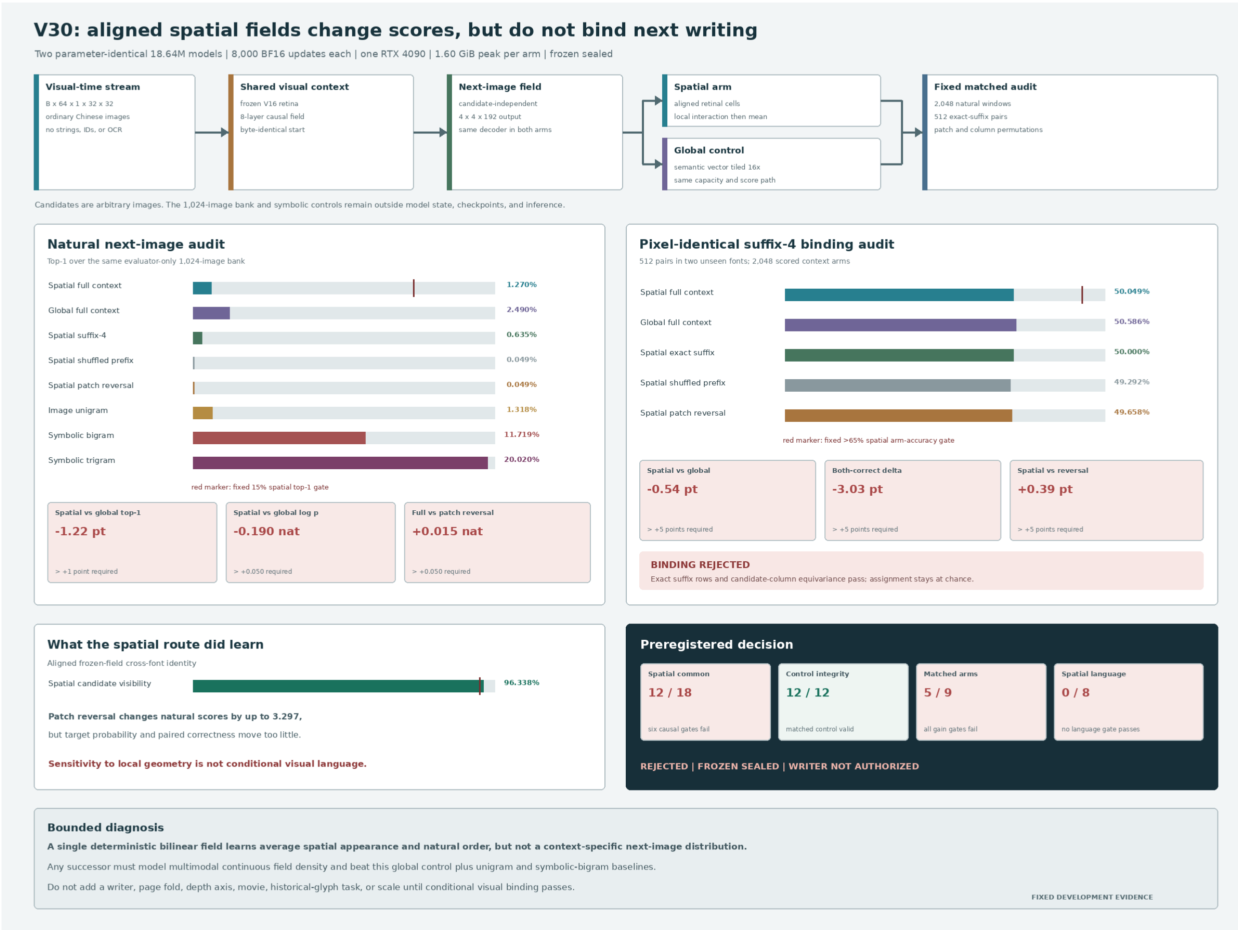

Figure 19: Measured V30 matched-arm result generated directly from the fixed audit. The spatial route is sensitive to local candidate alignment but is worse than the parameter-identical global route on natural top-1, target log probability, pair assignment, and pair both-correct. It passes 12/18 common gates and 0/8 language gates; the control passes all 12 integrity gates.

5.14 Conditional Visual Field Flow V31: context response without language

V31 executes the multimodal continuous-field test prescribed by V30. The candidate-independent causal reader remains image-only, but the deterministic field decoder is replaced by a context-conditioned velocity field. For normalized target retinal field Y, coherent Gaussian base field E, context state h, and $t \sim \mathcal{U}(0, 1)$, the interpolation and target velocity are

$$Z_t = (1 - t)E + tY, \qquad u^*(Z_t, t, h) = Y - E.$$

The learned field minimizes the conditional flow-matching objective

$$\mathcal{L}_{\text{flow}}(\theta) = \mathbb{E}_{h,Y,E,t}\left[\|v_\theta(Z_t, t, h) - (Y - E)\|_2^2\right].$$

One base vector is tiled coherently over the full retinal field; independent per-patch noise is forbidden. Thus one draw denotes one global visual alternative rather than a collage of independently sampled patches. The spatial arm predicts a 16×192 field. The matched global arm predicts one semantic vector tiled across the same 16 cells.

The audit reports both a path score and autonomous generation. For eight fixed probe times t_k, evaluator-only candidate ranking uses the negative velocity residual along each coherent base-to-candidate path,

$$s_\theta(Y \mid h, E) = -\frac{1}{K}\sum_{k=1}^{K} \|v_\theta((1-t_k)E + t_kY, t_k, h) - (Y-E)\|_2^2 .$$

Autonomous fields integrate

$$\frac{dZ_t}{dt} = v_\theta(Z_t, t, h), \qquad Z_0 = E,$$

with eight-step Heun integration. The 1,024-image bank is evaluator-only: it is absent from student inputs, parameters, checkpoints, and sampling. The displayed nearest glyph to a generated field is therefore a diagnostic proxy, not the model's generated pixel image.

Both arms contain 18,736,577 total and 17,107,201 trainable parameters. They start from byte-identical initialized states (`0682269c...2ad3d3`) and each completes exactly 10,000 finite BF16 updates. Spatial and global training take 4,217.57 and 4,470.01 seconds and peak at 0.99670 and 0.99661 GiB allocated CUDA memory. These are resource receipts for a rejected mechanism, not capability-normalized efficiency claims. The protocol, spatial checkpoint, global checkpoint, development audit, and comparison SHA-256 values begin `92b6f709`, `9808e996`, `485011a4`, `530c21d3`, and `2e97b732`, respectively.

Table 20 reports the fixed 2,048-window natural audit. The spatial route passes its order and local-layout log-probability interventions, and autonomous samples pass diversity and condition-sensitivity gates. Correct next-writing identity nevertheless falls below every useful baseline.

V31 natural development measure	Spatial	Global
Path full-context top-1	0.000977	0.018555
Path suffix-4 / shuffled top-1	0.001465 / 0.000000	0.013672 / 0.007812
Path spatial-permuted top-1	0.009277	0.018555
Autonomous full / shuffled top-1	0.001465 / 0.000000	0.005371 / 0.000000
Image-unigram top-1	0.016113	0.016113
Symbolic-bigram top-1	0.135254	0.135254
Symbolic-trigram top-1	0.209961	0.209961
Path full minus shuffled log probability	+0.125880	+0.131427
Path full minus spatial-permuted log probability	+0.052740	0.000000
Autonomous pairwise cosine distance	0.302390	0.933120
Same-noise full/shuffled displacement	5.736859	1.293999
Best-of-eight mean target rank	425.78	103.04
Cross-font candidate identity	0.963379	0.964355

Table 20: V31 fixed natural-Chinese development audit. Ordered and spatial interventions change the conditional field, but the spatial route identifies the correct next glyph in only two of 2,048 path rankings; autonomous nearest-bank matching identifies three of 2,048. Both are far below image-frequency and symbolic sequence baselines.

The 512-pair intervention changes earlier visual history and target while holding the final four context images pixel-identical. Table 21 shows that both path scoring and autonomous sampling remain at chance.

The spatial arm passes 14/19 common gates, the control passes 6/6 integrity gates, matched arms pass 4/8 gates, and spatial language and generation pass 0/10 gates. Frozen images remain uninstantiated; frozen evaluation and direct pixel writer training remain unauthorized. Figure 20 combines the architecture, measurements, and autonomous nearest-image diagnostic.

V31 suffix-4 pair measure	Spatial	Global
Suffix pixel equality / score-row max error	1.000000 / 0	1.000000 / 0
Path full-context arm accuracy	0.504883	0.514648
Path full both-correct rate	0.035156	0.078125
Path suffix-only arm accuracy	0.500000	0.500000
Path shuffled-prefix arm accuracy	0.492676	0.487793
Path full-minus-shuffle arm accuracy	0.012207	0.026855
Path full-minus-shuffle mean margin	0.005293	0.015111
Autonomous full-context arm accuracy	0.501953	0.505859
Autonomous full both-correct rate	0.008789	0.045898
Path spatial-permutation max score error	4.939047	0.000000

Table 21: V31 pixel-identical suffix intervention. Exact equality and candidate-column controls pass. Spatial flow responds to local permutation but does not bind earlier visual context to the correct target.

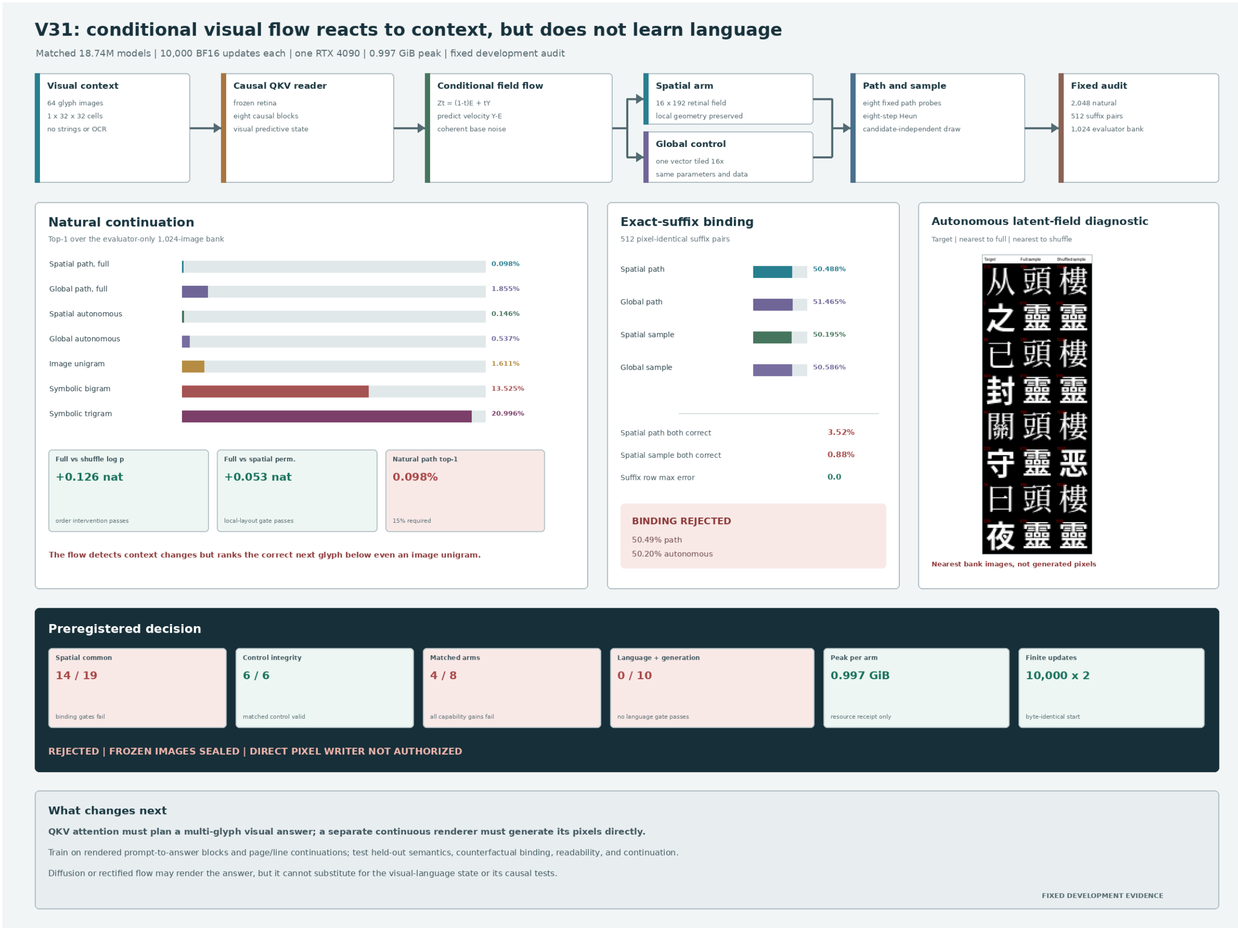

Figure 20: Measured V31 matched-arm result generated directly from the fixed audit. Conditional flow detects order, local layout, diversity, and context dependence while failing next-glyph identity and exact-suffix assignment. The embedded glyph columns are evaluator-nearest images to generated latent fields, not direct generated pixels.

V31 rejects one-glyph latent-field flow, not continuous visual generation in general. The next bounded proof changes both task and output variable. A causal QKV reader must form a visual predictive state from rendered prompts; a separate compact continuous head must directly render a multi-glyph answer block. A small reproducible curriculum should mix visual instruction-to-answer mappings with line/block continuation and hold out semantic compositions. Prompt counterfactuals, content swaps, paraphrases, generated-raster readability, and closed-loop continuation must all pass. Diffusion, rectified flow, or continuous autoregression may implement the renderer, but cannot substitute for causal language evidence. LocalLLM may author or audit fixed offline examples; it may not enter student batches, checkpoints, or inference. Open corpora provide the published benchmark, while local books enter only through a provenance-tracked preparation layer after this compact proof succeeds.

5.15 Continuous Glyph Representation Codec V34

V32–V33.1 showed that a language experiment is uninterpretable when the answer representation cannot preserve the target raster. V34 therefore isolates the visual interface before testing causality.

Figure 21: V31 autonomous latent-field diagnostic. Each row shows the target and the external evaluator's nearest glyph under full and shuffled context. Repeated wrong modes expose generation failure. The model did not render these nearest glyph pixels.

For a binary writing patch $p \in \{0,1\}^{1\times 32\times 32}$, the convolutional encoder and decoder are

$$z = \mathrm{LN}(E_\phi(p)) \in \mathbb{R}^{768}, \qquad \hat{p} = \sigma(D_\psi(z)).$$

The 7,423,361-parameter model uses four convolutional scales with channels $32, 64, 96, 192$, a continuous normalized latent, and no VQ, codebook, character label, OCR target, or external model. Its objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_{\text{weighted BCE}} + 0.5\,\mathcal{L}_{\text{Sobel}} + 0.5\,\mathcal{L}_{\text{ink Dice}}.$$

Half of training patches receive independently sampled latent noise with standard deviation in $[0, 0.05]$.

The public stream contains 7,017 public-domain Chinese book records with different Noto CJK files for train, development, and sealed rendering. The historical stream contains 84,626 local SVG records partitioned by modern character family, so all known forms of one modern character remain in one split. This local corpus is neither copied into the repository nor treated as a redistribution grant. V34 completes 6,000 BF16 updates in 321.31 seconds on one RTX 4090, processes 4,519,796 patches, and peaks at 2.38 GiB allocated memory.

V34 EMA measure	Development	Sealed
Rendered clean ink F1	0.997864	0.997793
Rendered clean edge F1	0.997270	0.997291
Rendered noisy-latent ink F1	0.997836	0.997749
Historical ink F1	0.998311	0.998068
Historical edge F1	0.997934	0.997747
Rendered OCR retention	1.002567	1.006088
Blank false-ink rate	0	0

Table 22: V34 development and once-opened sealed results. The sealed audit was authorized only after every development gate passed; every required sealed metric retained more than 97% of its development value. OCR is an evaluator, not a codec input or objective.

V34 is therefore `continuous-codec-qualified`. It establishes that a compact non-quantized interface can preserve held-font modern writing and held-family historical forms. It does not establish continuation, prompt following, factual knowledge, or autonomous language generation.

5.16 Causal Glyph Flow V35: complete raster loop, failed binding

Transferred foundation and student boundary. V35 tests whether a qualified visual interface and a transferred causal prior are sufficient for short Chinese continuation, copying, and instruction answering. The causal core is initialized from PIXAR's released 12-layer, width-768 transformer [18]. This prior is external work. The local source revision is MIT, while the separately hosted weight archive states no weight license; the derived weight artifact is consequently not authorized for redistribution. The V34 EMA codec is this project's frozen interface. At deployed inference, both foundations are packaged into one checkpoint and no PIXAR process, tokenizer, OCR engine, retrieval system, database, or language teacher is called. Thus *independent* denotes a self-contained visual runtime, not from-scratch provenance.

For visible patches $p_{\leq t}$, frozen codec E, D, aligned residual adapter A, and causal transformer T,

$$z_t = E(p_t), \qquad a_t = A(z_t), \qquad h_t = T(a_{\leq t}).$$

The deterministic writer gives a normalized anchor

$$m_{t+1} = \mathrm{LN}(M(h_t)).$$

The conditional writer instead learns a rectified flow. For target latent z, Gaussian base ϵ, and $\tau \sim \mathcal{U}(0,1)$,

$$x_\tau = (1-\tau)\epsilon + \tau z, \qquad v^\star = z - \epsilon,$$
$$\mathcal{L}_{\text{flow}} = \mathbb{E}\left[\|F_\theta(x_\tau, \tau, h_t) - v^\star\|_2^2\right].$$

The anchor supplies a one-pass route; the three-block width-512 flow head tests whether a conditional continuous distribution improves upon that route. A scalar visual-position head predicts when generation should stop.

The critical recurrence is visible rather than latent. Every predicted latent is decoded, thresholded into the emitted binary patch, and encoded again:

$$\widetilde{p}_{t+1} = \mathbf{1}[\sigma(D(\widehat{z}_{t+1})) \geq 0.5], \qquad z_{t+1}^{\text{feedback}} = E(\widetilde{p}_{t+1}).$$

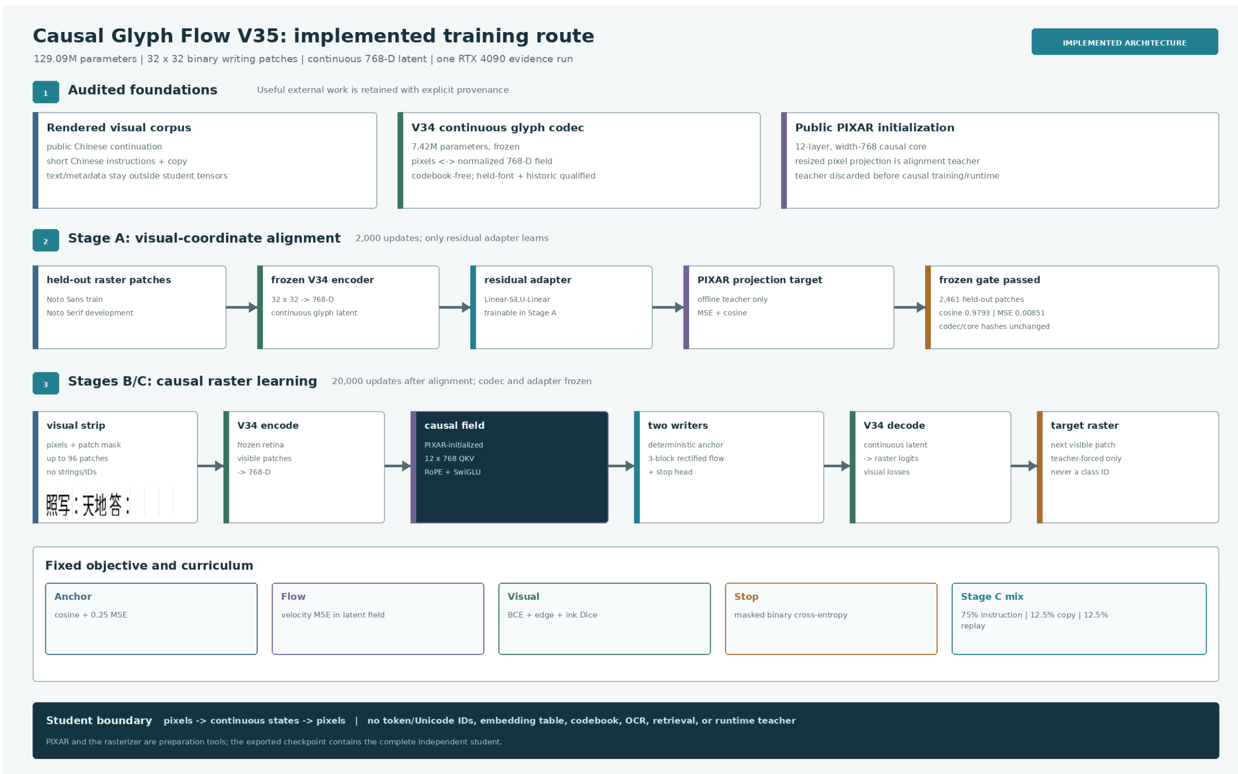

Figure 22: Implemented V35 training path. Stage A aligns the V34 latent interface to PIXAR's visual coordinate system while both the codec and causal core are frozen. The alignment teacher is then removed. Stages B and C train the causal field and continuous writers from raster-only student batches. PIXAR initialization and offline text rendering are explicitly outside the deployed student boundary.

No unrendered writer state crosses this recurrent boundary. The fixed causal objective is

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{flow}} &+ \big(1 - \cos(m, z) + 0.25\|m - z\|_2^2\big) \\ &+ 0.25\,\mathcal{L}_{\text{visual}} + 0.10\,\mathcal{L}_{\text{stop}},\end{aligned}$$

where $\mathcal{L}_{\text{visual}}$ combines pixel BCE, Sobel edge error, and ink Dice error. Flow and decoded-visual losses use at most 128 sampled valid positions per microbatch; anchor and stop losses use every supervised position.

Stage A runs for 2,000 updates. It passes with adapter MSE 0.008510 and cosine 0.979348 on 2,461 held-out patches while the causal core and codec hashes remain unchanged. Stage B runs 8,000 updates on public continuation. Stage C runs 12,000 updates with 75% Chinese Alpaca instructions, 12.5% deterministic visual-copy tasks, and 12.5% public replay. Physical batch 8, eight-way accumulation, BF16, gradient checkpointing, EMA, and maximum context 96 keep the 129,092,738-parameter model at 2.899 GiB peak allocated memory. All 22,000 updates are finite and complete in 9,972.46 seconds.

Fixed development audit. EMA is the preregistered primary state. The anchor writer is selected because its fixed selection-set OCR character accuracy is 0.4241%, versus 0 for the eight-step flow writer; neither is useful. The raw-weight diagnostic also selects anchor at 0.4464%, versus 0.2232% for flow. Raw public teacher-forced ink and edge F1 are 0.3398 and 0.4760, effectively the same as EMA's 0.3397 and 0.4754, so EMA selection does not conceal a capable raw route. The full audit evaluates every teacher-forced development record and autonomous correct, cyclically shuffled, blank, final-quarter, paraphrase, and copy-counterfactual conditions. Identical generation noise is reused across each intervention. OCR 5.3.4 with `chi_sim+chi_tra` is post-hoc only. In this audit, *readable* means only that OCR returned a nonempty string; it is not a human-readability claim.

V35 autonomous condition	Character accuracy	Nonempty OCR	Ink F1
Copy, correct	0.3125%	87.50%	0.09214
Copy, shuffled	0.1953%	87.50%	0.08645
Copy, blank	0	84.38%	0.07670
Public continuation, correct	0	71.88%	0.21265
Instruction, correct	0.1116%	75.00%	0.17346
Instruction, shuffled	0.5565%	75.00%	0.13709
Instruction, blank	0	100%	0.14888
Paraphrase, correct	0.3002%	87.10%	0.15300

Table 23: V35 autonomous EMA-anchor development results. All rows are nonblank, but nonempty OCR is not correct language. Correct instruction prompts perform worse than shuffled prompts, directly rejecting semantic binding.

The copy target-through-codec ceiling is 63.676%, already below its fixed 70% audit floor. Generated copy retains only 0.491% of that ceiling, versus the required 60%; correct prompts gain only 0.117 percentage point over shuffled and 0.313 point over blank prompts. Copy counterfactual target preference is 62.5%, below 75%, with zero mean character accuracy. The instruction ceiling passes at 78.858%, but autonomous correct-prompt accuracy is 0.1116%, against an 8% gate. It is 0.445 point worse than shuffled and only 0.112 point above blank, against required +2 and +3 point margins. The pixels change when prompts are shuffled or blanked, but the changes are not bound to the requested answer.

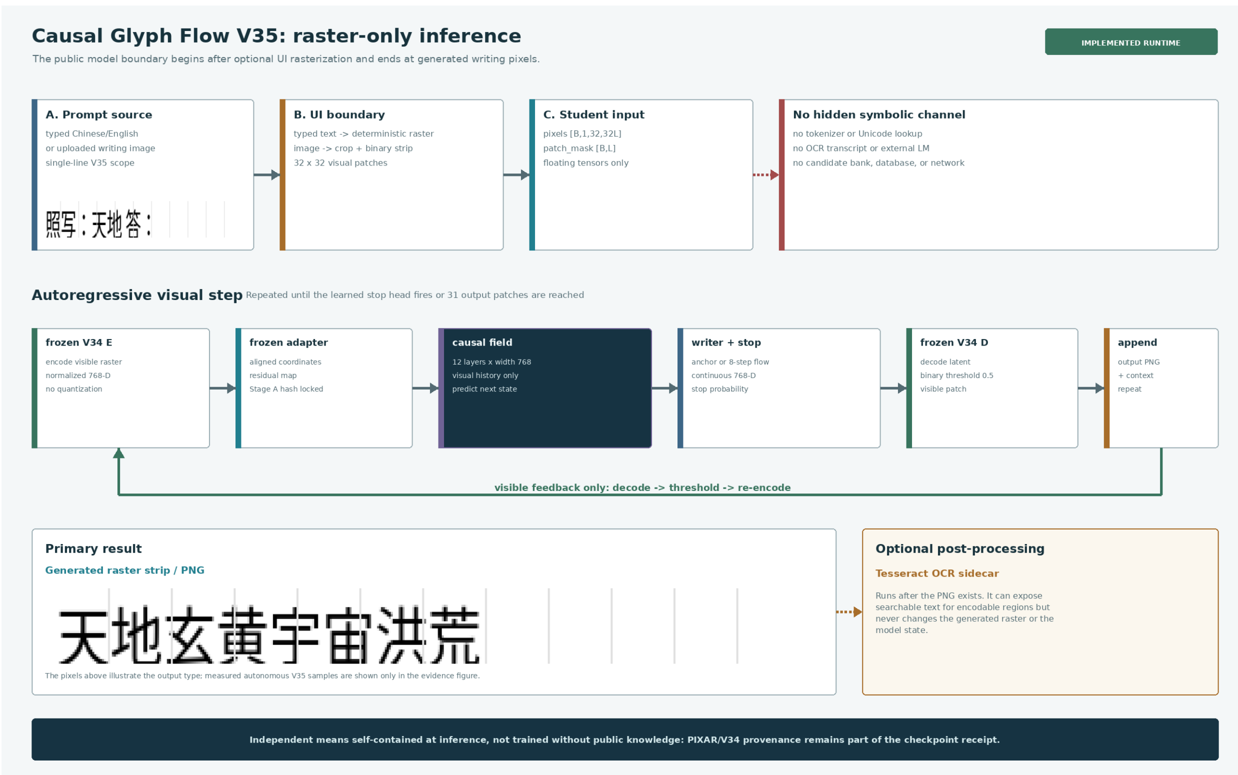

Figure 23: Implemented V35 inference path. A typed prompt is rasterized before the model or a writing image enters directly. The student sees pixels and a visual-position mask, generates raster patches through the selected continuous writer, and rereads only the thresholded visible result. OCR and coded text are optional post-hoc sidecars and cannot affect generation.

Figure 24: Measured V35 development decision generated from the hash-pinned 22,000-row metric log, report JSON, and immutable EMA galleries. Targets and generated rasters are shown directly; OCR strings never replace model pixels. The model closes the visible feedback loop and responds to prompt interventions, but fails copy, continuation, counterfactual, and instruction-binding gates.

V35 passes 5/12 visual-causal checks and 5/9 semantic-raster checks. Its exact decision is `not-qualified`. The sealed 139 public and 202 instruction records were not rendered, inspected, OCR-scored, or opened. A standalone EMA artifact was exported only under an explicit unqualified diagnostic override; typed-raster and native-image prompts both execute with finite closed-loop states, but their outputs remain incorrect. This is a functioning experimental system and a failed language mechanism. It establishes neither useful autonomous Chinese generation nor parity or compute superiority over a text language model.

The failure localizes the next test. A reconstruction-qualified local glyph latent and a transferred causal transformer can produce prompt-responsive marks without learning answer-level binding. The next model should retain the audited raster boundary while separating a prompt-conditioned multi-patch semantic plan from local raster realization, train that plan with answer-level contrastive and counterfactual objectives, and promote it only if correct prompt interventions move output toward the paired target. External visual or language foundations may be used when they improve this test, provided their provenance, license, training role, and runtime absence or presence remain explicit.

5.17 Visual Semantic Plan V36-P: weak relation, failed semantic gate

Candidate-free deployed boundary. V36-P tests the necessary semantic mechanism before attaching another writer. A Pixel-Linguist-v0-initialized visual reader receives a $3 \times 16 \times 1024$ prompt strip and 64-position visual mask. Five learned queries cross-attend to its visual memory and emit one global plus four ordered 768-dimensional plans. A separate scalar head estimates visual answer length. The deployable method is

$$(x, m) \longmapsto (z_0, z_1, z_2, z_3, z_4, \hat{\ell}), \qquad \|z_k\|_2 = 1.$$

It contains no strings, token or Unicode IDs, OCR, glyph lookup, answer teacher, candidate bank, or visual codebook, and it does not yet generate a raster. Candidate retrieval is evaluator-only.

During training, a frozen copy of Pixel-Linguist sees answer images and produces detached global and spatial targets. For normalized global predictions z_i and answer targets q_i, V36 combines symmetric in-batch InfoNCE,

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2B} \sum_i \left[\log \frac{e^{z_i^\top q_i/\tau}}{\sum_j e^{z_i^\top q_j/\tau}} + \log \frac{e^{q_i^\top z_i/\tau}}{\sum_j e^{q_i^\top z_j/\tau}} \right],$$

with cosine regression, a hard-negative margin, active spatial-chunk alignment, two-view consistency, and visual-length Huber loss. The answer teacher is removed after detached target construction. Pixel-Linguist is external prior work [13]; its released checkpoint states no license, so the derived V36 weights remain local-research-only.

The 93,473,281-parameter student completes 2,000 frozen-reader alignment updates and 4,000 final-two-block adaptation updates over 5,832 train pairs. The run consumes 512,000 rendered examples, takes 1,537.63 seconds, and peaks at 1,654,311,424 allocated bytes (1.541 GiB) on GPU 0 of one RTX 4090. All states, losses, gradients, targets, and reports are finite, and the checkpoint contains no target tensors or answer teacher.

The EMA route passes 13/23 conjunctive checks. Exact source and external hashes, strict upstream mapping, runtime boundary, resource limits, shuffled-prompt degradation, counterfactual assignment, paraphrase top-5, and answer-teacher cross-font stability pass. Absolute retrieval, improvement over the untrained head, cyclic separation, blank-prompt degradation, held-font transfer, paraphrase consistency, and length fail. The exact decision is `not-qualified`; the sealed split remains unopened and V36-R is not implemented.

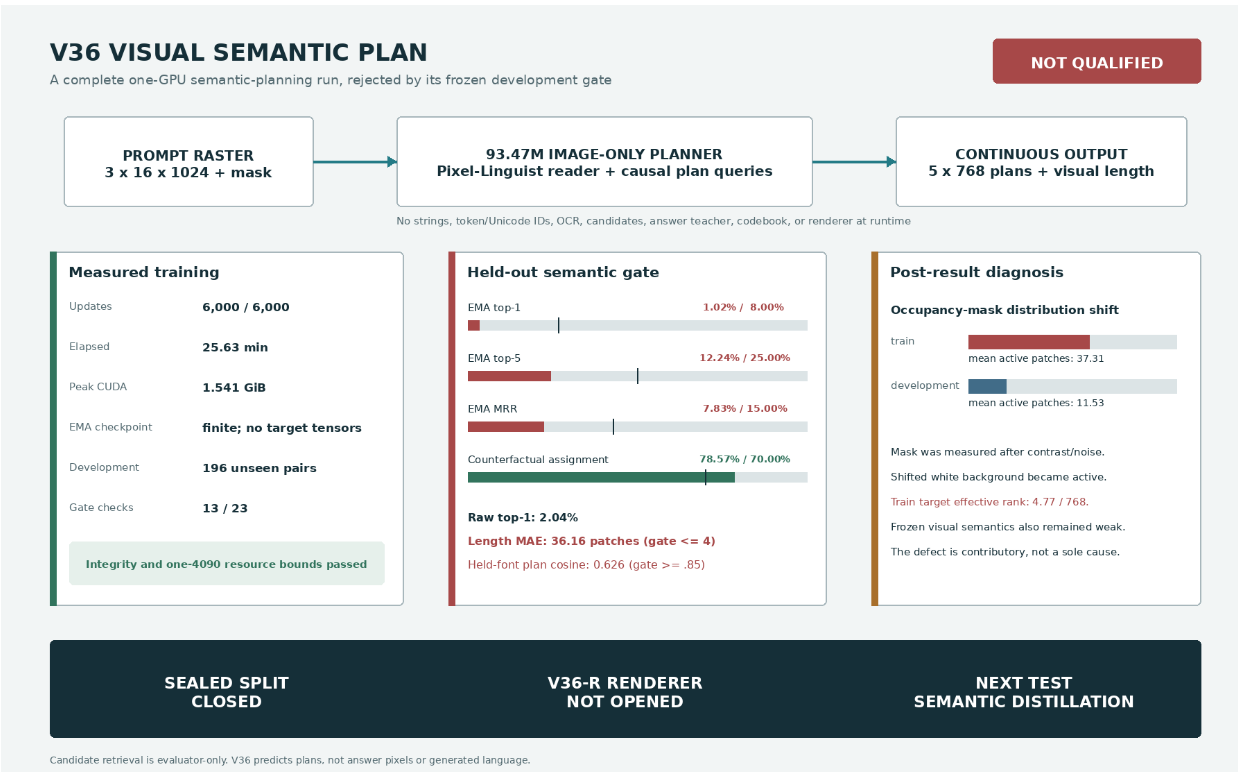

Figure 25: Measured V36-P development decision generated from hash-pinned training, EMA/raw evaluation, target-bank, and protocol artifacts. Retrieval is an evaluator-only probe of the autonomous continuous plans. V36 does not retrieve at runtime and does not emit answer pixels.

V36 development metric	EMA	Raw	Frozen EMA gate
Top-1 retrieval	1.02%	2.04%	$\geq 8\%$
Top-5 retrieval	12.24%	11.22%	$\geq 25\%$
Mean reciprocal rank	0.0783	0.0808	≥ 0.15
Correct–cyclic cosine margin	0.0427	0.0418	≥ 0.05
Correct beats cyclic	67.35%	65.82%	$\geq 70\%$
Counterfactual assignment	78.57%	76.53%	$\geq 70\%$
Held-font plan cosine	0.6265	0.6100	≥ 0.85
Paraphrase top-5	33.33%	33.33%	$\geq 20\%$
Original–paraphrase cosine	0.6578	0.6473	≥ 0.75
Visual-length MAE	36.16	36.53	≤ 4

Table 24: V36-P development result. Counterfactual assignment and paraphrase top-5 pass individually, but the preregistered decision is conjunctive. Raw weights do not reveal a hidden qualified route.

Post-result diagnosis. The train renderer measured patch occupancy after contrast and noise. When a nominally white background shifts below exactly white, blank patches cross the ink threshold. Mean stored train answer length is consequently 37.31 patches, versus 11.53 on clean development, although median character lengths are 19 and 18.5. The normalized train targets have centered effective rank only 4.77 in 768 dimensions, mean pairwise cosine 0.777, and median nearest-neighbor cosine 0.967. This defect directly compromises length and target pooling, but it is not the sole cause: the frozen visual teacher's direct development prompt-to-answer top-1 is only 4.59%.

A nonsealed directional probe reused the exact local BGE-M3 multilingual embedding model [14]. Its text prompt-to-answer ceiling on the same 196 pairs is 83.16% top-1, 93.37% top-5, and 0.8772 MRR, with answer effective rank 77.27. A train-fitted closed-form map from clean frozen visual features to that semantic space reaches only 2.04% top-1. This is not V36 evidence and does not put a text teacher in deployment. It motivates a successor that fixes clean occupancy, adapts the visual reader end to end, distills multilingual semantic geometry offline, and explicitly controls dimensional collapse with variance/covariance regularization [15]. A raster writer remains gated on that semantic result.

5.18 Visual Semantic Distillation V37: useful reading, failed complete gate

Offline targets and image-only runtime. V37 implements the bounded successor prescribed by the V36 diagnosis. For each selected training pair, the pinned BGE-M3 model [14] prepares raw prompt and answer vectors offline. With the arithmetic mean over all train prompt and answer vectors denoted by μ, the detached targets are

$$u_i = \mathrm{norm}(e(p_i) - \mu), \qquad v_i = \mathrm{norm}(e(a_i) - \mu).$$

The development bank reuses the immutable train μ, so no development-fitted centering enters the result. The target-space sanity ceiling on 196 development pairs is 82.65% top-1, 92.86% top-5, and 0.8723 MRR, with answer effective rank 76.53. Target construction stores neither strings nor token IDs in the tensor bank, and BGE-M3 is absent from student training and inference processes.

The deployed student begins with the exactly pinned Pixel-Linguist-v0 `vit.*` visual state [13]. It adapts a 12-layer, 768-wide, 12-head reader end to end over 64 clean visual patches. Masked mean pooling feeds a 1,024-dimensional normalized semantic head. A bounded residual head converts prompt semantics into a normalized answer plan, while a separate head estimates the clean answer length:

$$F_\theta(x,m) \longmapsto \Big(s_\theta(x,m), z_\theta(x,m), \hat{\ell}_\theta(x,m)\Big), \qquad s, z \in \mathbb{S}^{1023}, \quad \hat{\ell} \in [0,64].$$

The resulting 89,768,706-parameter artifact contains no text, token or Unicode IDs, OCR, vocabulary logits, candidate bank, lookup table, visual codebook, target tensor, BGE component, or network client. It emits continuous states, not a raster.

Patch occupancy is measured from clean black-on-white geometry before blur, contrast, or noise, so augmentation cannot reproduce V36's mask shift. Two independent prompt views and two answer views are trained with detached 512-candidate InfoNCE and cosine terms. If p, a, z denote prompt state, answer-image state, and prompt-conditioned answer plan, respectively, the principal loss is

$$\begin{aligned}\mathcal{L} = {} & 0.70\mathcal{L}_{\text{prompt}} + 0.70\mathcal{L}_{\text{answer}} + \mathcal{L}_{\text{plan}} + 0.50\mathcal{L}_{\text{margin}} \\ & + 0.20\mathcal{L}_{\text{relation}} + 0.20\mathcal{L}_{\text{view}} + 0.05\mathcal{L}_{\text{variance}} + 0.005\mathcal{L}_{\text{covariance}} \\ & + 0.02\mathcal{L}_{\text{length}} + 0.01\mathcal{L}_{\text{residual}}.\end{aligned}$$

The relation term matches within-batch student and teacher pair geometry; VICReg-style variance/covariance terms discourage collapse; and the hard negative is the highest-scoring detached candidate below a frozen teacher similarity ceiling. Candidate tensors supervise training and scoring only. They are not a deployed retrieval path.

Completed one-GPU result. The fixed run uses 5,822 train pairs, 500 frozen-reader projection updates, and 7,500 full-reader updates with effective batch 64. It consumes 512,000 rendered examples and completes all 8,000 BF16 updates in 3,695.48 seconds (61.59 minutes) on GPU 0 of one RTX 4090 D. Peak allocated CUDA memory is 2,951,057,920 bytes (2.748 GiB). Every reported loss, gradient, parameter, optimizer state, EMA state, target, checkpoint, prediction, and metric is finite. The deterministic answer-plan rank probe rises from 9.23 after warmup to 36.44 after full adaptation.

V37 development metric	EMA	Raw	Frozen EMA gate
Prompt paired cosine	0.2374	0.2347	≥ 0.70
Prompt top-1 / top-5	47.45 / 77.55%	47.45 / 77.04%	$\geq 25/60\%$
Answer top-1 / top-5	20.41 / 45.41%	20.41 / 46.94%	$\geq 30/60\%$
Answer MRR	0.3377	0.3340	≥ 0.40
Paired answer cosine	0.1634	0.1621	≥ 0.35
Correct–cyclic margin	0.1590	0.1572	≥ 0.20
Counterfactual assignment	93.88%	93.88%	$\geq 85\%$
Held-font plan cosine	0.4089	0.4088	≥ 0.85
Paraphrase top-5	73.33%	73.33%	$\geq 50\%$
Original–paraphrase cosine	0.4558	0.4523	≥ 0.75
Plan effective rank	55.37	56.83	≥ 32
Visual-length MAE	3.682	3.678	≤ 3

Table 25: V37 development result. Prompt retrieval, counterfactual assignment, paraphrase top-5, and plan rank pass individually, but absolute answer alignment and invariance remain below the conjunctive gate. Retrieval is an evaluator-only probe of candidate-independent continuous states.

The EMA route passes 20/33 conditions. It passes all integrity and resource checks, prompt top-1/top-5, the tenfold gain over an untrained head, counterfactual assignment, paraphrase top-5,

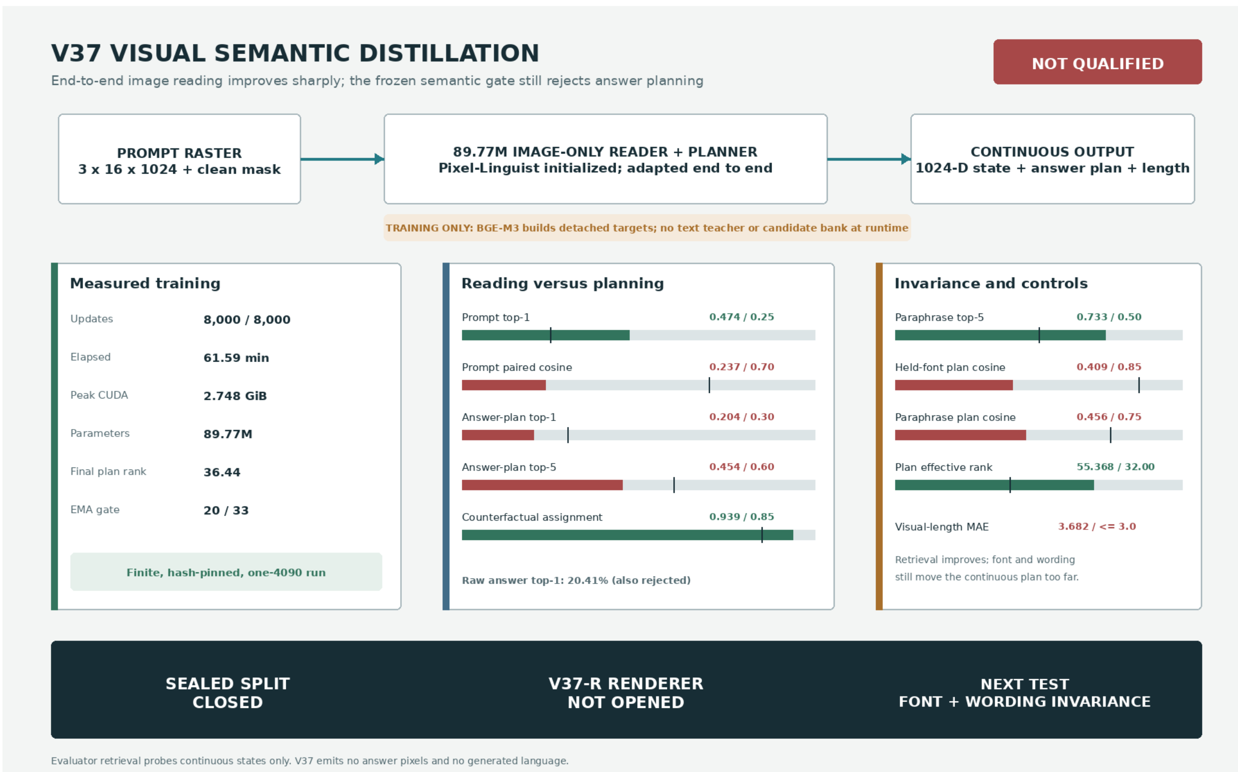

Figure 26: Measured V37 development decision generated from tracked, hash-pinned training and EMA/raw evaluation receipts. Pixel-Linguist is external initialization and BGE-M3 is an offline target builder; neither is claimed as project work or called by the deployed student. V37 predicts continuous states and does not emit answer pixels.

and both rank conditions. It fails prompt paired cosine, absolute answer top-1/top-5/MRR, the 0.20 absolute gain by 0.00102, answer cosine and cyclic margins, shuffled/blank cosine drops, held-font consistency, original–paraphrase consistency, and length. Raw weights pass 19/33 and do not repair the mechanism. The exact decision is `not-qualified`; the sealed split remains unopened and V37-R remains forbidden.

V37 is nevertheless a substantive bounded advance over V36: answer-plan top-1 rises from 1.02% to 20.41%, prompt top-1 reaches 47.45%, and the learned plan no longer fails the effective-rank controls. The remaining failure is not target availability but stable visual-to-semantic transfer. The next bounded experiment should train explicit canonical/held-font and original/paraphrase positive pairs, use semantically nearest detached relation negatives, and improve clean-length supervision. Stronger external visual foundations are acceptable when provenance, licensing, training role, and runtime absence or presence are exact. No favorable external component or subset metric may bypass the same complete semantic gate.

5.19 Visual Path Alignment V38: better reading, failed answer generalization

Paired visual paths and full answer map. V38 directly tests the three V37 diagnoses. Each training item contributes five image paths drawn from five distinct fonts: two prompt views, a validated paraphrase view when available, and two answer views. Clean patch occupancy is computed before augmentation. The reader and prompt head start from the exactly pinned V37 EMA state; a train-only orthogonal prompt-to-answer fit initializes a full linear map, and a zero-initialized nonlinear adapter can depart from that map without V37's angle cap. For pooled visual state (h), prompt feature (q), and normalized states (s,z), the deployed computation is

$$q = P_{\mathrm{p}}(h), \qquad s = \mathrm{norm}(q), \qquad z = \mathrm{norm}(Ws + A(q)), \qquad \hat{\ell} = 64\,\sigma(P_{\ell}(h)).$$

The 90,753,281-parameter checkpoint accepts only a $3 \times 16 \times 1024$ prompt raster and a 64-element clean mask. It returns $s, z, \hat{\ell}$; it contains no strings, token or Unicode identifiers, OCR, vocabulary logits, codebook, candidate bank, BGE, Qwen, teacher, or network client, and it does not generate a raster.

Training uses detached BGE-M3 prompt and answer states, explicit cross-font and paraphrase consistency, and 16 semantically nearest answer negatives under a fixed 0.85 teacher-cosine ceiling inside each 512-candidate set. Qwen-family models prepare and independently audit 1,024 training-only paraphrases before the tensor boundary. Pixel-Linguist, BGE-M3, and Qwen are exactly attributed external foundations or offline tools, not project contributions or deployed services. Strong external work is admissible when its provenance, license, training role, and runtime status are explicit.

Completed one-GPU result. The frozen schedule uses 500 head-realignment updates and 7,500 full-path updates with effective batch 64. It consumes 512,000 rendered examples and completes all 8,000 BF16 updates in 5,982.31 seconds (99.71 minutes) on GPU 0 of one RTX 4090 D. Peak allocated CUDA memory is 3,188,168,704 bytes (2.969 GiB). All training and evaluation routes are finite. The complete checkpoint and evidence hashes are retained in the tracked V38 result receipt.

The EMA route passes 25/39 conditions. Relative to V37, prompt top-1/top-5 rises from 47.45/77.55% to 60.71/86.73%, prompt paired cosine rises from 0.2374 to 0.3789, and canonical-to-held prompt/answer consistency rises from 0.4130/0.4089 to 0.7925/0.7323. Mean prompt-answer cosine falls from 0.9972 to 0.5769 while transition-direction cosine reaches 0.3401 and answer-state effective rank reaches 49.02. Thus the reader and transition geometry improve in the intended directions.

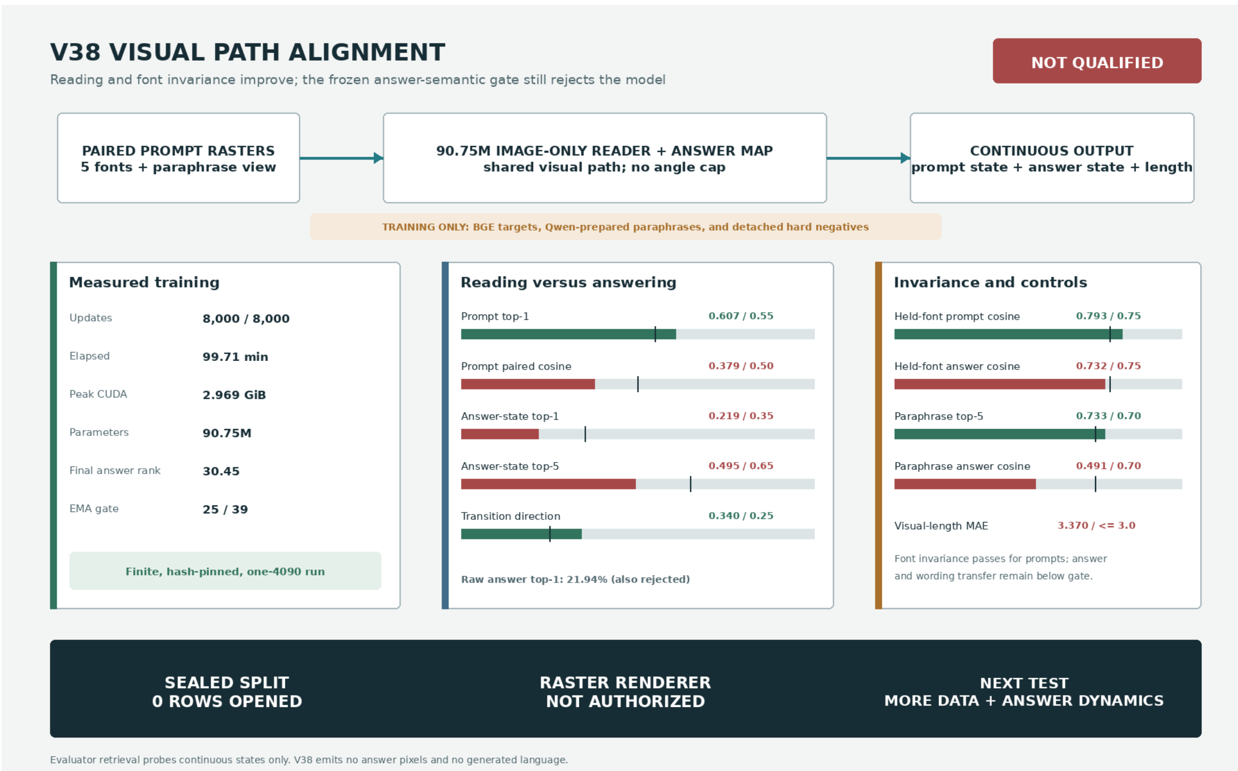

Figure 27: Measured V38 result generated from hash-pinned training and EMA/raw development receipts. Paired paths improve reading and prompt font invariance, while absolute and transferred answer semantics remain below the complete gate. Zero sealed rows were rendered and no writer is authorized.

V38 development metric	EMA	Raw	Frozen EMA gate
Prompt top-1 / top-5	60.71 / 86.73%	60.71 / 85.71%	$\geq$ 55/80%
Prompt paired cosine	0.3789	0.3791	$\geq$ 0.50
Direct answer-read top-1	44.39%	43.37%	$\geq$ 45%
Conditional answer top-1 / top-5	21.94 / 49.49%	21.94 / 49.49%	$\geq$ 35/65%
Conditional answer MRR / cosine	0.3460 / 0.2441	0.3436 / 0.2428	$\geq$ 0.45/0.30
Answer cyclic margin / win	0.2050 / 83.67%	0.2037 / 84.18%	$\geq$ 0.15/85%
Counterfactual assignment	89.80%	89.80%	$\geq$ 90%
Held-font prompt / answer cosine	0.7925 / 0.7323	0.7944 / 0.7321	$\geq$ 0.75/0.75
Paraphrase top-5	73.33%	73.33%	$\geq$ 70%
Paraphrase prompt / answer cosine	0.5408 / 0.4908	0.5403 / 0.4871	$\geq$ 0.70/0.70
Transition direction / prompt-answer cosine	0.3401 / 0.5769	0.3392 / 0.5693	$\geq$ 0.25/ $\leq$ 0.95
Answer-state effective rank	49.02	49.28	$\geq$ 32 and 40% of target
Visual-length MAE	3.370	3.370	$\leq$ 3

Table 26: Frozen V38 development result on 196 pairs. Retrieval is an evaluator-only probe of candidate-independent continuous states after all student inference. It is not generated text or a deployed candidate system.

The central answer relation does not improve enough: top-1 moves only from 20.41% to 21.94%, and held-font and paraphrase answer consistency fail. The last training batch reports 96.35% answer-plan top-1 while development remains 21.94%; this is diagnostic evidence of poor generalization, not a capability claim. Raw weights pass 24/39 only because the frozen primary route is EMA and otherwise match the rejected answer metrics. The exact decision is `not-qualified`; zero sealed rows were rendered and the raster writer remains closed.

The bounded diagnosis is curriculum and answer dynamics rather than reader invariance alone. The 5,822-pair stream supplies roughly 88 effective passes, and one vector must represent the complete conditional answer. The next test should enlarge deduplicated relation diversity, reserve held-operation and held-composition probes, and predict a small ordered set of continuous answer states or recurrent latent dynamics. It must retain image-only deployment and the same complete gate before any answer-pixel writer is trained.

5.20 V39.1 trajectory correction and V41 image-conditioned glyph motor

A corrected trajectory still fails semantic order. V39 first extends V38 from one answer state to one global answer state and up to 16 ordered answer-span states. A 512-record pilot detects bounded visual conditioning, but independent stop-threshold decoding selects all 16 positions and a cyclic last-to-first negative corrupts the order objective. V39.1 corrects these mechanics: it normalizes a categorical first-stop process, uses categorical stop and count-moment losses, removes the wraparound negative, initializes a realistic length prior, strengthens order supervision, and uses a warm-start all-parameter EMA. The matched 150-update rerun completes in 184.23 seconds with 2,984,609,280 peak allocated CUDA bytes on one RTX 4090 D.

The raw route reaches only 0.07352 answer MRR and gives the same decision. Correct, shuffled, and blank prompts remain distinguishable, but the generated states do not select the right held-out content. V39.1 is therefore `not-qualified`; it is not scaled to the 44,637-record bank, no sealed data are opened, and no answer writer is authorized by this result.

Separate content choice from the visible motor. A language field and a writing motor solve different problems. Let x_t be a visible canonical glyph cell, E and D the V34 image encoder and

Held-out development metric	V39 pilot EMA	Corrected V39.1 EMA
Answer top-1	0.04339	0.03562
Answer MRR	0.08302	0.07406
Answer cosine	0.18048	0.14486
Segment top-1	0.00523	0.00441
Segment MRR	0.01264	0.01234
Transition-direction cosine	0.00016	0.00648
Expected span count / target	7.244 / 6.313	6.303 / 6.313
Count-mode accuracy	0.00389	0.15220
Visual-length MAE	9.025	4.915

Table 27: Matched V39 correction result. The count mechanics improve, but the semantic trajectory does not bind the visual prompt to the right ordered answer. The corrected final-answer route is 0.01049 MRR below its inherited V38-compatible baseline.

decoder, and T_θ a future causal visual field:

$$c_t = E(x_t), \qquad h_t = T_\theta(c_{\leq t}).$$

Because the next glyph is multimodal, the planned language head is a continuous conditional flow rather than deterministic mean regression. For Gaussian base ϵ and $\tau \sim \mathcal{U}(0,1)$,

$$u_\tau = (1-\tau)\epsilon + \tau c_{t+1}, \qquad \mathcal{L}_{\text{flow}} = \mathbb{E}\left\| F_\psi(u_\tau, \tau, h_t) - (c_{t+1} - \epsilon)\right\|_2^2 .$$

This follows the codebook-free continuous autoregressive motivation of MAR [34]. A sampled state is decoded to a canonical raster; a separate image-conditioned form projector may then control font or handwriting before the visible raster is re-encoded. No nearest-character lookup appears in the deployed equation.

V41 audits only that last visual-motor bridge. The external MX-Font [35] generator has 22.76 million parameters and accepts a source glyph image plus style-reference glyph images. Its source revision and checkpoint are pinned; the checkpoint SHA-256 begins `dcbcb6438d9b1e32`. The V34 EMA checkpoint has 7,423,361 parameters and SHA-256 prefix `a138c9cb3b05`. Ten supported Chinese glyphs are rendered in a source font, passed through exact, 32-pixel, clean-codec, and latent-perturbed routes, then projected using four held-style image references. Character strings are host-side evaluation labels and never model inputs. The model receives no token or Unicode IDs, OCR, retrieval, or candidate bank.

Source raster supplied to motor	Input ink F1	Motor ink F1	Gain
Exact 128-pixel source	0.43944	0.57863	+0.13919
Coarse 32-pixel source	0.43968	0.56481	+0.12513
V34 clean projection	0.43182	0.54794	+0.11612
V34 projection, $\sigma = 0.03$	0.43124	0.54441	+0.11317
V34 projection, $\sigma = 0.05$	0.43170	0.54610	+0.11440
V34 projection, $\sigma = 0.10$	0.42941	0.54324	+0.11383

Table 28: Hash-pinned V41 motor audit. Ink F1 is measured against a held target style. The motor improves every imperfect source condition; all ten outputs are finite and nonblank.

The complete V41 audit takes 1.650 seconds with 771,669,504 peak allocated CUDA bytes on one RTX 4090 D. It passes the bounded motor gate: every route is finite and nonblank, every

Figure 28: Measured V41 image-conditioned motor result generated from the tracked receipt and contact sheet. The $\sigma = 0.05$ route retains 99.66% of clean projected-motor F1 and gains 0.1144 F1 over its projected input. This is a motor qualification, not a language result: the intended source glyph raster is supplied to the motor in every row.

route improves target-style F1, and noise levels $\sigma = 0.03$ and 0.05 retain more than 99% of clean projected-motor F1. This closes a practical raster-quality component but does not rescue V39.1 or establish language understanding. The next experiment must predict the canonical content state from raster history without receiving the evaluator's target glyph, then beat visual unigram, bigram, shuffled-history, and blank-history controls under autonomous pixel feedback.

5.21 Canonical Glyph Language V42–V45

A canonical image field isolates language before historical form. V42 deliberately removes style variation and answer-level instruction tuning. Each visible cell is a single canonical-font raster $x_t \in [0,1]^{1\times32\times32}$. A fixed orthonormal two-dimensional DCT maps thresholded signed pixels into a normalized continuous image field,

$$\phi(x_t) = \mathrm{norm}\big[\mathrm{DCT}_2\big(2\mathbf{1}[x_t \geq 0.5] - 1\big)\big] \in \mathbb{R}^{1024}.$$

A learned linear projection and eight causal visual Transformer blocks consume 64 such fields. The final state predicts a normalized anchor $a_t \in \mathbb{R}^{1024}$. All 64 shifted positions in each training stream participate in dynamic image contrast, anchor cosine, and inverse-field pixel losses. Candidate images exist only inside the train/evaluation loss; no finite bank is present in `language`, `sample_next`, or recurrent generation. The production model has 24,346,497 parameters, completes 10,000 BF16 updates in 1,374.70 seconds, and peaks at 0.63194 GiB allocated CUDA memory on one RTX 4090 D.

The fixed 2,048-window development audit uses a 1,024-image evaluator bank only to identify predictions after the model emits a field. It also computes image unigram and symbolic bigram baselines, suffix-four and last-cell ablations, suffix-preserving prefix shuffles, blank history, 512 exact-suffix pairs, and 256 direct generated rasters. Table 29 reports the predeclared decision. V42 passes every ordered-language, runtime-boundary, and resource condition. Its full top-1 is 19.9707%, compared with 12.2559% bigram and 18.3594% shuffled history; ordered target log probability gains 0.17391 nat over shuffled history. These controls provide bounded evidence that the model uses earlier writing images to predict a natural Chinese next image. Counterfactual assignment and pixel F1 remain below gate, so the evidence does not imply general understanding or readable autonomous writing.

Development metric	V42	V43	Fixed requirement
Full ordered top-1	0.19971	0.17822	report
Image-unigram top-1	0.01416	0.02295	control
Symbolic-bigram top-1	0.12256	0.13721	full gain > 0.01
Shuffled-history top-1	0.18359	0.14063	full gain > 0.015
Ordered–shuffled log probability	+0.17391	+0.27946	> 0.05 nat
Exact-suffix arm accuracy	0.53027	0.54492	> 0.60; both fail
Generated identity top-1	0.08203	0.04297	> own unigram; both pass
Generated pixel F1	0.37308	0.44507	> 0.55; both fail
Generated blank rate	0	0	< 0.02
Peak allocated CUDA memory	0.63194	1.19858	< 18 GiB

Table 29: Preregistered canonical glyph-language results. V42 and V43 use independently seeded fixed development samples, so cross-version absolute differences are descriptive; every within-version control and gate is matched.

V43 tests binding and rendering as separate mechanisms. Stage A initializes the reader from the pinned V42 SHA-256 `a5ce2ff2...a9abe870`. In addition to ordinary streams, each update contains eight train-only context pairs with pixel-identical final four glyphs but different earlier histories and next-glyph images. Candidate columns are deterministically permuted. For pair b, context arm i, and raster-candidate field c_{bj},

$$S_{bij} = s\, a_{bi}^{\mathsf{T}} c_{bj}, \qquad \mathcal{L}_A = \mathcal{L}_{dynamic} + 0.25\mathcal{L}_{anchor} + 0.20\mathcal{L}_{pixel} + 2.0\mathcal{L}_{pair}.$$

The 3,000-update stage completes in 207.89 seconds and peaks at 0.61394 GiB.

Stage B freezes that reader and trains a 5,693,697-parameter convolutional spatial writer. Final hidden state h, anchor a, and its fixed inverse-DCT ink plan $p(a)$ condition a rectified flow. For signed target pixels x, Gaussian noise ϵ, and $\tau \sim \mathcal{U}(0,1)$,

$$y_\tau = (1-\tau)x + \tau\epsilon, \qquad u^* = \epsilon - x, \qquad \mathcal{L}_B = \mathcal{L}_{stroke\text{-}weighted\ flow} + 0.10\mathcal{L}_{endpoint}.$$

At inference, four independent fields use 16 Heun steps and guidance 1.25. The reader re-encodes those generated pixels and its own anchor selects one; no glyph bank participates. The chosen pixels are thresholded and become the only feedback for another step. Stage B completes 5,000 updates in 272.09 seconds. The combined model has 30,040,194 parameters and writer-checkpoint SHA-256 `f2f142f3...e6fe0539`.

V43 preserves all four ordered-language wins, emits no blank fields, and raises generated pixel F1 to 0.44507. Exact-suffix arm accuracy reaches only 0.54492, below 0.60, and pixel F1 remains below 0.55. Thus V43 passes 8/10 conditions and is `rejected-or-partial`; the frozen partition remains sealed.

Post-result diagnostics localize the bottleneck. The corpus contains about 215,340 eligible train suffix-four pairs. V43 uses only 5,000 and revisits them about 4.8 times. On matched 512-pair samples, V43 scores 0.99219 on sampled training pairs but only 0.58301 on unseen train pairs and 0.54688 on development. V42 scores 0.55078, 0.57422, and 0.53613 on the same sets. The large seen-only gain diagnoses pair-pool memorization rather than a generalized earlier-history rule.

The writer diagnostic points in the opposite direction. With the autonomous predicted plan, ordinary anchor selection reaches 0.46110 pixel F1; even target-side best-of-four pixel selection reaches only 0.52006. If the evaluator supplies the exact target ink plan solely to test motor capacity, the unchanged writer reaches 0.88240 F1 with ordinary anchor selection and 0.92032 with target-side selection. The direct V43 anchor inverse is 0.46315. Writer capacity and sample selection are therefore secondary to the predicted visual language field. This diagnosis fixes the V44 intervention: freeze V42, train a small residual long-history path over a nonrepeating 24,000-pair sample, and align pairwise anchor differences with pairwise target-image differences before reopening the V43 writer.

V44 removes repeated-pool memorization but not visual binding. V44 keeps every V42 parameter byte-exact and adds a two-block, width-192, six-head cross-attention residual. Its query combines the frozen full-history state, a second frozen suffix-four state, and their difference; its memory contains only the preceding 60 image fields and hidden states. If a_0 is the frozen V42 anchor and r is the residual, V44 permits only a tangent update,

$$r_\perp = r - (r^{\mathsf{T}} a_0)a_0, \qquad a = \mathrm{norm}(a_0 + 0.5r_\perp).$$

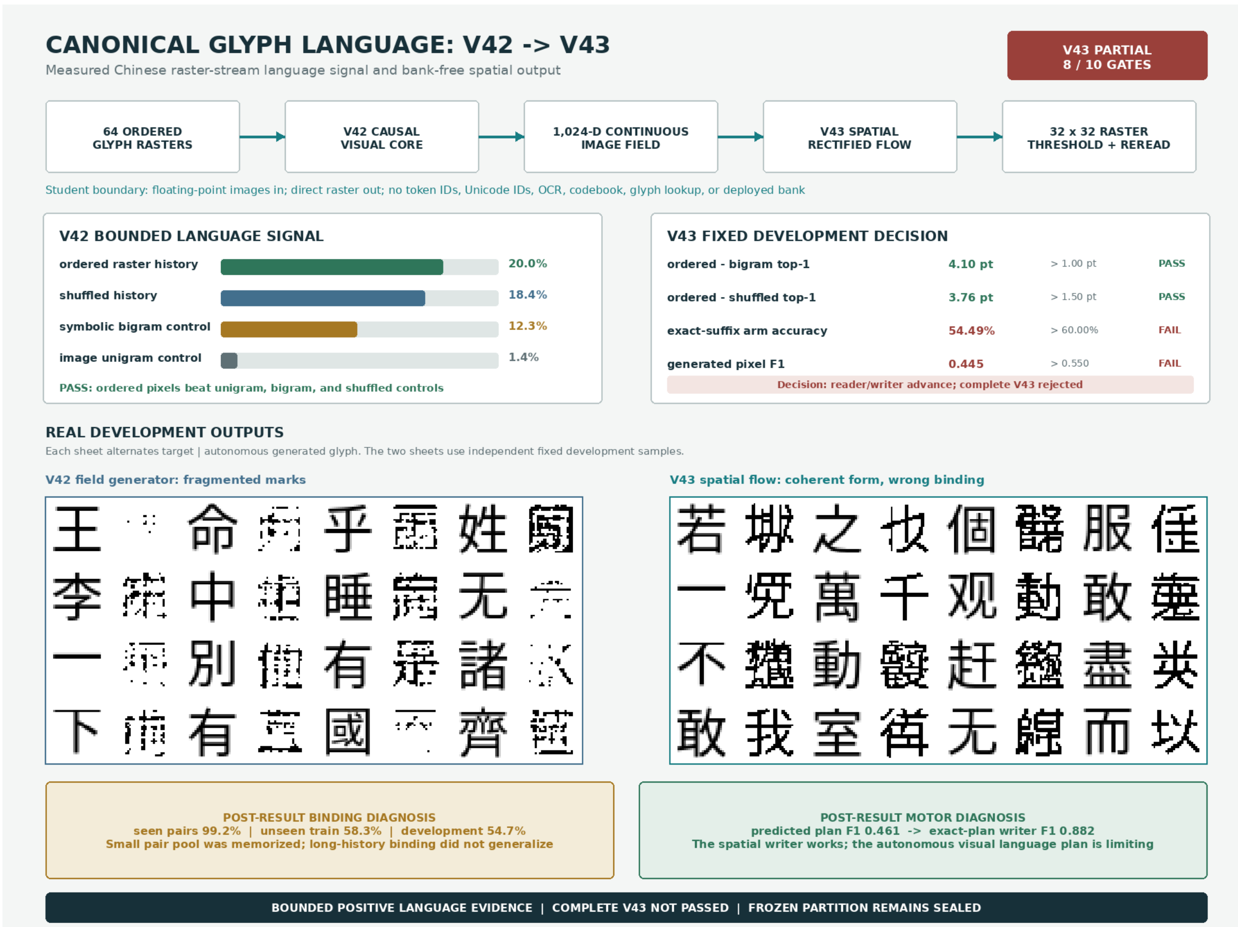

Figure 29: Measured V42/V43 result rendered from hash-pinned receipts and real target/generated raster sheets. V42 establishes a bounded ordered-image language signal. V43 makes autonomous marks more glyph-like but misses its exact-suffix and generated-pixel gates. The bottom panels are explicitly post-result diagnostics and do not change that decision.

The output projection starts at zero, contexts of four images or fewer bypass the adapter, and no candidate bank enters the model. The fixed loss combines pair cross-entropy, signed target-displacement alignment, displacement-norm and order margins, ordinary natural-image contrast, target cosine, and frozen-base distillation.

The 1,735,936-parameter adapter consumes each of 24,000 unique training pairs once; the next 1,024 unique suffixes are untouched training-partition holdouts. Its 3,000 BF16 updates take 210.87 seconds and peak at 0.19848 GiB allocated CUDA memory. The complete 26,082,433-parameter runtime remains image-only, the V42 state is exact after training, and the frozen evaluation partition is not instantiated.

Matched development metric	V42	V44	Fixed requirement
Natural full top-1	0.19434	0.17432	$\geq$ V42$-$0.015; fail
Natural full log probability	-5.23837	-5.77730	$\geq$ V42$-$0.10; fail
Natural shuffled top-1	0.18604	0.15430	full gain $>$ 0.015; pass
Symbolic-bigram top-1	0.13672	0.13672	full gain $>$ 0.01; pass
Development pair arm accuracy	0.52441	0.53418	$>$ 0.60; fail
V44 shuffled-pair arm accuracy	–	0.51855	full gain $>$ 0.04; fail
Consumed train-pair accuracy	–	0.57471	diagnostic
Unseen train-pair accuracy	–	0.57422	$>$ 0.60; fail
Consumed–unseen gap	–	0.00049	$<$ 0.10; pass

Table 30: V44 fixed matched-base and counterfactual audit. V44 retains the four absolute ordered-language controls and removes the V43 seen-pair gap, but fails both matched-base retention gates and all required binding improvements. It passes 8/14 conjunctive conditions.

The failure is representational, not a residual-scale choice. A fixed post-result interpolation evaluates residual strengths from zero to two. Development arm accuracy peaks at only 0.53809 for strength 0.75; no tested strength reaches 0.60. The learned update-difference cosine with the true target-image difference is -0.02289 on development and -0.03395 on unseen training pairs. On natural windows, anchor cosine with the corpus-mean image field rises from 0.77712 at the frozen V42 anchor to 0.88183 at V44. Raw target cosine also rises from 0.60298 to 0.65413, yet exact top-1 falls from 0.19434 to 0.17432 and target log probability falls by 0.53893 nat. The normalized raw DCT field lets shared background and common-stroke modes improve cosine while erasing discriminative displacement. Another larger residual on that target is therefore unjustified. The next bounded test must center and variance-balance a reversible continuous raster field and qualify its held-out image geometry before another causal reader or the V43 writer is opened.

V45 qualifies a reversible noise-limited retinal field. Motivated by noise-limited efficient coding [16] and symmetric whitening that preserves the original coordinate orientation [17], V45 fits only fixed statistics from the 8,000 most frequent training-partition canonical rasters, covering 99.9249% of retained Han instances. Let d be the unnormalized orthonormal DCT of a signed 32×32 raster, let w_i be normalized corpus counts, and define

$$\mu = \sum_i w_i d_i, \qquad C = \sum_i w_i (d_i - \mu)(d_i - \mu)^\mathsf{T} = U \operatorname{diag}(\lambda) U^\mathsf{T}, \qquad \bar{\lambda} = \frac{\operatorname{tr}(C)}{1024}.$$

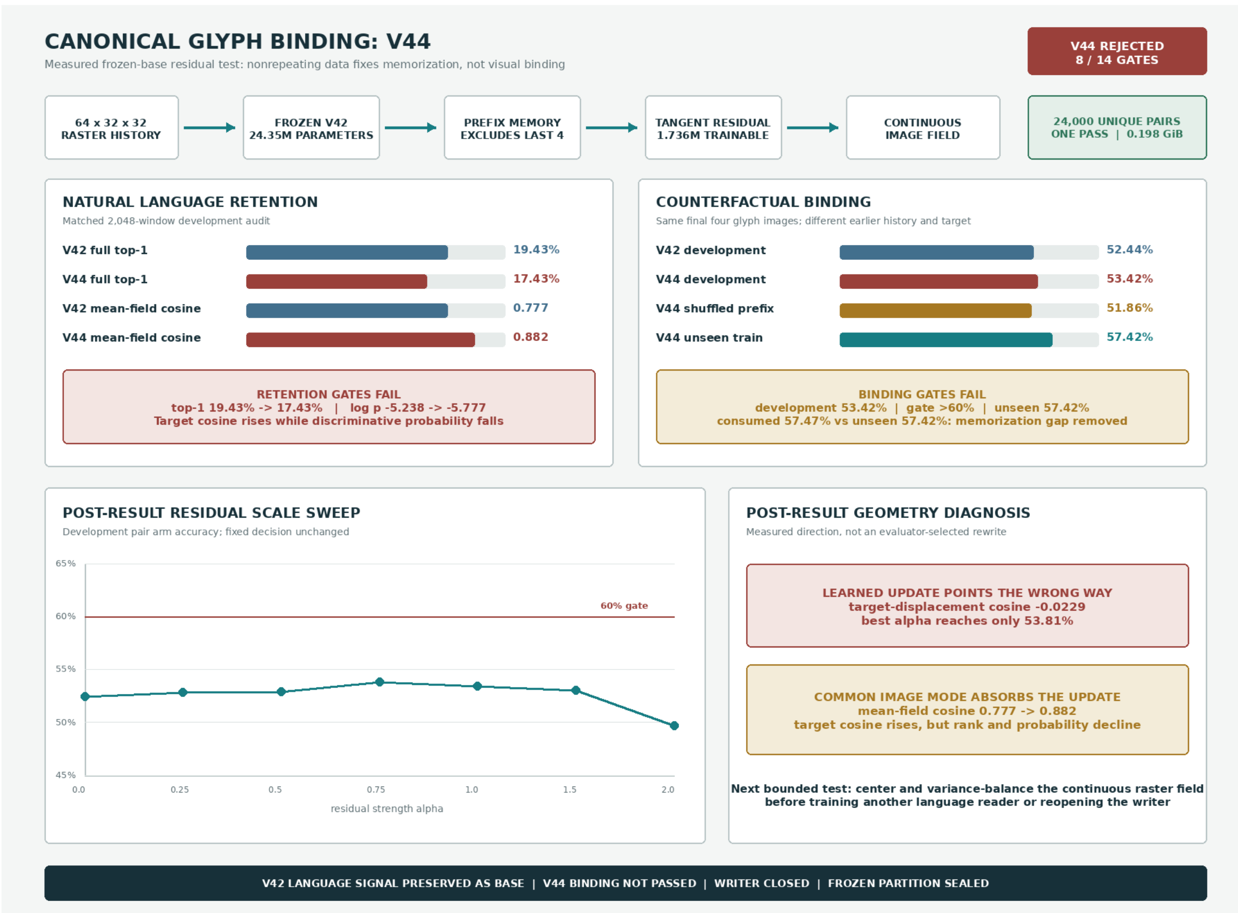

Figure 30: Measured V44 result generated from hash-pinned training, development, and diagnostic receipts. A one-pass frozen-base residual removes repeated-pool memorization but neither passes held-out binding nor preserves matched V42 natural calibration. The lower panels are explicitly post-result diagnostics and do not alter the preregistered 8/14 rejection.

The preregistered transform uses a noise ridge equal to half the mean variance and a fractional matrix power,

$$A = U\,\mathrm{diag}\big[(\max(\lambda, 0) + 0.50\bar{\lambda})^{-0.10}\big]\,U^{\mathsf{T}}, \quad z = A(d - \mu), \quad u = \frac{z}{\|z\|_2}, \quad q = \log\|z\|_2.$$

The continuous state is (u, q), not an index. Its exact inverse is $d = \mu + A^{-1}(e^q u)$. The deployed field has zero trainable parameters and no token, Unicode or character ID, OCR engine, visual codebook, quantizer, glyph lookup, external language model, or candidate bank.

The fixed reconstruction audit spans all 8,000 fit rasters, two held fonts over 1,024 identities, and four cardinal one-pixel translations over the same identities. Across these 14,144 rasters, maximum FP64 DCT reconstruction error is 4.2633×10^{-14}; binary pixel accuracy and ink F1 are both 1.0; every field is finite; and blank rate is zero. Table 31 reports the geometry and continuity gates.

Fixed audit metric	Raw DCT	V45	Requirement / result
Weighted common resultant	0.71900	0.01524	< 0.05 and $< 0.1\times$; pass
Fit covariance effective rank	145.08	185.38	$\geq 1.20\times$; pass
Bold held-font top-1	0.96582	0.96582	loss ≤ 0.01; pass
Serif held-font top-1	0.82129	0.83984	loss ≤ 0.01; pass
Mean cardinal-shift top-1	0.75781	0.76660	no loss > 0.02; pass
Held-pair candidate cosine	0.56319	0.06402	reduction ≥ 0.25; pass
Held-pair displacement norm p05	0.36976	0.77724	$\geq 1.50\times$; pass
Held-pair displacement effective rank	122.80	144.95	$\geq 1.10\times$; pass
Held-pair displacement stable rank	60.82	67.79	$\geq 1.08\times$; pass

Table 31: V45 preregistered representation audit. Pair metrics use the exact 1,024-pair V44 holdout receipt. Font and shift rows compare retrieval against the same 1,024 canonical identities. Reconstruction, image-only-boundary, and resource gates also pass, yielding 13/13 conjunctive conditions.

V45 completes in 66.51 seconds with 0.30211 GiB peak allocated CUDA memory on one RTX 4090 D, reconstructs its on-disk field state with the recorded SHA-256, and passes all 13 gates. The frozen evaluation images and V43 writer remain unopened. A preregistered, non-gating coordinate retrofit applies V45 after the already-trained V42 reader; natural top-1 falls from 0.19434 to 0.15039 and target log probability from -5.23837 to -7.13590. V42 optimized anchors in raw normalized-DCT geometry, so this decline rejects a post-hoc calibration bridge rather than the new geometry. The authorized next experiment must be separately preregistered and train a causal raster reader from initialization with V45 targets and feedback before any writer composition is reconsidered.

6 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

Item	Count
Characters	9,055
Glyph records	84,642
Oracle	32,216
Bronze	23,500
Liushutong	21,923
Seal	6,996

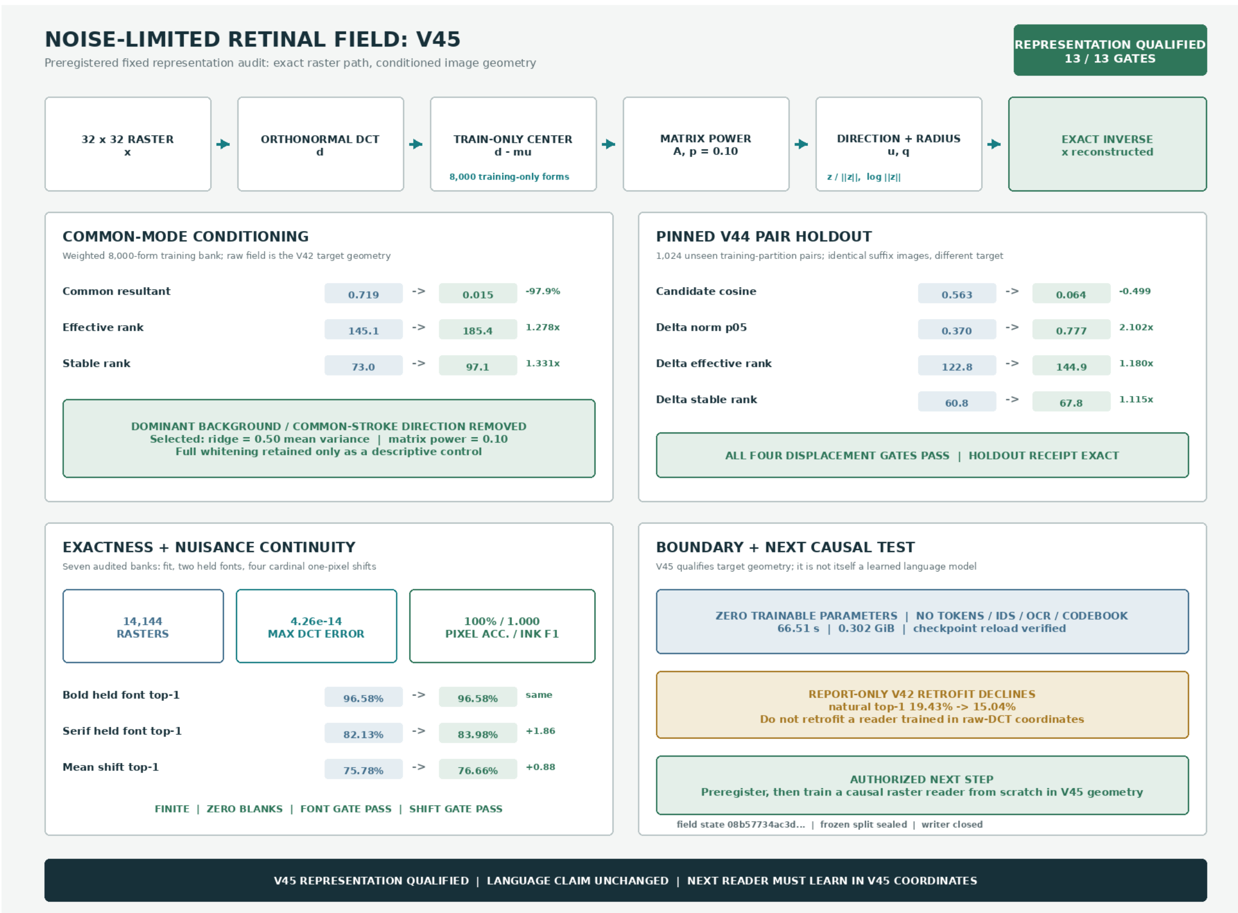

Figure 31: Measured V45 result rendered from hash-pinned report and checkpoint receipts. The fixed direction-plus-radius field is exactly invertible, suppresses the common raster mode, increases held-pair displacement magnitude and rank, and preserves held-font and translation retrieval. The lower-right retrofit is explicitly report-only: V45 qualifies a representation, not a language model.

This supports future evidence-bearing historical answer pages but was not used as a character-label channel in the RFLM run. The language trajectory corpus is a 7,017-record manifest built from 16 public-domain Chinese Wikisource book packages. The contribution layer is recorded as CC BY-SA, and the manifest stores source artifacts and rights. Strings exist only in the offline manifest and renderer; model batches contain image tensors.

Eight Noto Sans/Serif CJK fonts provide independent visual views. The run stores the exact font paths, byte counts, and SHA-256 hashes in checkpoints and receipts. Cmap intersection filtering leaves 8,282 supported visible characters; 107 unsupported unique forms account for only 0.0074% of visible corpus instances and are skipped instead of rendered as missing-glyph boxes.

For word-origin and book-layout research, 11 additional local works are registered by SHA-256 in `references/source_books/manifest.json`. They total 963,986,234 bytes and include 9,361 PDF pages plus one EPUB: English and Chinese encyclopedias, an English word-origin reference, and Chinese character- origin and oracle-form references. Relative symlinks avoid duplicating or committing the source binaries. Their rights are unverified, so they are private reference, source-comparison, and evaluation material only; they are not part of the releasable training corpus. PDF/OCR extraction may create offline sidecars, but those strings cannot enter a student batch.

7 Training Objectives

Each 48-fixation page supplies eight sampled causal positions, making the language objective denser than one endpoint per page. The complete training loss is

$$\mathcal{L} = \lambda_f \mathcal{L}_{\text{flow}} + \lambda_e \mathcal{L}_{\text{energy}} + \lambda_r \mathcal{L}_{\text{retina}} + \lambda_v \mathcal{L}_{\text{view}} + \lambda_p \mathcal{L}_{\text{endpoint}} + \lambda_s \mathcal{L}_{\text{stroke}} + \lambda_c \mathcal{L}_{\text{cycle}} + \rho(t) \mathcal{L}_{\text{roll}}.$$

After step 5,200, V7 adds a ramped correction

$$\mathcal{L}_{\text{V7}} = \mathcal{L} + \kappa(t) \left(0.5 \mathcal{L}_{\text{ctx}} + 0.2 \mathcal{L}_{\text{sample}}\right).$$

The energy term uses 512 cross-render candidate views per batch in the final run. Visual duplicate positives are discovered by image similarity rather than labels. The view term aligns independent renderings in both retinal and candidate-projection spaces. Endpoint and stroke terms improve sparse-ink geometry. The cycle term is weighted toward high-noise examples, where a writer must use recurrent context rather than visible target pixels.

For V6, $\rho(t)$ ramps from zero to one over 400 updates. A rollout subset of eight sequences uses two generated steps, two candidates, and two flow steps per candidate. This follows the distribution-shift lesson of scheduled sampling and DAgger [52, 51]; it remains image-only because the visited states, selected observations, and recovery targets are pixels and continuous visual fields. The final rollout state cosine reaches 0.961, but selected-image target cosine is only 0.103. This distinction predicts the measured outcome: state recovery becomes robust while sampled visual identity remains weak.

V7 resumes V6 for 800 updates with a 512-form, four-view image-anchor curriculum and a two-step differentiable endpoint on four examples per batch. The anchor retina is refreshed every 200 updates. The selected step 5,800 checkpoint is chosen from preregistered validation checkpoints because it has the strongest combination of independent-anchor context gain (+0.1798) and sampled pixel F1 (0.3636). The continuation adds approximately 261 seconds of training and peaks near 3.0 GiB allocated VRAM on one RTX 4090.

7.1 PVF state objectives

V8–V16 reuse the proven retina but remove the pixel writer from the state ablation. The causal GRU, proposal, and hyperspherical field are trainable. For proposal μ, target image state z^+, and image-derived candidate states $\{z_j\}$, visual identity uses

$$\mathcal{L}_{\text{proposal-NCE}} = -\log \frac{\sum_{j\in\mathcal{P}} \exp\{s\mu^\top z_j\}}{\sum_j \exp\{s\mu^\top z_j\}},$$

where positives $\mathcal{P}$ are inferred by frozen retinal similarity. A detached last-image condition supplies the calibrated context objective

$$\mathcal{L}_{\text{proposal-ctx}} = \max\{0, m - \ell(z^+ \mid h_{\text{full}}) + \ell(z^+ \mid h_{\text{last}})\}.$$

Equivalent identity and context losses are applied to states numerically sampled from the flow. Direct sampled geodesic loss prevents inference-path drift, while contrastive loss prevents collapse to the retinal mean.

V15 has 10,470,273 total parameters, 9,137,345 trainable parameters, and zero classifier or pixel-actuator parameters. It trains in BF16 with 48-image sequences and peaks at 1.181 GiB allocated CUDA memory. The full V14–V15 trajectory records 1,895.24 seconds on one RTX 4090. Step 2,000 was selected on the development image bank before running the frozen evaluator.

V16 resumes selected V15 step 2,000, adds 6,001,536 causal-memory parameters, and resets the optimizer with a new 100-step warmup. It has 16,471,809 total parameters, 15,138,881 trainable parameters, and zero classifier or actuator parameters. The 1,200-update continuation records 2,047.63 seconds, about 92–114 sequences per second, and 1.479 GiB peak allocated CUDA memory. Step 2,200 was selected before the frozen evaluator by maximum development proposal top-1 subject to full-context over last-only, positive normalized context gain, and target cosine above 0.10.

8 Image-First Training and Inference Targets

The next two figures specify the intended full-system contract; they do not claim that the rejected V30 reader can yet generate answer pages. The measured comparison is Figure 19: local spatial alignment changes scores, but exact-suffix assignment remains at chance and writer training stays closed. Presenting the target and the experiment in one figure sequence keeps the research direction visible without turning a concept rendering into evidence.

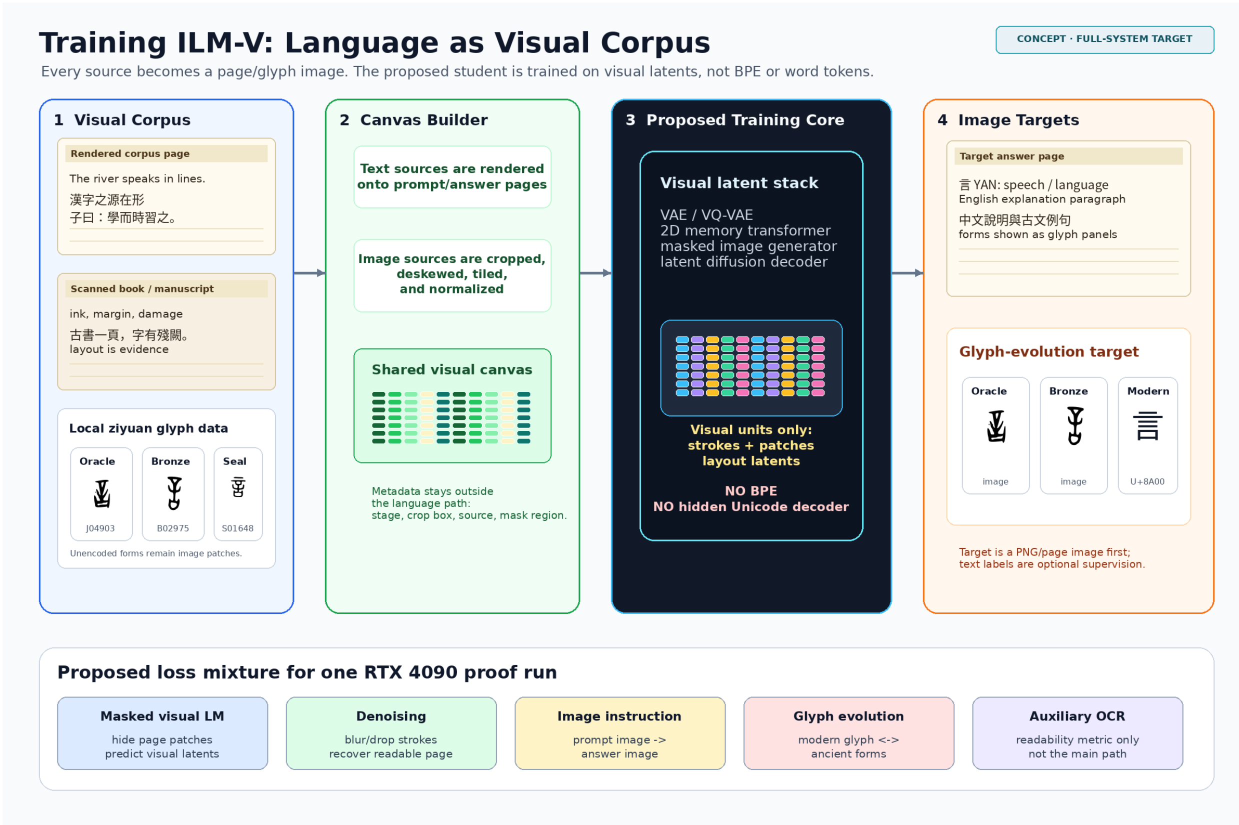

Figure 32: Proposed full-system training target for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks become visual canvases. The intended objective predicts visual latents and images, not BPE tokens. This is a design figure; Figure 19 reports the measured reader experiment.

Figure 32 makes the proposed training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the student path must learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

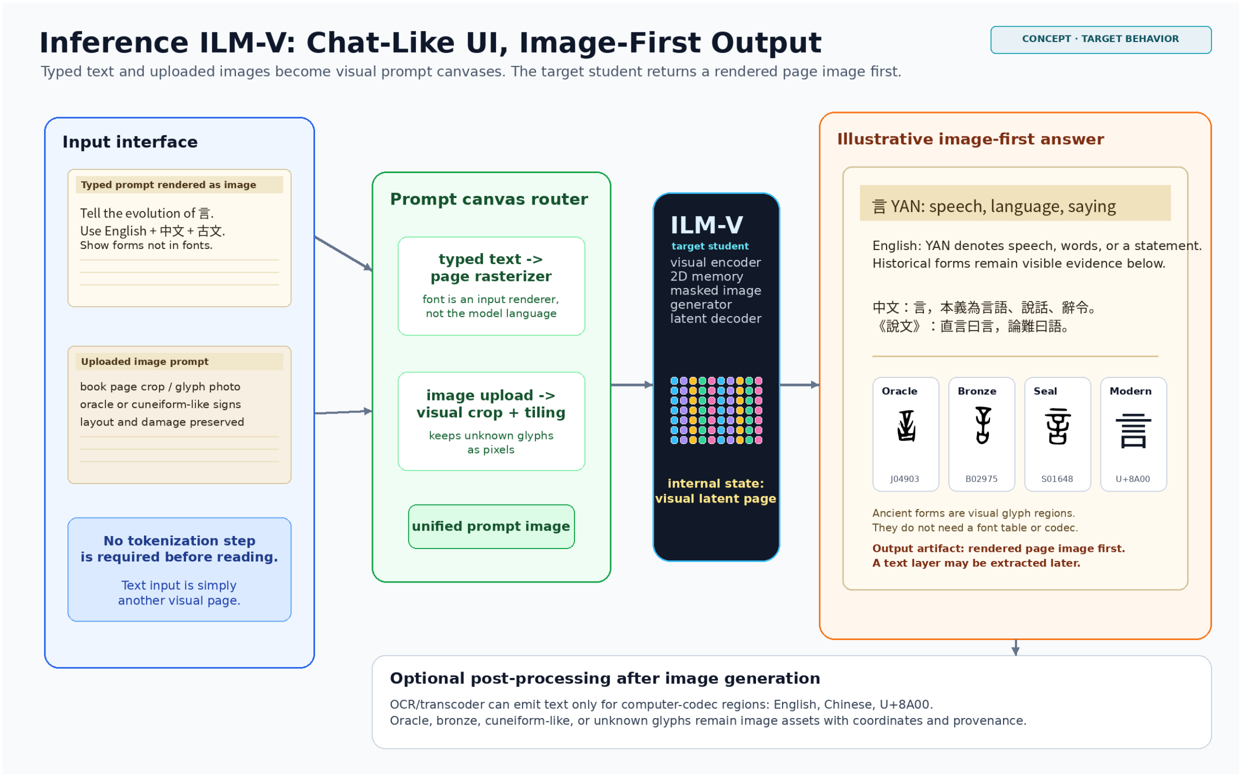

Figure 33: Proposed inference contract for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The target primary output is a rendered page image. OCR or transcoding is optional and applies only to output regions represented by computer codecs. The displayed answer is illustrative, not a V30 output.

Figure 33 shows the target user-facing path. The interface can behave like a normal LLM prompt box. Typed text is rasterized into a clean prompt band; an optional book page, inscription, or glyph crop is normalized and placed beside or below it on the same canvas. Borders, whitespace, and coordinates make the relation visible rather than encoding it in hidden IDs. A successful ILM should answer a typed question alone or a grounded question such as "What does the marked form mean, and how did it evolve?" about an attached page. It returns a page image containing modern language and historical glyph regions only after the corresponding visual-language gates pass. A later OCR/transcoder may attach a searchable layer for encodable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

The formal contract is:

$$\begin{aligned}
r_\alpha(\text{text prompt}) &\to x_{\text{text}},\\
g_\beta(\text{uploaded page or glyph}) &\to x_{\text{source}},\\
\text{Compose}(x_{\text{text}}, x_{\text{source}}) &\to x_{\text{prompt}},\\
f_\theta(x_{\text{prompt}}) &\to y_{\text{page}},\\
o_\psi(y_{\text{page}}) &\to \text{optional text layer} + \text{glyph-region metadata}.
\end{aligned}$$

Here r_α and Compose are deterministic UI adapters, not language components. Either visual region may be absent. The core f_θ receives the resulting pixels and maps them to answer image y_{page}. The optional reader o_ψ may recover coded text where possible, but unencoded historical forms remain

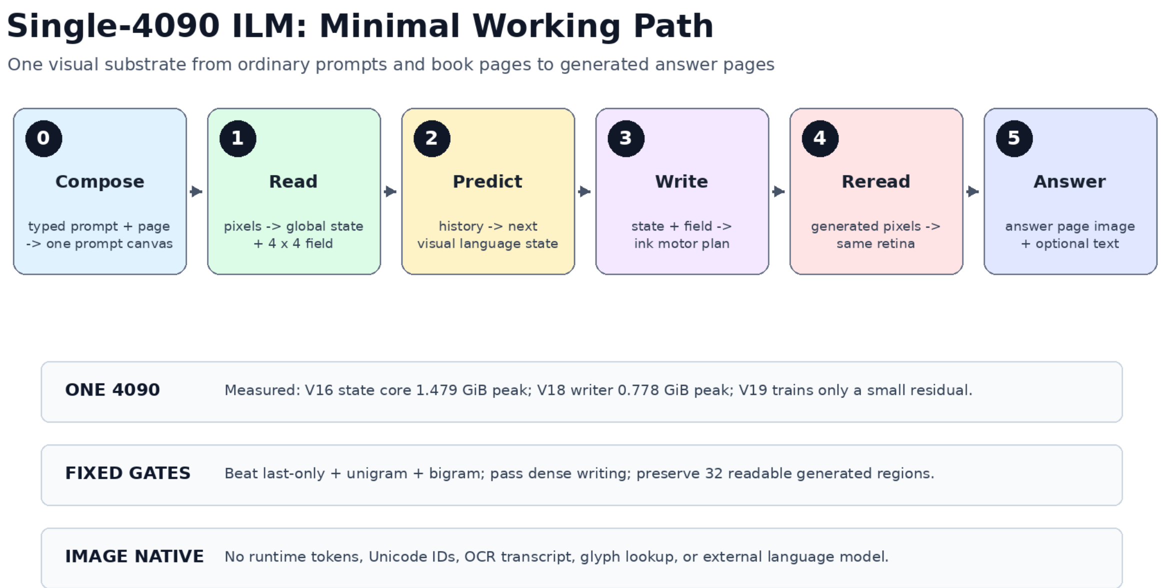

Figure 34: A staged curriculum that keeps each causal proof small enough for one RTX 4090 before any page-scale expansion.

valid image regions.

A useful answer canvas should resemble a page excerpt from a book or corpus:

- a title region rendered as text image,
- one or more modern-language paragraphs, for example English and Chinese,
- a classical-style line or quotation when appropriate,
- glyph panels for forms that may not exist in computer fonts,
- a provenance strip with source IDs or nearest retrieved exemplars.

This schema is intentionally different from "text answer plus illustrations". The generated page image is the primary output object, and the text layer is secondary.

9 Single-4090 Curriculum

Figure 34 records the broad curriculum. The executable path is intentionally simpler than the complete research space:

1. Read 32×32 writing images into one global state and one small 4×4 spatial field. V16 already provides the retina and causal state.
2. Predict the next continuous visual state from image history and require it to beat last-only, unigram, and bigram controls.

3. Draw that state with a continuity-preserving local route. V19 shows that a small optional residual is ignored; V20 makes field detail causal; V21 makes both occupancy and detail field-causal but rejects disjoint patch rasterization. The successor may overlap only bounded neighboring patches.
4. Render typed prompts and optional source pages into visual frames, then predict local/global answer fields from open question–answer trajectories. Require held-out prompt counterfactuals and prompt-shuffle controls.
5. Feed generated answer pixels back through the same retina and require a readable, prompt-dependent multi-frame autonomous rollout.
6. Add model width or a second GPU only after the preceding fixed gate passes; use the second GPU for throughput or independent replication, not to hide an unresolved objective.

This uses one retina, one causal field, one deterministic motor planner, and one write–reread loop. Stochastic surface generation, billion-parameter scaling, and broad multimodal alignment are deferred until this minimal system works. The same Visual Language Stream can later expose geometric depth and time, but 3D glyph strings and character movies are also deferred until the 2D write–reread gate passes.

The 16.47M PVF V16 state model, 5.73M V17 actuator, and 2.36M V18 motor planner are deliberately small enough for one RTX 4090. Their measured peaks are 1.479, 1.588, and 0.778 GiB, respectively, leaving substantial capacity headroom for controlled corrections. V20 reduces the causal routing test to two exactly matched 0.506M-parameter arms, peaking at 0.323 and 0.397 GiB, and cheaply rejects the writer despite positive field causality. This is not an end-to-end efficiency victory: the causal language core remains below a symbolic bigram, dense topology remains imperfect, and autonomous continuation has not passed. "More efficient than an LLM" remains an unsupported hypothesis until quality is matched on a named task.

10 Target Demonstration: Visual Word-Origin Book

The first demo should answer either a normal rendered prompt such as "Explain the evolution of the Chinese character yan" or the same question composed with a photographed book page. It must also support English word-origin questions. The target answer image should contain:

- a title and concise modern explanation,
- English and Chinese paragraphs rendered as page text,
- a classical-style line such as "yan is the visible trace of speech",
- a historical timeline with real oracle, bronze, seal, and modern glyph exemplars,
- stage labels and citations as rendered image text.

The existing Zhong figure in Figure 35 is a compact stage-timeline example; the Yan figures in Figures 1 and 33 show the fuller answer-page style. Together they define a future benchmark for combining prompt semantics, visual page composition, and historical-script generation; they are not measured V35 outputs.

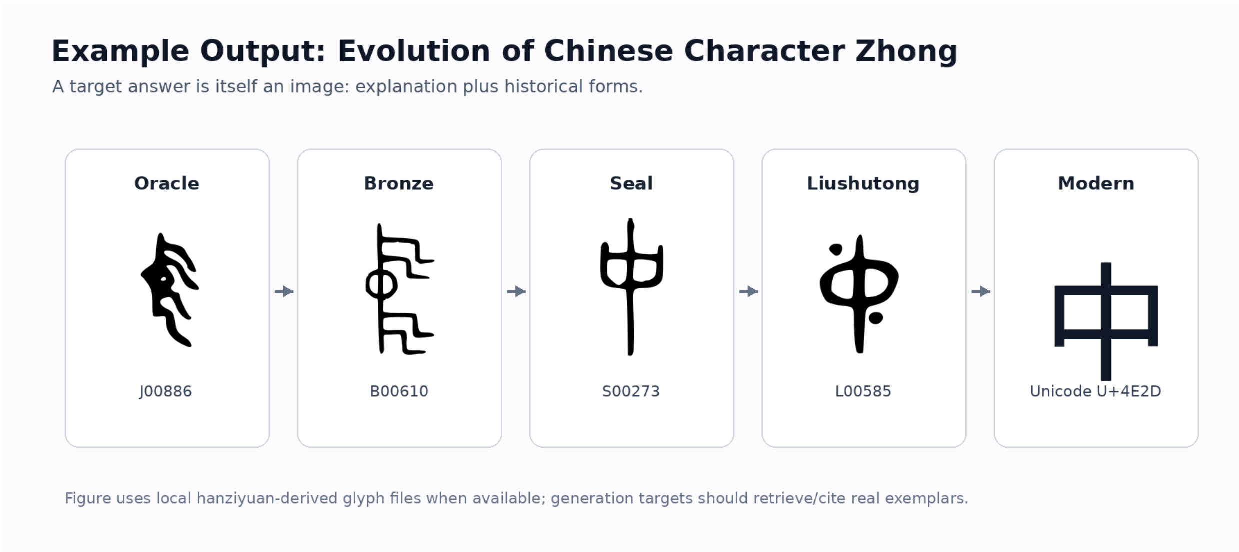

Figure 35: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

11 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

12 Evaluation

The model should be evaluated as both a language model and a visual generator:

- Fixed-bank top-1, top-5, and reciprocal rank for full context and a last-fixation ablation.
- Random-weight, chance, unigram, and symbolic bigram baselines on identical contexts.
- Cross-font retina-oracle retrieval, separated from recurrent language prediction.
- Generated-image target similarity, pixel F1, ink occupancy, variance, and energy-reranked hit rate.
- Autonomous rollout length, throughput, VRAM, and blinded human readability.
- Student-boundary audit for token IDs, Unicode IDs, OCR, codebooks, labels, and external models.
- Visual Word-Origin Book tasks: factual proposition accuracy, modern-text readability, prompt-image grounding, attested glyph-stage and source accuracy, unencoded-region preservation, and explicit synthetic-form labeling.
- Optional output transcription scored separately from the native answer image so an OCR sidecar cannot conceal a failed visual answer.

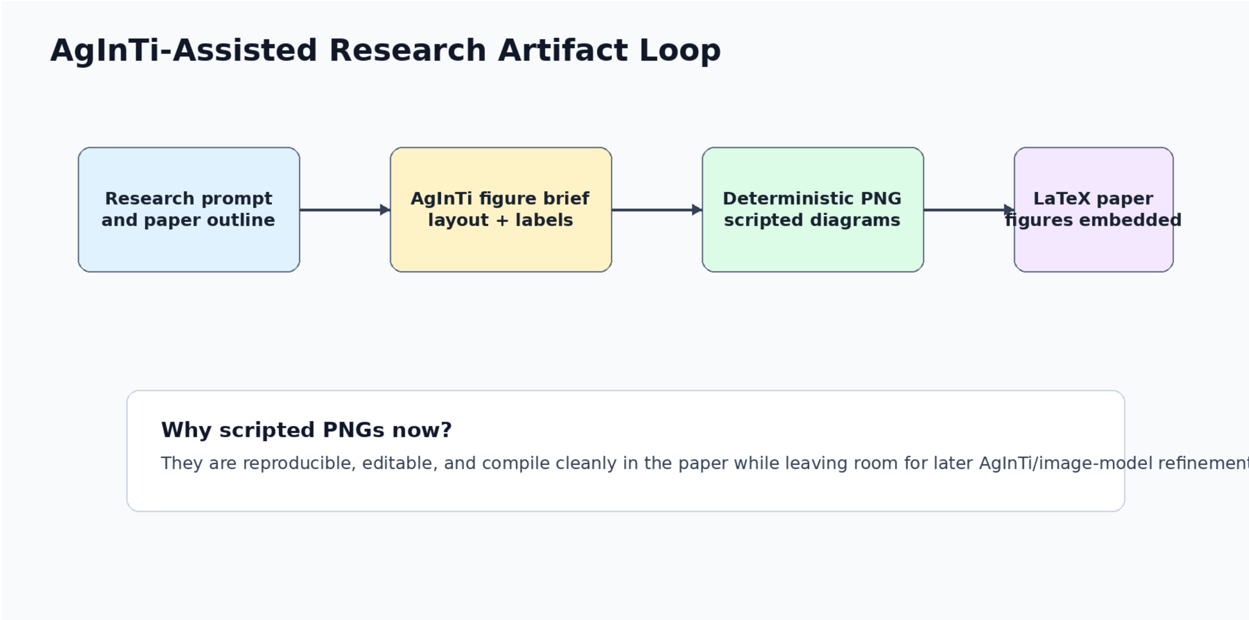

Figure 36: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

System	Measured signal	Interpretation
Whole-page flow	velocity MSE 0.851	Unreadable page texture; negative baseline
Global visual memory	top-1 2.23% over 2,016	Above random, but 0/16 historical paraphrases
Causal InkStream	validation ink F1 0.639	Autonomous repeated-grey collapse; negative baseline
RFLM V5 retina oracle	top-1 97.65%	Strong cross-font visual alphabet
RFLM V5 recurrent state	top-1 0.91%	Below last-only 1.36%, unigram 1.86%, and bigram 13.58%
RFLM V6 recurrent state	top-1 1.20%	Improved, but below last-only 1.69% and both language baselines
RFLM V6 pixel writer	context gain +0.0077	Positive but weaker than V5's +0.0211
RFLM V6 autonomous loop	ink ratio 1.168	Stable character-scale marks remain unreadable
RFLM V7 recurrent state	top-1 2.31%	Above last-only 2.02% and unigram 1.86%, far below bigram 13.58%
RFLM V7 calibration	log-probability gain −0.2155	Much better than V6's −0.9066, but full history still lowers mean target probability
RFLM V7 pixel writer	context cosine +0.0303	Passes generated-signal gate; autonomous writing remains unreadable

The fixed evaluation contains 512 common Han candidates with four independent font views

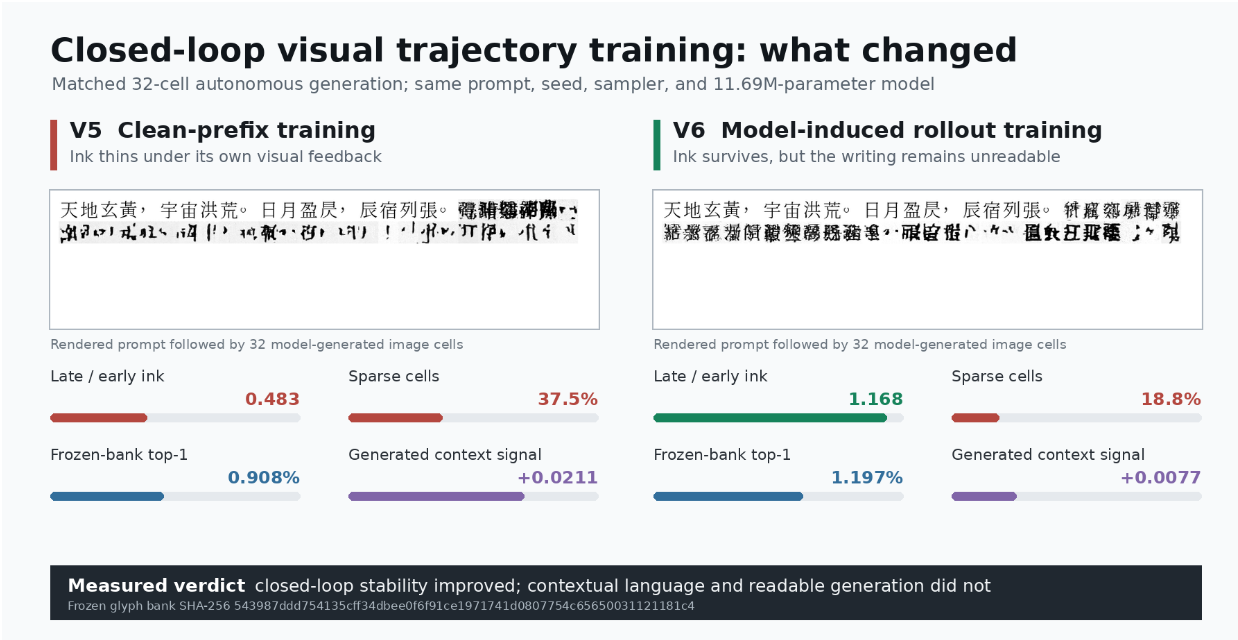

Figure 37: Matched autonomous evidence for V5 and V6. The prompt, seed, sampler, model size, and 32-cell horizon are fixed. Model-induced rollout training prevents the earlier loss of ink and halves the sparse-cell fraction, but neither continuation is readable and generated target signal weakens.

and 2,423 eligible held-out contexts. V6 and V7 use the identical bank SHA-256. Selected V7 raises full-context top-1 from 1.197% to 2.311% and top-5 from 3.219% to 5.613%, while the retina oracle remains near 98.27%. Full context now ranks above V7 last-only (2.022%) and unigram (1.857%) but remains far below symbolic bigram (13.578%). Its normalized target-log- probability gain is -0.2155, improved from a corrected V6 value of -0.9066 but still negative. Raw target-score gain stays positive in both models, showing that raw energy was a misleading metric. The formal acceptance result remains false, and retina-oracle accuracy cannot be cited as language accuracy.

PVF uses a separate frozen evaluation with the same 512 objects and four views for V14 through V16, bank SHA-256 `543987dd...1181c4`, and 1,788 eligible held-out contexts. V15 proposal top-1 is 5.872%, compared with 4.418% last-only, 1.734% unigram, 0.168% random dynamics, and 13.143% symbolic bigram. Proposal top-5 is 12.192%, and normalized context gain is +0.07069. The stochastic field obtains 3.412% top-1, 7.662% top-5, +0.03032 normalized context gain, and +0.08053 sampled context cosine gain. Random-dynamics sampled target cosine is 0.03180, versus 0.21576 for V15. The retina oracle is 98.546%.

V16 was selected at step 2,200 without reading that frozen bank and evaluated once. Its proposal obtains 6.264% top-1 (112/1,788), versus 3.971% last-only, 1.734% unigram, 0.224% random dynamics, 0.195% chance, and 13.143% symbolic bigram. Its proposal context gain is +0.07732. The stochastic field obtains 3.691% top-1 versus 3.244% last-only, normalized context gain +0.03579, sampled target cosine 0.19506, and sampled context cosine gain +0.07947. V16 adds seven correct proposal contexts over V15, while proposal top-5 and target cosine decline. This is directional evidence and not a statistically established memory advantage; paired predictions on a new frozen bank are required for attribution.

Thus the proposal and flow state gates are accepted, while the bigram language gate remains

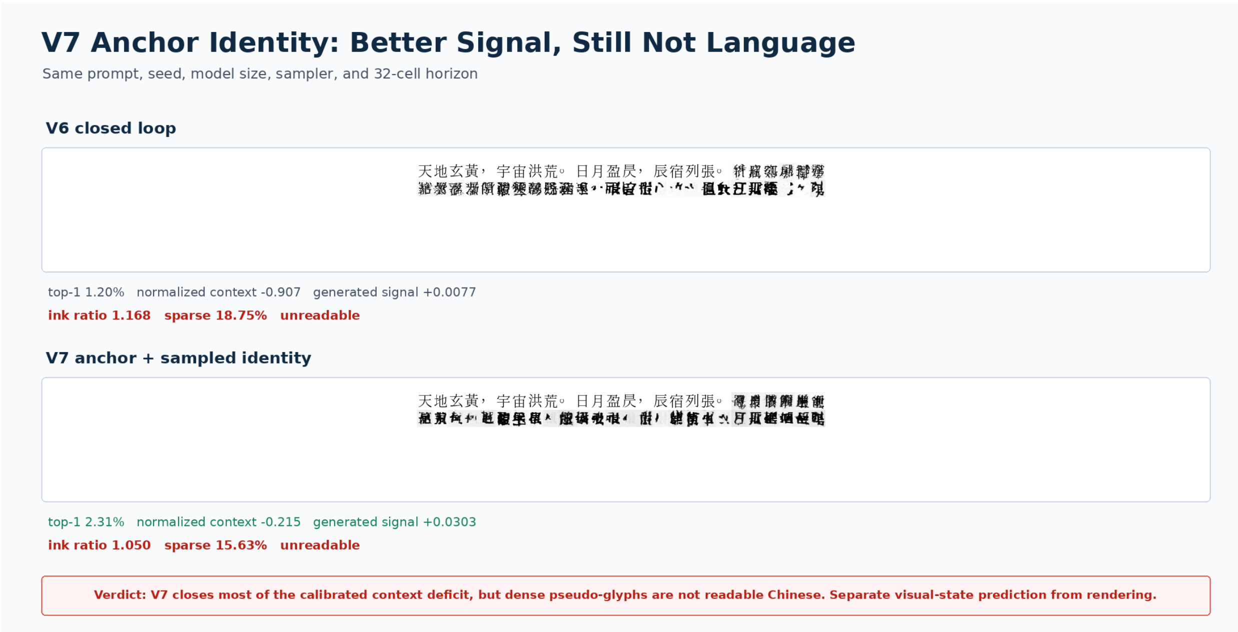

Figure 38: Matched autonomous evidence for V6 and selected V7 step 5,800. V7 improves calibrated retrieval and generated target signal while retaining stable ink, but visual inspection still finds dense unreadable pseudo-glyphs.

false. Separately, V17 accepts causal visual actuation and rejects readable topology. V18 then passes every automatic development topology gate and makes many held-out forms readable, but leaves dense-form and prospective human-readability gates unresolved. V19 then rejects an additive local-field repair: dense F1 is 0.7278, but field shuffling reduces it by only 0.0088 and field removal by only 0.0054. V20 then forces unavailable within-block detail through the field. Correct dense F1 exceeds shuffled-field by 0.1218 and zero-field by 0.3613, with exact quadrant locality, so the local-causality hypothesis passes. Writer selection still fails, and endpoint dense advantage over the equal-capacity control is only 0.0179 against a fixed 0.03 margin. V21 removes the remaining global spatial route. At its best diagnostic snapshot, correct dense F1 exceeds shuffled-field by 0.1739 and zero-field by 0.3465, identity top-1 is 79.10%, target cosine is 0.8331, locality is 1.0, and every exact-basis invariant passes. The tiled-global control collapses to repeated patches. Overall, simple, and medium F1 remain below their fixed gates, so no candidate selects and no paired, human, or frozen stage is authorized. V22 then tests the first prompt-to-answer image stream. Its candidate changes answer identity correctly for only 0.78% of query-counterfactual pairs, versus 0% for the query-blind control, while identity and writer quality are nearly equal. Every candidate selector argmax lands on the operation image. Thus the implemented stream boundary works, but an entropy-sharpened single-frame selector cannot perform the required multi-frame relation. V23 then qualifies and freezes an image canonicalizer before learning the relation. Its candidate reaches 99.80% minimum query/operation switch on a fresh paired audit; the corresponding blind controls are exactly invariant. The opaque visual review and single frozen evaluation also pass, with 99.46% frozen identity top-1 and 0.7848 pixel F1. This turns V22's interface into the first accepted bounded image-prompt-to-image-answer circuit in the project. V24 then replaces six absolute frame roles with visible headers in a 15–24-frame packet stream and closes a two-frame visual autoregressive loop. The fresh candidate reaches 99.22% query, 99.32% operation, and 99.61% generated-history switch accuracy; each corresponding blind control is exactly invariant.

PVF branch	Frozen signal	Interpretation
V14 proposal	top-1 2.52%	First proposal gate pass; above last-only 2.24% and unigram 1.73%
V14 state flow	top-1 2.35%	First stochastic state gate pass
V15 proposal	top-1 **5.87%**	Above last-only 4.42%, unigram 1.73%, and random 0.17%; context gain +0.0707
V15 state flow	top-1 **3.41%**	Above last-only 2.68% and unigram; sampled context cosine gain +0.0805
V16 proposal	top-1 **6.26%**	112/1,788; above last-only 3.97% and V15 by seven contexts; context gain +0.0773
V16 state flow	top-1 **3.69%**	Above last-only 3.24% and unigram; sampled context cosine gain +0.0795

Table 32: Frozen Predictive Visual Field results. V15 and V16 branches pass their visual-state gates but remain below the 13.14% symbolic bigram.

V17 frozen actuator metric	Correct state	Shuffled state	Gate
Global identity top-1	**58.594%**	0.977%	pass
Target cosine	**0.71301**	0.08612	pass
Pixel F1	**0.43849**	0.28001	fail (< 0.50)
Ink fraction	0.15947	0.15967	controlled
Human readability	rejected	rejected	fail

Table 33: Untouched V17 evaluation with identical style and noise in each paired branch. Causal visual identity succeeds while readable topology fails.

The opaque audit scores 47/48 and 48/48 for the two output frames. On the single frozen run, frame-1 unseen-identity accuracy is 98.67%, frame-2 label accuracy is 99.73%, and held-out-length scores are 98.47% and 100%. Every fixed gate passes. V25 then removes the packet grammar and exposes the larger gap. Full natural- Chinese visual history reaches 1.123% top-1, above 0.146% last-only and 0.342% shuffled history, but below 1.611% image unigram and 12.158% symbolic bigram. Six fixed gates reject the language model. The exploratory writer's nonblank images preserve density while obtaining zero identity top-1, so appearance stability cannot substitute for semantic continuation. The frozen partition remains sealed. V26 then separates last-glyph appearance from the preceding visual history and replaces one-point prediction with continuous future particles. It passes the full-versus-last log-probability gate, the perfect retina-oracle gate, the image-only boundary, and the resource gate. It fails the more specific causal tests: full history gains only 0.0197 nat over suffix-4 and 0.00369 nat over a suffix-preserving shuffle. Pixel-identical suffix pairs produce a 4.628 mean history-residual difference but exactly 50% correct target ranking. Thus the representation reacts to history while the predictive distribution does not use that reaction correctly. V26 rejects the proposed factorization and loss; the frozen partition remains sealed and no writer is trained. A post-hoc frozen-state probe further shows that this is not solely a particle- proposal bottleneck: a deterministic nonlinear scorer reaches 50.684% from history and 50.342% from fused state while a raw-retina identity control reaches 99.951%. This motivates jointly learning visual context representation and candidate compatibility rather than preserving V26's encoder unchanged. V27 performs that joint test. Its full-context pair assignment is 50.708%, versus 50.562% after shuffling the earlier prefix, while natural top-1 is 1.611%, below 2.051% image unigram and 12.598% symbolic bigram. The raw retina still gives 99.951% cross-font identity

V18 development motor-plan metric	Correct intent	Shuffled intent	Gate
Global identity top-1	**73.633%**	0.977%	pass
Target cosine	**0.84618**	0.07162	pass
Pixel F1	**0.65772**	0.31290	pass (> 0.60)
Ink fraction	0.17505	0.17505	controlled
Human readability	simple/medium clear	unrelated	dense forms fail
Frozen images instantiated	no		sealed

Table 34: Fresh V18 development audit with style held fixed and only continuous intent shuffled. Automatic topology gates pass, but frozen promotion is withheld because the human rubric was not numerically prespecified.

and candidate permutation error is exactly zero in the two-candidate pair control. Separately, learned 1,024-bank identity is 94.873% and misses its 99% gate; the two retrieval sizes are not directly comparable. V27 therefore rejects the global query/key compatibility route; its frozen partition remains sealed. V28 then freezes the raw retina and distributes supervision across 16 causal positions and three future horizons. The same-scope 1,024-way EMA semantic identity reaches 96.436%, 4.395 percentage points above the raw-retina route, and full target log probability gains 0.215 nat over the shuffled-prefix arm. Nevertheless, natural top-1 is 1.416%, below 1.855% unigram and 13.135% bigram, while full-context pair assignment is 49.561% versus 49.951% after prefix shuffling. The pair mean-margin gate passes, but the rank gates fail. V28 therefore rejects dense global future energy as a sufficient binding mechanism; its frozen partition remains sealed and no writer is trained. V29 then makes each candidate image cross-attend to retained per-cell context states and trains the centered full-minus-suffix increment directly. Full target log probability gains 3.102 nat over suffix-preserving prefix shuffle, but natural top-1 is only 2.344% versus 13.867% for bigram. Exact-suffix incremental assignment is 50.708%, versus 49.561% shuffled, and both-correct is 8.984%. Raw two-candidate identity is 99.951%, frozen-semantic 1,024-way identity is 96.436%, and all equality and permutation controls pass. V29 thus separates a strong global order effect from candidate-specific binding. It also shows algebraically and empirically that a shared suffix baseline cancels from the aggregate two-by-two assignment margin. The critic itself must learn the missing interaction; changing only the baseline cannot supply it. V30 makes that context state predict a candidate-independent 4×4 retinal field and tests aligned local scoring against a parameter-identical tiled-global control. Patch reversal changes spatial scores, but natural top-1 is 1.270% versus 2.490% for the global route, and exact-suffix assignment is 50.049% versus 50.586%. The spatial route is therefore sensitive to local geometry without learning a useful conditional next-image distribution. The combined result establishes that a small image-only student can learn causal language signal, that a small continuous field can control topographic generated ink, and that several visual roles can be composed into a rendered answer stream whose second frame depends on rereading the first. It does not establish useful autonomous language. V25 shows that simply replacing the hand-authored grammar with a generic causal visual field is insufficient. V26 shows that separating appearance from history is also insufficient when the objective permits changed residuals with nearly unchanged target preference. V27 shows that joint dot-product candidate compatibility is also insufficient: it preserves raw target visibility in the pair control but neither passes the separate learned full-bank identity gate nor makes earlier order useful. V28 shows that freezing that retina, improving semantic identity, and adding dense ordered future supervision can make earlier order affect probability and margin without producing correct binding. V29 shows that candidate-conditioned cross-attention and explicit prefix-increment contrast can amplify the natural order effect while exact-suffix assignment remains at chance. V30 shows that one aligned deterministic next-image field is also insufficient and is weaker than its matched global route. V31 then models a coherent multimodal continuous field distribution. Its spatial route detects order and local permutation and produces diverse context-dependent fields, yet natural path top-1 is 0.098% and exact-suffix path and autonomous assignment remain at 50.488% and 50.195%. Thus adding continuous transport to the same one-glyph target does not by itself repair language binding. V42 changes that narrow conclusion with a simpler canonical full-field objective: ordered-raster top-1 finally exceeds image unigram, symbolic bigram, and shuffled-history controls, and ordered target log probability improves by 0.17391 nat over shuffle. This is bounded positive natural-language evidence, even though exact-suffix assignment remains 0.53027. V43 then shows that a compact spatial writer can turn an exact continuous plan into 0.88240-F1 pixels, but the autonomous reader supplies an inadequate plan and a small pair pool is memorized. V44 removes that repeated-pool defect with a one-pass, frozen-base residual: consumed and unseen-pair accuracy differ by only 0.00049. It still reaches only 0.53418 on development binding and reduces matched natural top-1 from 0.19434 to 0.17432. Its learned update is negatively aligned with target displacement and moves anchors toward the corpus-common image field. The next experiment therefore changes the continuous raster geometry before changing reader or writer capacity. Only after a centered, variance-balanced, reversible field passes held-out representation and language gates should variable page length, multiple generated lines, historical form control, or raw-book scale expand the task. This order is also consistent with measured error accumulation in iterative diffusion chains [53].

13 Conclusion

Visible writing is a legitimate modeling substrate, but replacing symbolic tokens with pixels does not make language learning easy or automatically more efficient. The experimental sequence separates several requirements that are often conflated: preserving glyph form, using ordered visual context, binding a prompt relation, writing pixels, and rereading generated output. V23 and V24 establish bounded visual relation following and generated-pixel causality under fixed grammars. V25–V31 then show repeatedly that context-sensitive continuous states can fail to select the correct natural-language future.

V34 resolves one interface bottleneck. Its 7.42M-parameter continuous codec preserves held-font modern Chinese and held-family historical forms with about 0.998 ink and edge F1, without a discrete visual vocabulary, and trains in 321 seconds with 2.38 GiB peak allocated memory. This is a strong visual representation and reconstruction result. It is not evidence that the latent organizes meaning, grammar, or likely futures.

V35 tests the stronger closed-loop hypothesis using that codec and an explicitly credited PIXAR initialization. The 129.09M-parameter system completes 22,000 finite BF16 updates in 2.77 hours with 2.899 GiB peak allocated memory on one RTX 4090. Its packaged inference path receives pixels and a visual-position mask, emits binary raster patches, and feeds only decoded, thresholded, and re-encoded visible patches into later context. It contains no tokenizer, Unicode or character IDs, OCR, retrieval, codebook, lookup table, or runtime teacher. This makes V35 a complete and reproducible raster-input/raster-output experimental runtime.

The language result is nevertheless negative. Autonomous copy character accuracy is 0.3125%, correct copy prompts exceed shuffled prompts by only 0.117 percentage point, and copy target preference is 62.5%. Autonomous instruction accuracy is 0.1116%, lower than the shuffled-prompt condition. Generated strips are finite, nonblank, and responsive to pixel interventions, but their marks are corrupted and not bound to the requested target. V35 passes 5/12 visual-causal gates and 5/9 semantic-raster gates and is `not-qualified`. The sealed split remains unopened. Training loss, nonblank pixels, and nonempty OCR are not substitutes for counterfactual semantic evidence.

V36 then tests the isolated answer-level plan before another raster writer. Its 93.47M-parameter image-only runtime completes 6,000 updates in 25.63 minutes with 1.541 GiB peak allocated memory. Counterfactual assignment reaches 78.57% and paraphrase top-5 reaches 33.33%, but EMA top-1 is only 1.02%, raw top-1 is 2.04%, and the conjunctive gate passes 13/23 checks. The mask audit additionally exposes a 37.31-versus-11.53-patch train/development length shift and effective target rank 4.77/768. V36 is therefore also `not-qualified`; neither its sealed split nor its renderer is opened.

V37 implements the resulting clean-mask route with end-to-end semantic distillation. Its 89.77M-parameter image-only runtime completes 8,000 updates in 61.59 minutes with 2.748 GiB peak allocated memory. Prompt top-1/top-5 rises to 47.45/77.55%, and candidate-independent answer plans reach 20.41/45.41% with 0.3377 MRR. Counterfactual assignment is 93.88%, paraphrase top-5 is 73.33%, and plan effective rank is 55.37. These are substantial improvements over V36, but paired cosine, absolute answer retrieval, cyclic and intervention margins, font and wording invariance, and length still fail. EMA passes 20/33 conditions and raw weights pass 19/33. V37 is `not-qualified`; sealed evaluation and V37-R remain closed.

V38 then tests paired visual-path alignment and removes the angle-limited answer transition. Its 90.75M-parameter image-only runtime completes 8,000 updates in 99.71 minutes with 2.969 GiB peak allocated memory. Prompt top-1/top-5 reaches 60.71/86.73%, held-font prompt consistency reaches 0.7925, and the mean prompt-answer cosine falls from V37's 0.9972 to 0.5769. The actual conditional answer remains weak: top-1/top-5 is 21.94/49.49%, paired cosine is 0.2441, and held-font and paraphrase answer consistency remain below threshold. EMA passes 25/39 conditions; raw weights do not repair the result. V38 is `not-qualified`; zero sealed rows were rendered and no raster renderer is authorized.

V39.1 tests whether an ordered answer trajectory repairs this binding failure. Its corrected first-stop process calibrates expected span count to 6.303 against 6.313 target, raises count-mode accuracy to 15.22%, and reduces visual-length MAE to 4.915. Those mechanical gains do not become language: answer MRR falls to 0.07406, segment MRR is 0.01234, and transition-direction cosine is 0.00648. The full-bank run is therefore rejected before compute is spent.

V41 then separates the visual motor from semantic selection. A pinned 22.76M-parameter MX-Font generator projects exact and imperfect V34-decoded source images into a held style. On ten glyphs, clean V34 projection improves from 0.4318 to 0.5479 target ink F1; a $\sigma = 0.05$ latent perturbation improves from 0.4317 to 0.5461 and retains 99.66% of clean projected-motor F1. Every output is finite and nonblank, and the audit uses 0.772 GB peak allocated CUDA memory. This qualifies an image-conditioned glyph motor. Since the source glyph image is supplied, it does not qualify language, continuation, or question answering.

V42 performs the missing autonomous canonical-content test. Its 24.35M-parameter image-only causal core trains from scratch on public-domain Chinese raster order in 22.91 minutes with 0.63194 GiB peak allocated memory. Full next-image top-1 reaches 19.9707%, above image unigram (1.4160%), symbolic bigram (12.2559%), and shuffled earlier history (18.3594%). Ordered target log probability gains 0.17391 nat over shuffled history. This is the project's first bounded positive natural-language result. Exact-suffix arm accuracy (53.0273%) and generated pixel F1 (0.37308) fail, so V42 does not establish reliable binding or readable autonomous writing.

V43 tests those failures without changing the gates. A 3,000-update pair stage and 5,000-update spatial-flow stage produce a 30.04M-parameter image-input, image-output runtime in eight minutes total, peaking at 1.19858 GiB. It preserves every ordered-language control win and raises generated pixel F1 to 0.44507, but exact-suffix assignment is only 54.4922%; V43 passes 8/10 and is `rejected-or-partial`. Post-result evidence shows 99.22% seen-pair accuracy but only 58.30% on unseen train pairs, while an evaluator exact ink plan lets the unchanged writer reach 0.88240 F1. Pair memorization and the autonomous visual plan, not writer capacity, are the immediate limitations.

V44 tests the prescribed frozen-reader correction with a 1.736M-parameter long-history tangent residual and one nonrepeating pass over 24,000 raster pairs. Training takes 210.87 seconds and 0.19848 GiB peak allocated memory. Consumed and unseen-train arm accuracy is 57.47% and 57.42%, so the V43 memorization gap is removed, but fixed development binding is only 53.42% and matched natural top-1 falls from 19.43% to 17.43%. V44 passes 8/14 and is `rejected-or-partial`. No residual-strength interpolation reaches the 60% gate; the learned displacement has negative cosine with target displacement and increases alignment to the corpus-mean image field. This localizes the next bottleneck to target representation rather than adapter scale, pair count, or spatial-writer capacity.

V45 executes that representation test with a fixed, invertible fractional matrix-power field fitted from 8,000 training-only rasters. It passes all 13 preregistered gates: common resultant falls from 0.7190 to 0.0152, fit-field effective rank rises from 145.08 to 185.38, all four pinned held-pair displacement gates pass, held-font and one-pixel-shift retrieval are retained, and all 14,144 audited rasters reconstruct exactly at binary resolution. The zero-parameter audit takes 66.51 seconds and 0.30211 GiB peak allocated memory. This qualifies V45 as target geometry. A report-only post-hoc V42 retrofit reduces natural top-1 from 19.43% to 15.04%, rejecting a coordinate shortcut and requiring the next causal reader to learn in V45 geometry from initialization.

The strongest supported claim remains narrow but is now positive at the language-control level: one consumer GPU can train from scratch a 24.35M- parameter image-only causal model whose ordered Chinese raster history predicts the next glyph image better than matched unigram, bigram, and shuffled-history controls. A 5.69M-parameter spatial writer can render a supplied continuous ink plan at high quality. Their current autonomous composition does not pass counterfactual binding or readable-pixel gates. No Qwen, GPT, or general-LLM parity is shown, and no matched evidence supports compute superiority over token models. Runtime independence does not erase provenance: V34, V42, V43, and V45 are project work, as is the rejected V44 diagnostic; PIXAR and Pixel-Linguist provide external initialization to other branches; BGE-M3 supplies offline target geometry; Qwen-family models prepare V38 training paraphrases; and MX-Font supplies the V41 external motor. None is an uncredited ILM contribution or a deployed language teacher. Unstated upstream weight licenses prevent redistribution of the derived V35–V39 artifacts; V41 provenance is recorded separately.

The next bounded experiment does not enlarge or retrofit the old reader and does not open the writer. It must be preregistered, initialize a new causal raster reader, replace V42's raw normalized DCT target and feedback with V45 direction plus radius, and use the same public-domain stream and matched V42 language controls. Only a development result that preserves the ordered-raster signal and improves counterfactual binding may reopen the V43 writer or frozen partition. Stronger external foundations and offline teachers remain admissible when provenance, licensing, training role, and deployed absence are exact.

A Visual Word-Origin Book remains the product target: rendered questions should produce readable answer images containing modern explanation and provenance-linked historical forms, including forms absent from computer encodings. V42–V45 still predict or qualify only one canonical next-glyph interface and do not answer such questions. Page-scale reading, historical answer composition, 3D writing, motion, speech, and broad multilingual generation remain future extensions of the same sensory interface and require separate evidence.

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